C9555 A Practical Guide to Simulations: Visualization with Blender

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Motivation - Figures for Articles, Conference Presentations

- High-impact papers increasingly feature state-of-the-art visualizations

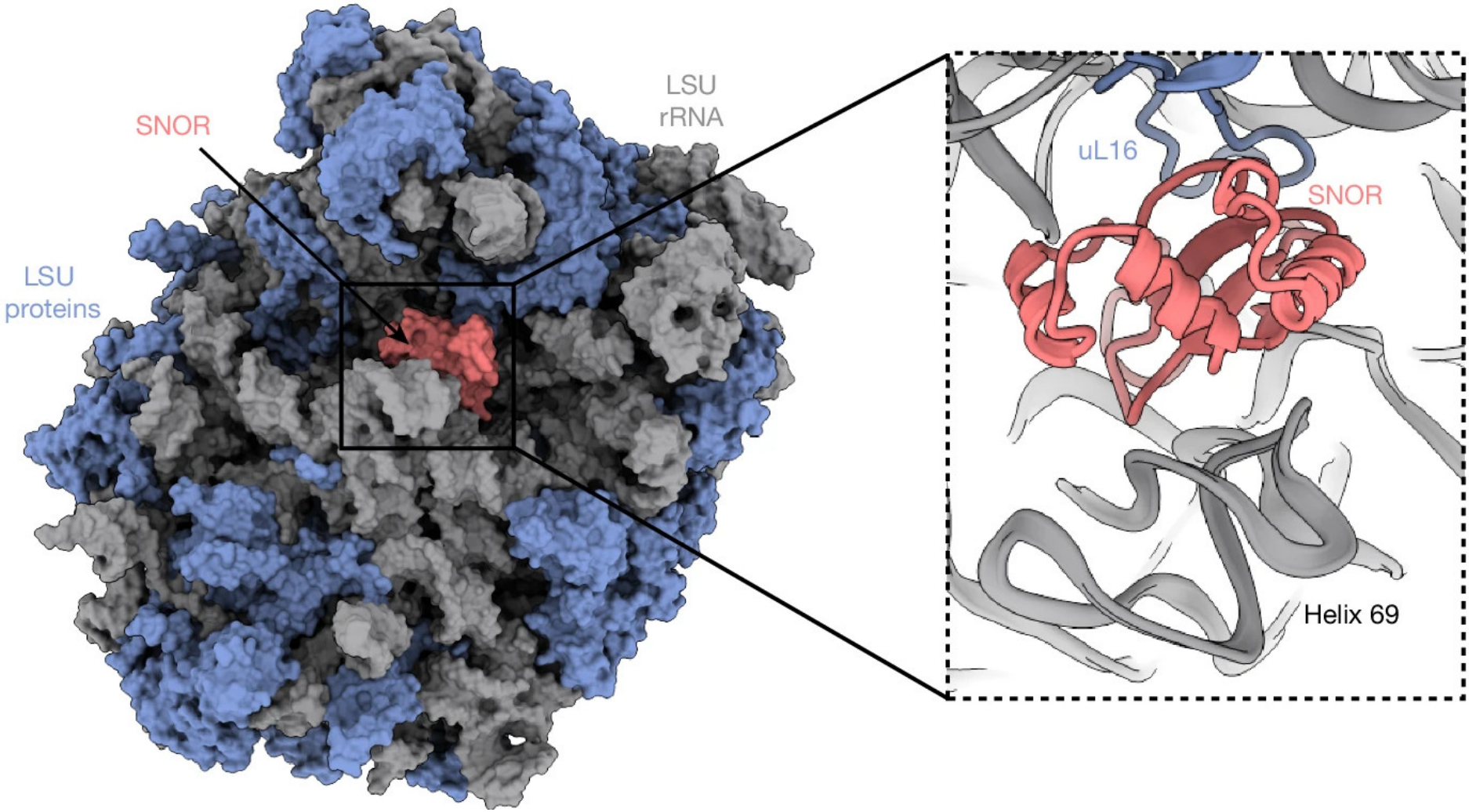


Figure: SNOR, an SBDS domain-containing ribosome-associated factor. Made in chimeraX. DOI:10.1038/s41586-026-10530-7

Motivation - Covers

- A journal cover boosts the visibility of your research

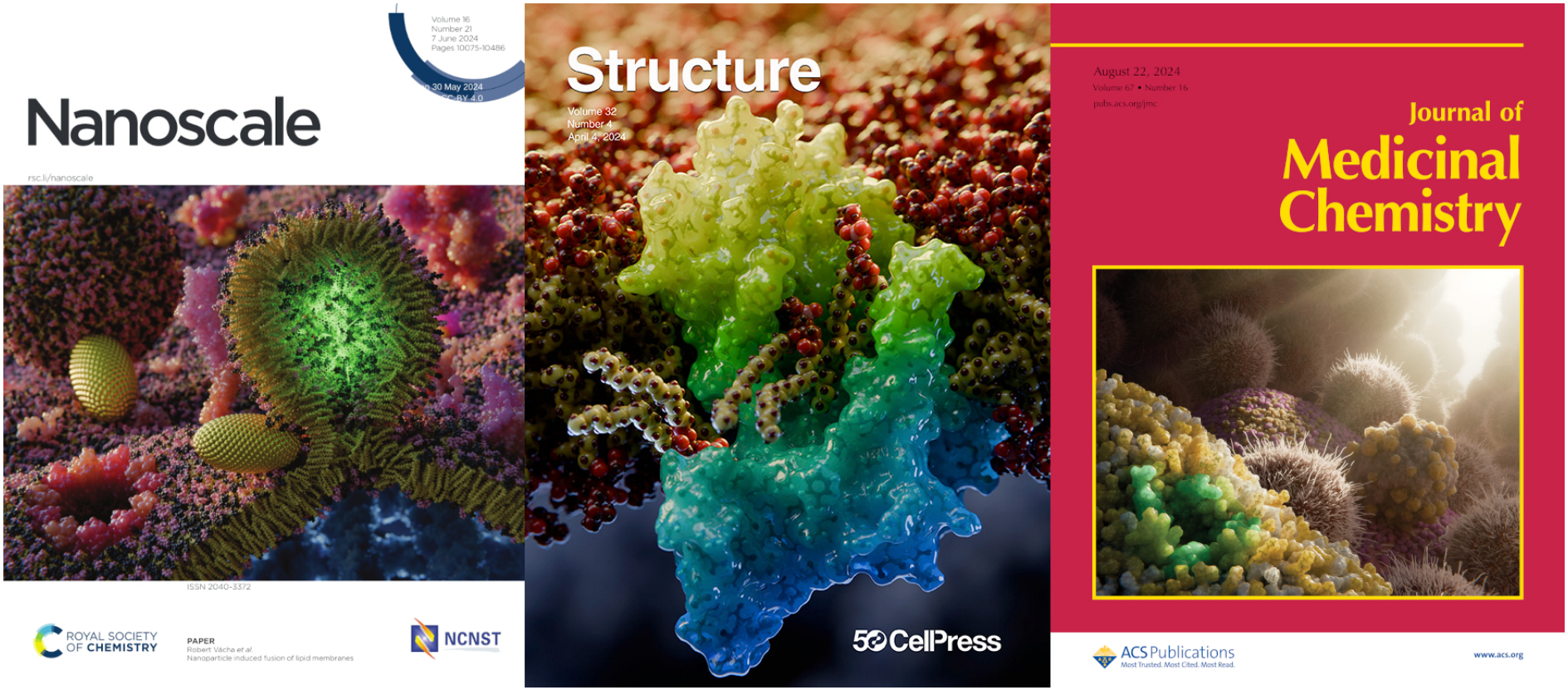


Figure: Nanoparticle induced fusion of lipid membranes - 10.1039/d4nr00591k
Insertases scramble lipids - 10.1016/j.str.2024.01.012
Computational Design of Pore-Forming Peptides - 10.1021/acs.jmedchem.4c00912

How to Create This Image?



Lecture Outline

Steps

- Setup remote desktop on sokar cluster
  - Only necessary in Wolf teaching room, since PCs lack a GPU
- Download and install blender
  - Open Default.blend, Enable and setup EasyHDRI.py
  - Setup Optix, Install molecular nodes extension
- Example Sphere :: vmd, chimeraX, blender
- Example Connext 46/50 (pdb 7JM9) :: vmd, chimeraX, blender
- Example SNOR (pdb 9PHC) :: vmd, chimeraX, blender
- Example Human TRiC (pdb 7LUM) :: blender
  - Structure of a scene in blender
  - Molecule model, Lighting, Camera setting, Shaders, Post-processing
- Render Setup - Resolution, Max Samples

Setup Remote Desktop on Sokar Cluster 1

In Wolf Teaching Room - to run blender remotely

- `site activate sokar`
- `mkdir workdir`
- `cd workdir`
- `psubmit ncbr_workshop gui ngpus=1 ncpus=12 mem=50gb`
- `pinfo`
- `pgo`
  - This command will write out instructions how to run vncviewer
- Open a new terminal
  - The instructions will specify `<VNCID>`:
  - `module add tigervnc`
  - `vncviewer <VNCID>`

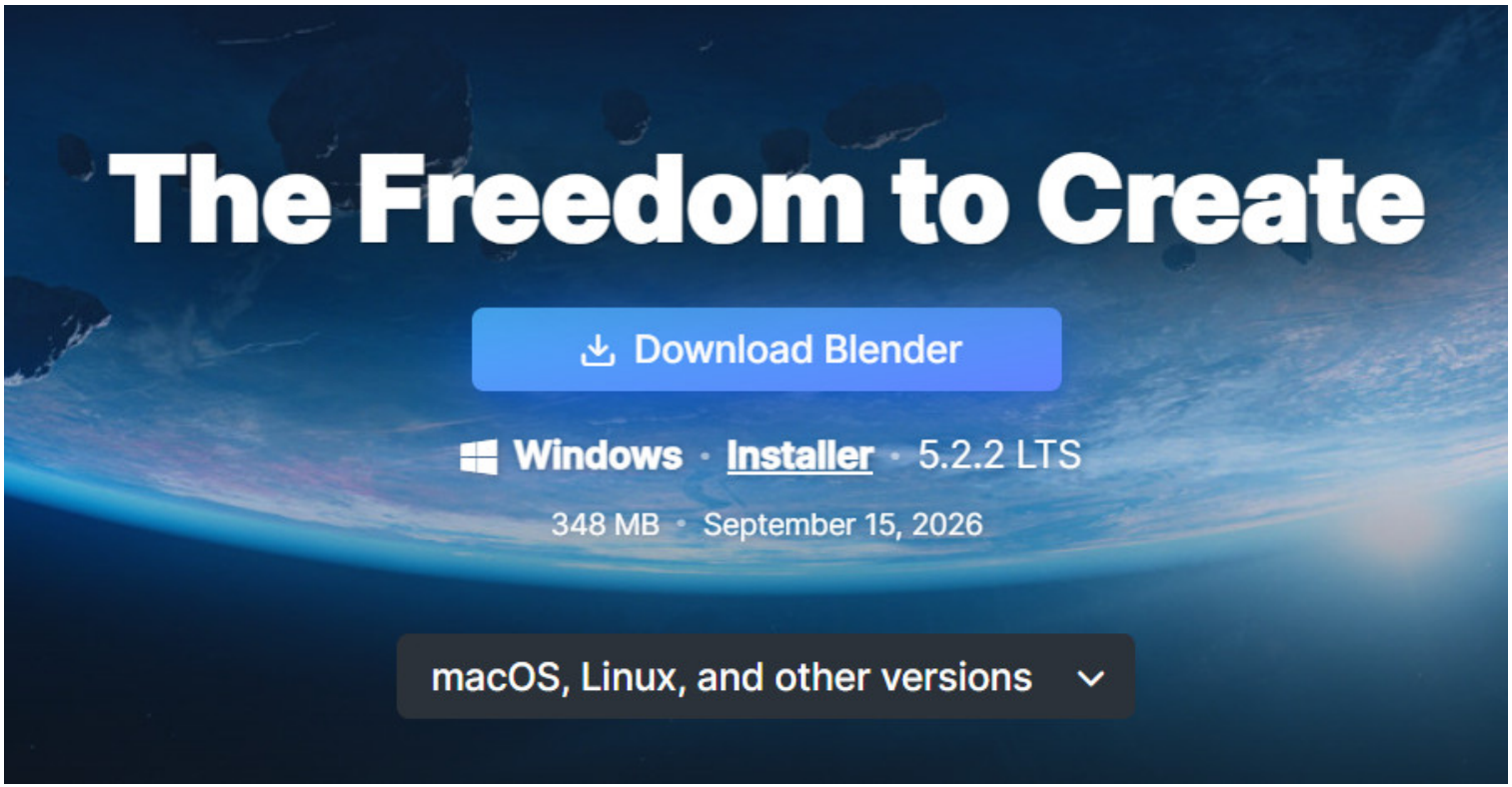
Setup Remote Desktop on Sokar Cluster 2

vncviewer

- open xterm
- `pwd`
- Your directory is scratch on a sokar machine
- You need to copy setup_blender.sh and run_blender.sh
- from `/storage/wolf/home/xsukenik/`
- execute setup_blender.sh once
- run blender: `./run_blender.sh`

Download Blender

- Download blender 5.2.2
- `https://www.blender.org/download/`
- Untar
- cd `blender-5.2.2-linux-x64`
- run blender: `./blender`



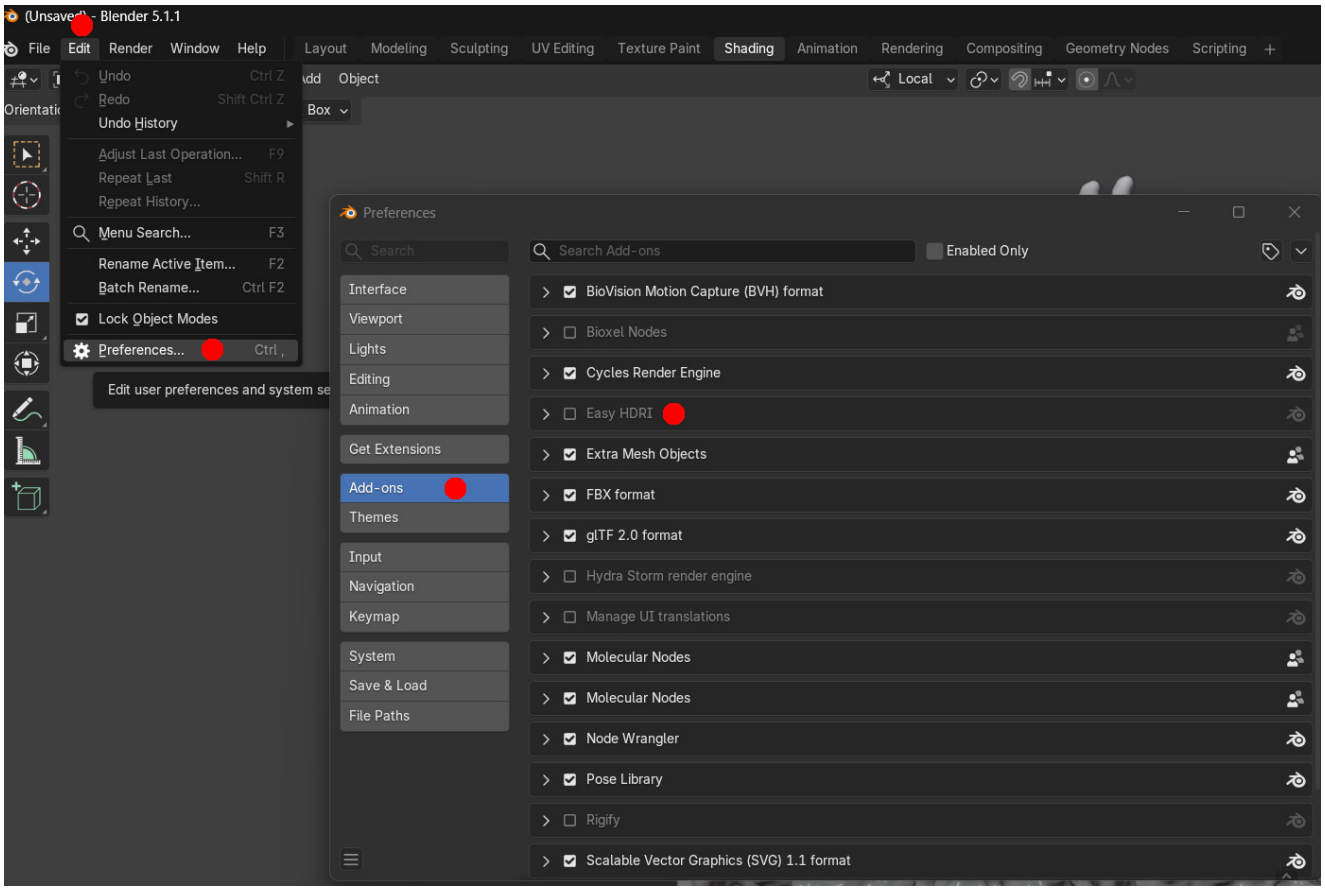
Install Blender

Steps

- Copy EasyHDRI.py, Default.blend, other stuff
- Run blender
- Enable Easy HDRI
- Open Default.blend
- Setup EasyHDRI path
- Configure use of optix on GPU
- Install Molecular Nodes

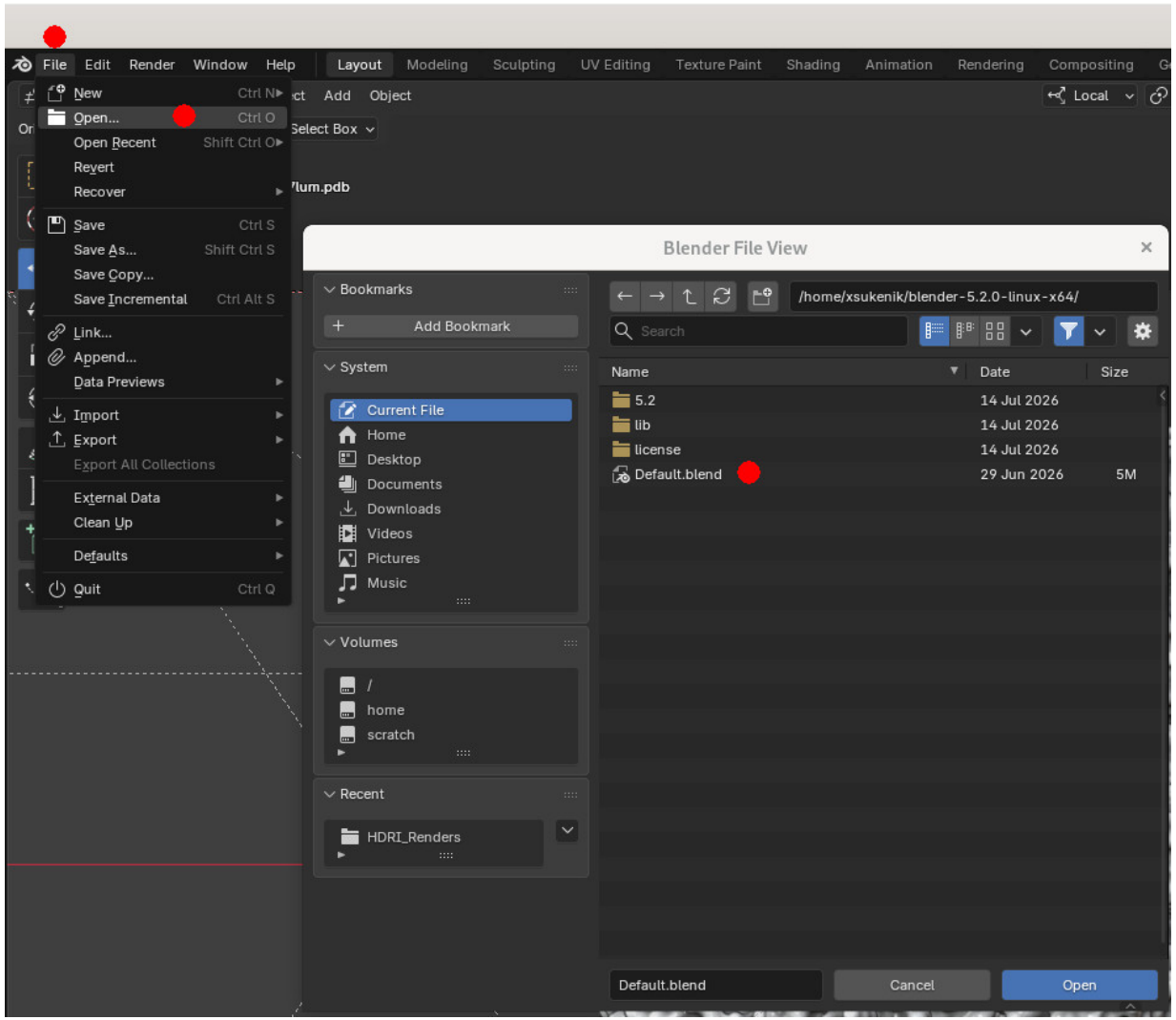
Install Blender: Setup EasyHDRI

- $ copy lecture files /storage/wolf/home/xsukenik/C9555/*
- $ move EasyHDRI.py to blender-5.2.2-linux-x64/5.2/scripts/addons_core/
- in blender: Edit/Preferences - Add-ons : enable Easy HDRI



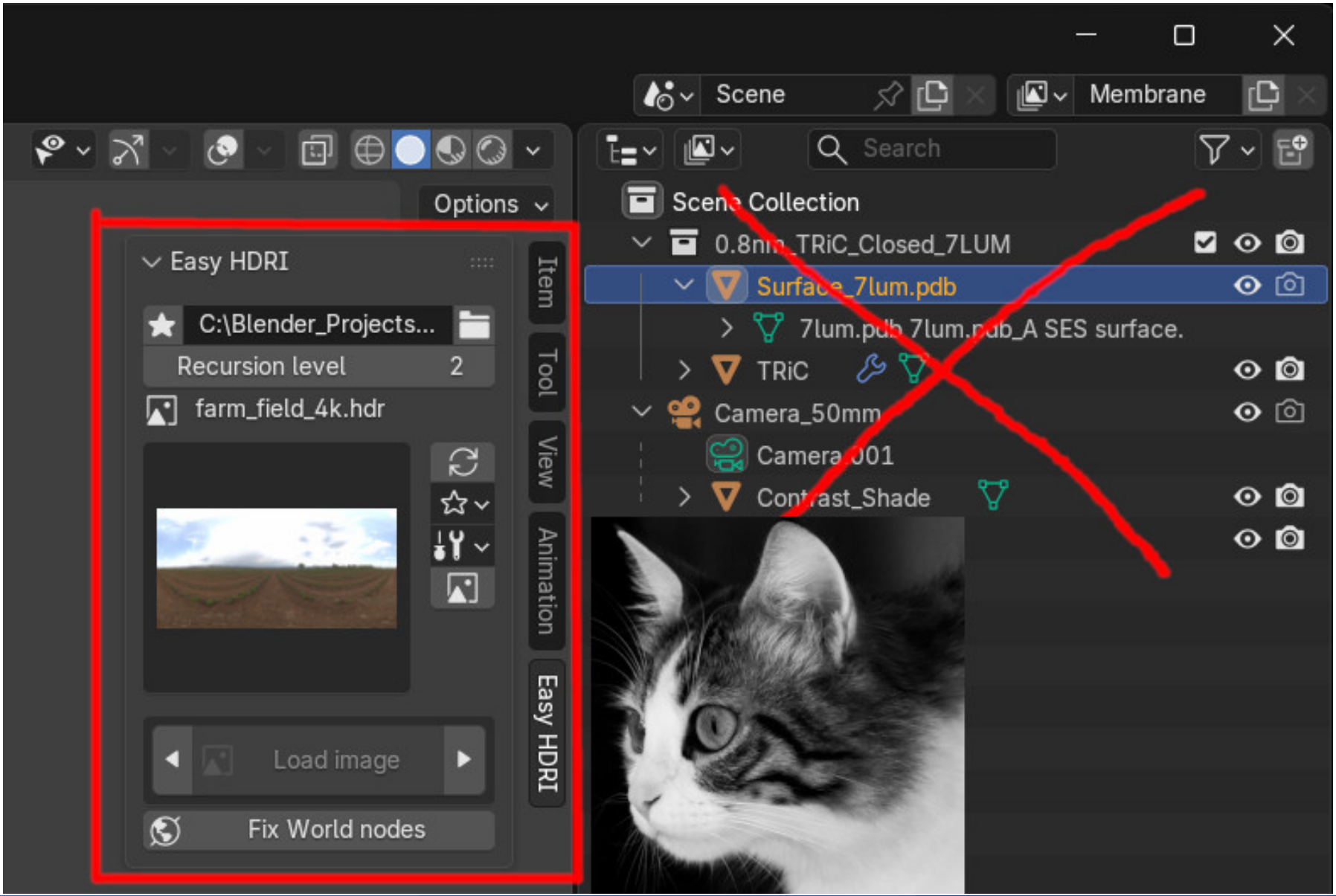
Install Blender: Open Default.blend

- in blender: File/Open locate your local copy
- The path and files in the folder will be different than in the image



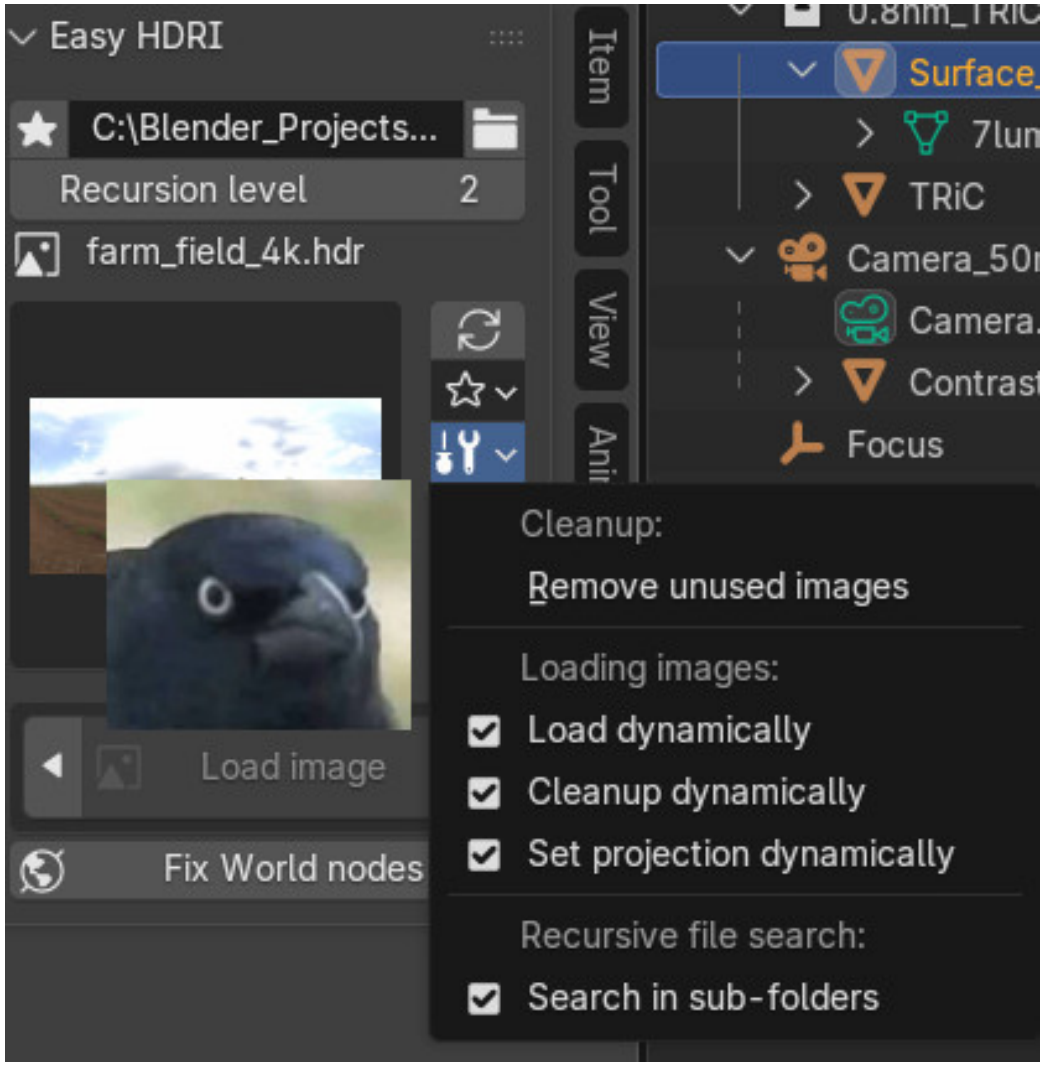
Install Blender: Where is EasyHDRI

- Locate your folder with HDRIs - start by clicking next to `C:\...`

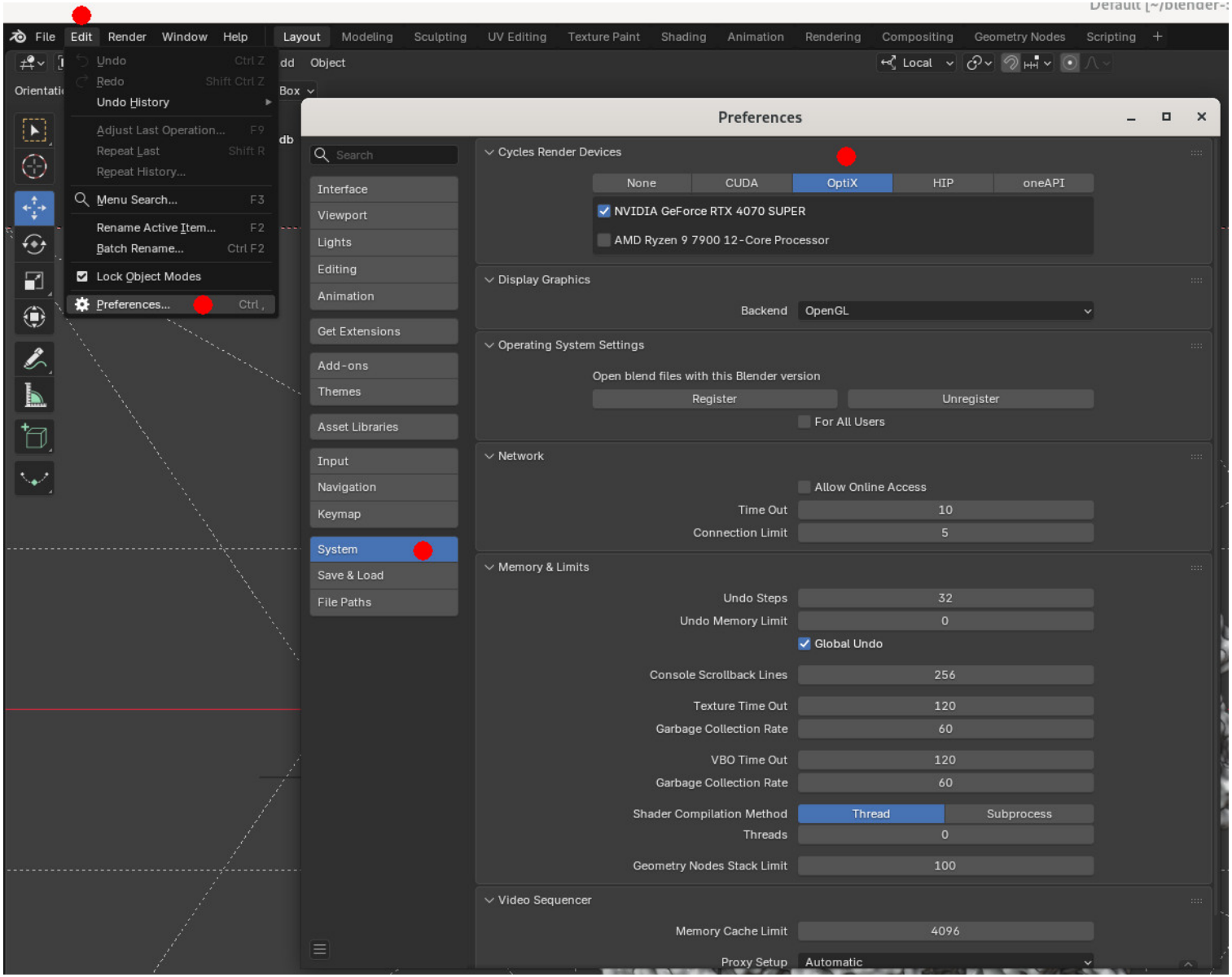


Install Blender: Annoying buggy EasyHDRI

- EasyHDRI sometimes doesnt load the HDRIs, just uncheck and check these 4 things
- Load, Cleanup, Set projection dynamically + Search in sub-folders

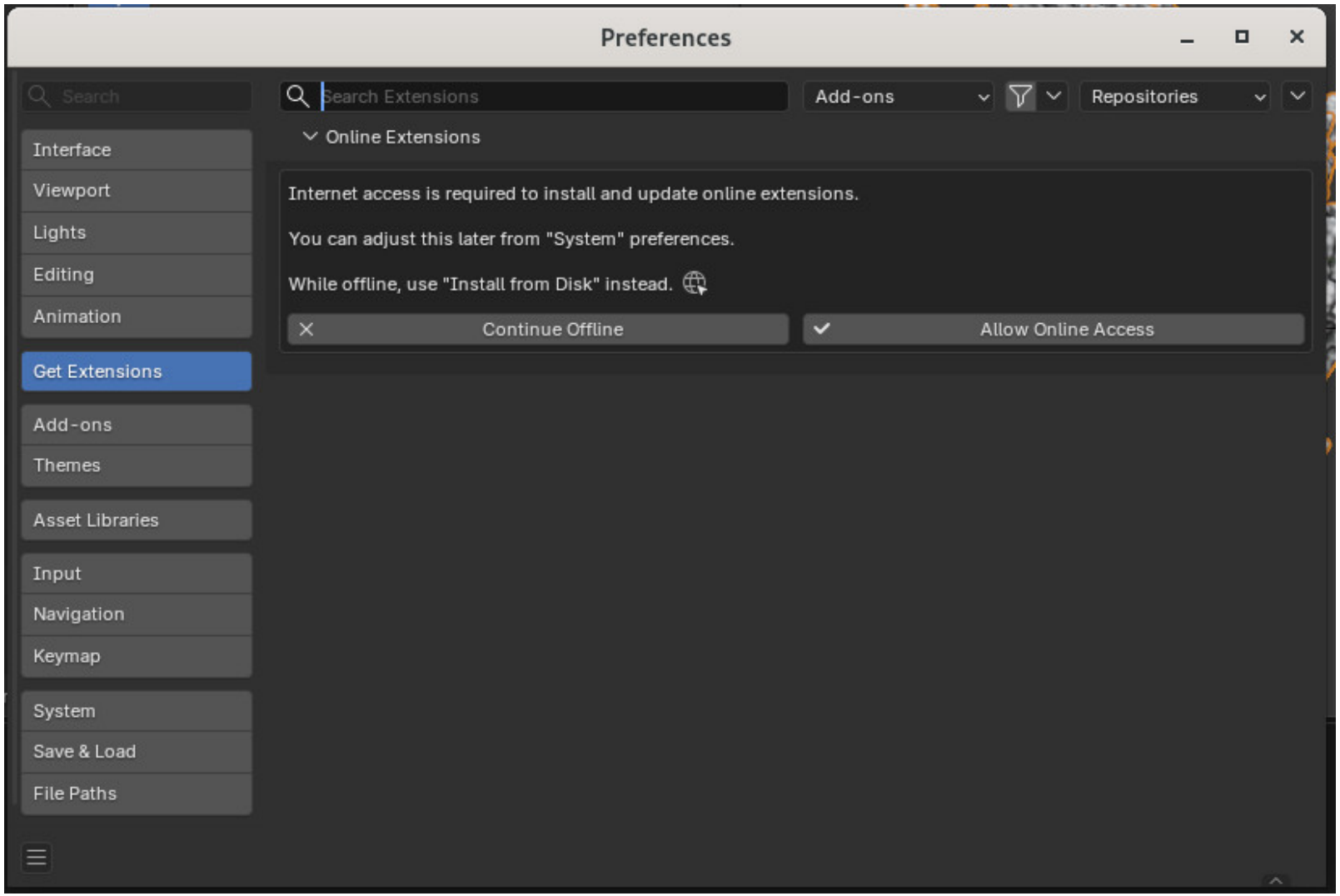


Install Blender: Configure Optix

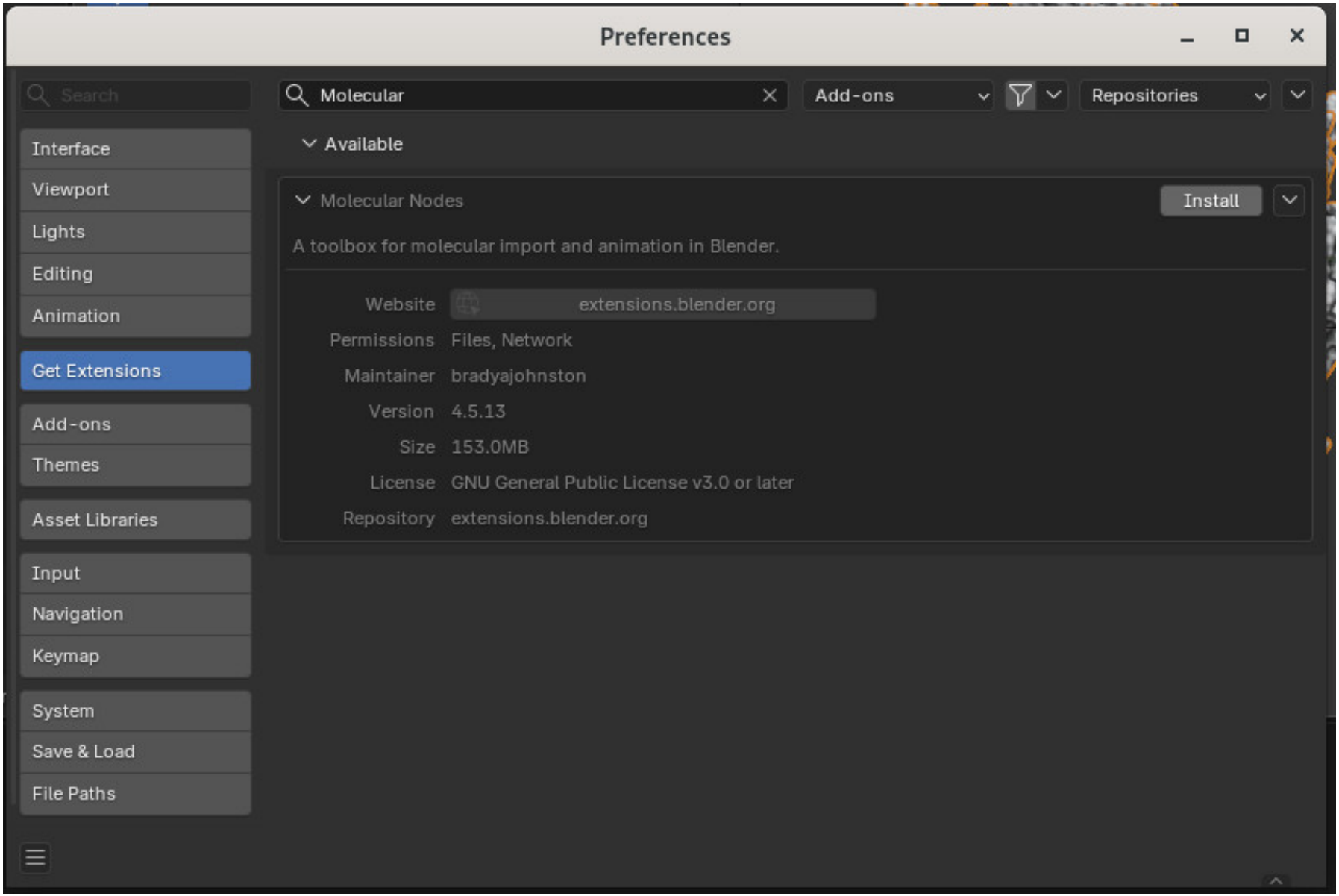


Install Blender: Molecular Nodes

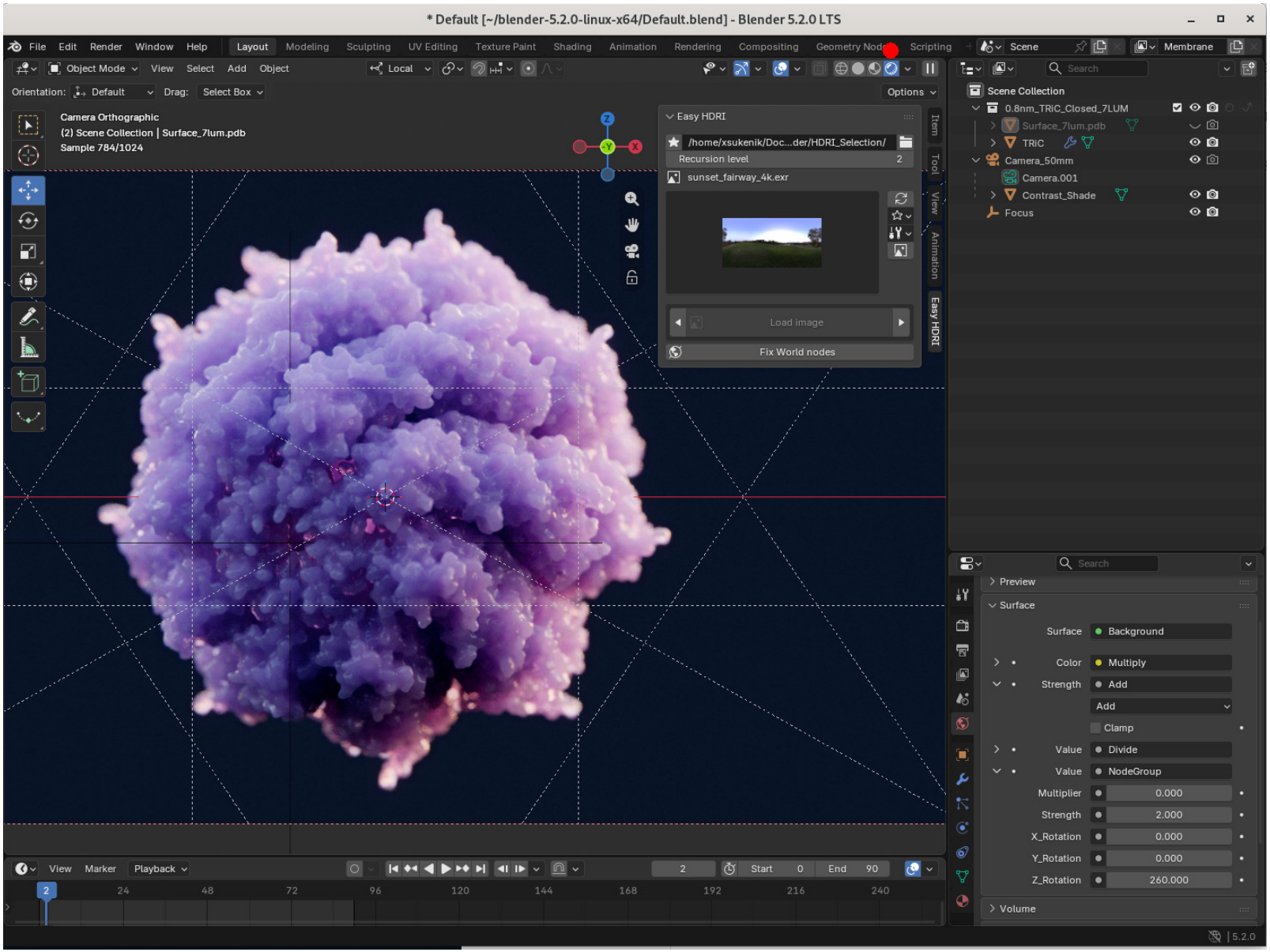
- Same as previous slide, go Edit - Preferences - Get Extensions
- Allow Online Access
- Search molecular nodes



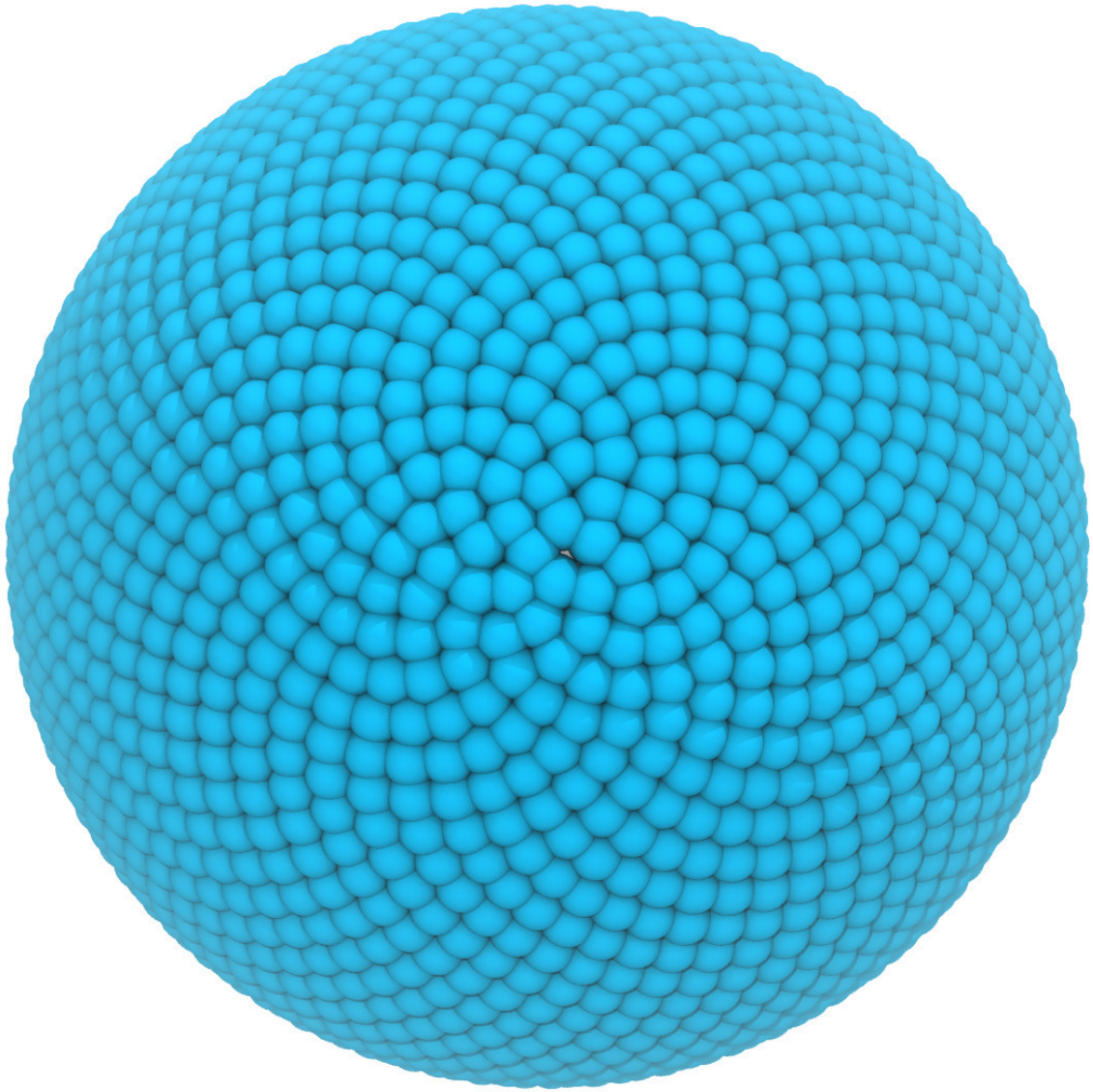
Install Blender: Install Molecular Nodes



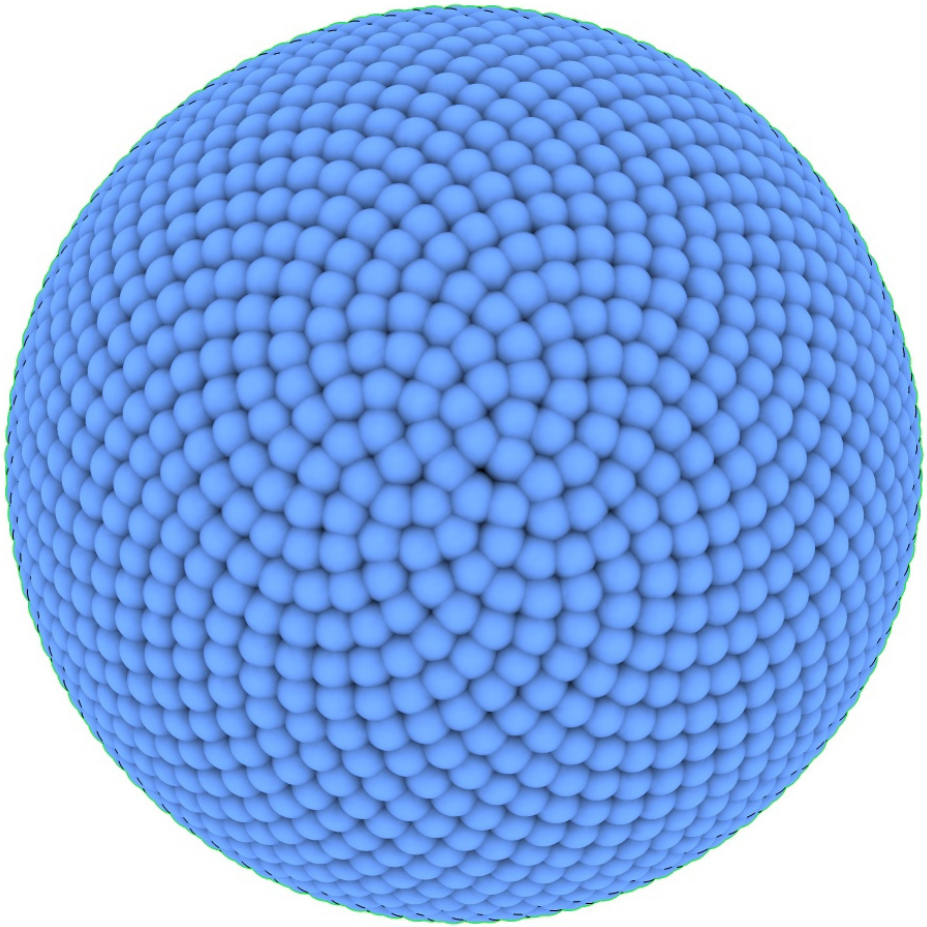
Install Blender: Success

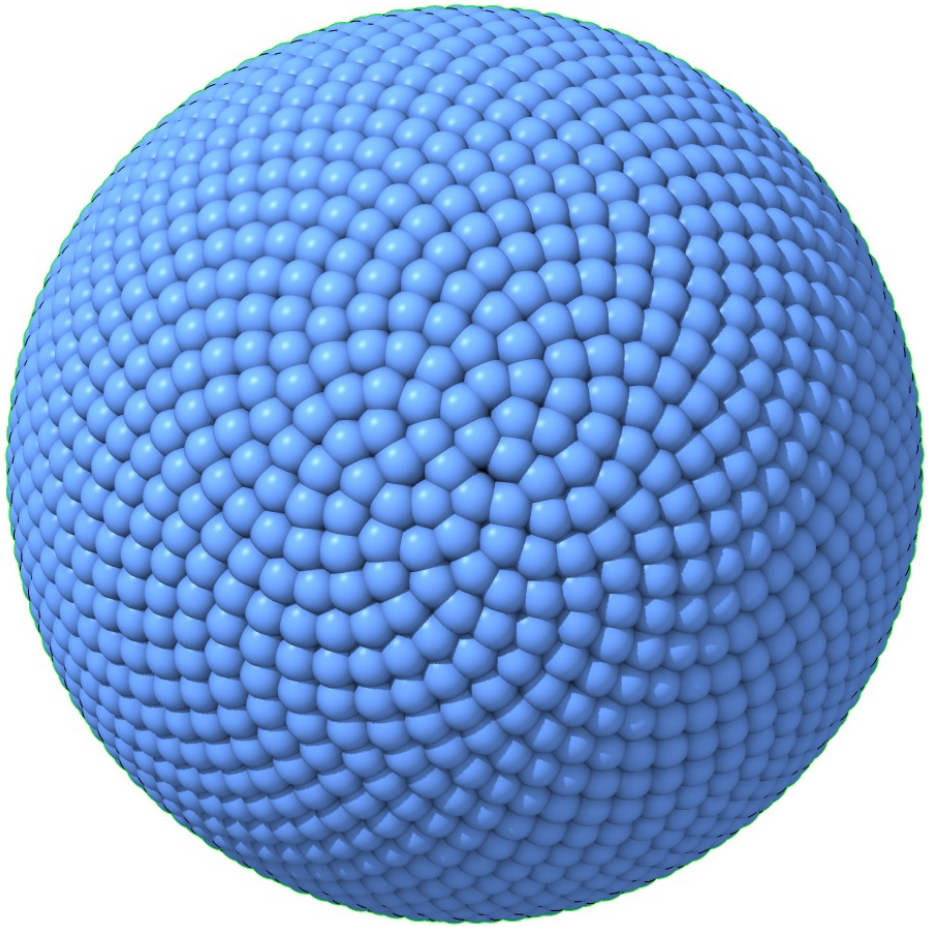


Sphere in vmd

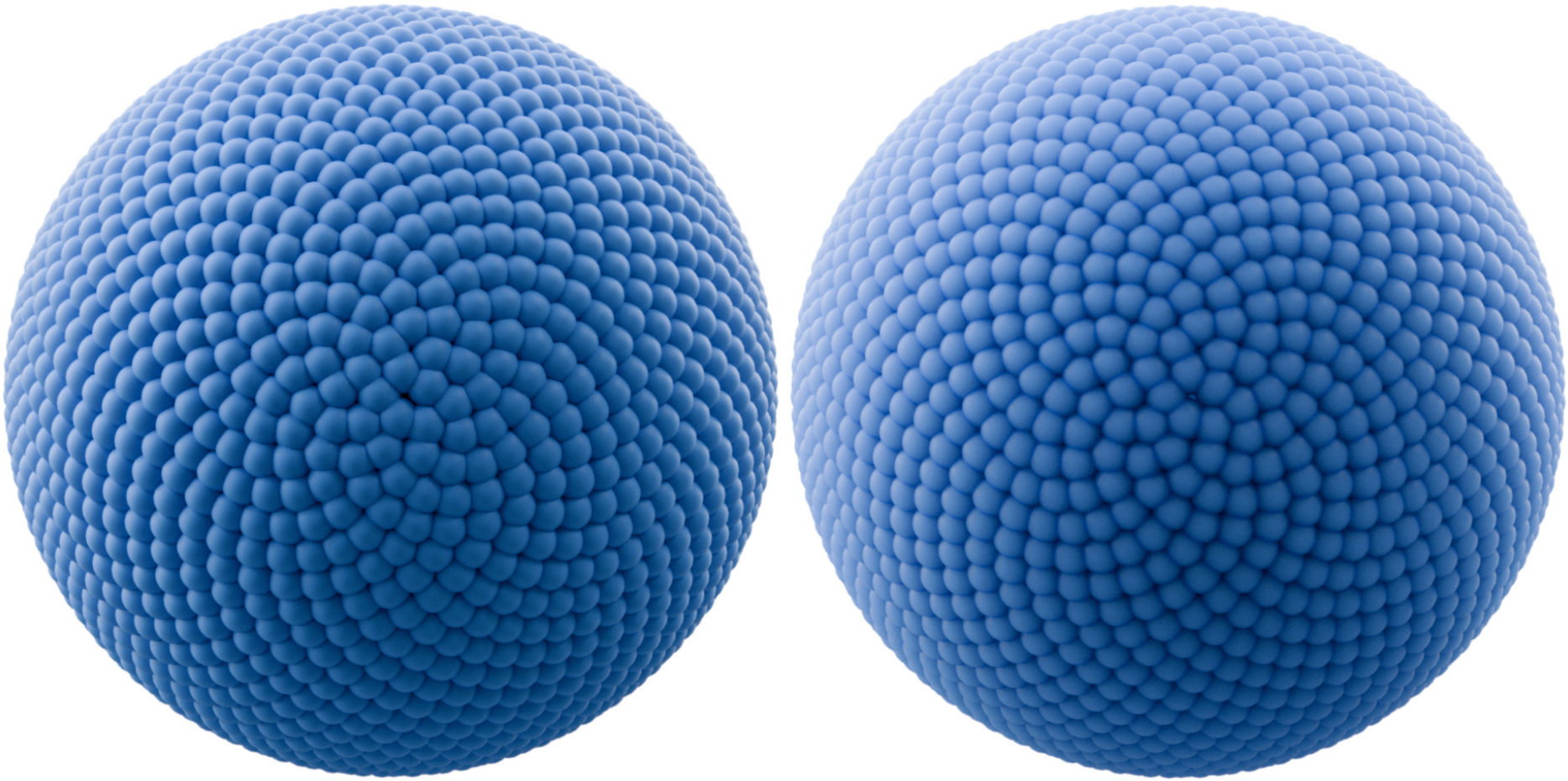


Sphere in chimeraX - Soft vs Full Lighting





Sphere in Blender - Simple vs Subsurface scattering shader



Sphere in vmd

Exercise 1 - vmd

- Display
  - Orthographic
  - Rendermode - Tachyon RTX RTRT
  - Display Settings... - Shadows ON, Amb. Occl. ON
  - Axes - Off
- Graphics - Colors - Display - Background - 8 White
- Graphics - Representations
  - Coloring method - ColorID
  - Material - AOChalky
  - Drawing Method - VDW
  - Color - 22 Cyan3
- File - Render... , alternatively just printscreen
  - TachyonL-Optix (internal, in-memory, GPU-accelerated)
  - vmdscene.ppm
  - Render Command - empty
- in terminal: convert vmdscene.ppm vmdscene.jpg

Sphere in chimeraX

Exercise 1 - chimeraX

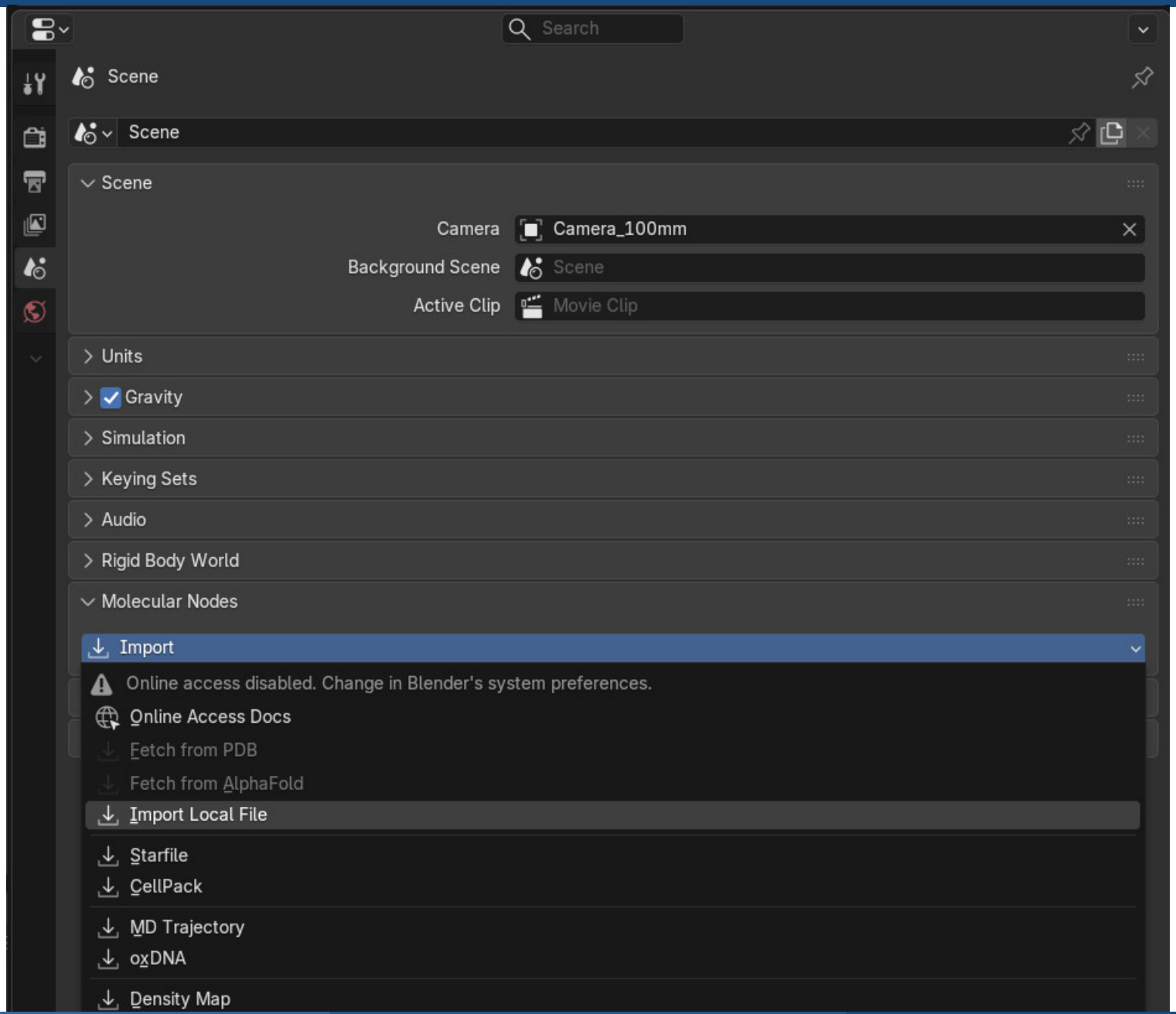
- Drag and drop —or— File - Open sphere.pdb
- Molecule Display tab
  - Styles :: Sphere
- Select - All
- Action - Color
- Graphics tab
  - Background :: White
  - Lighting & Effects - Soft/Full
- File - Save
  - As jpg
  - Setup resolution as 1920x1080
  - Turn off preserve aspect

Sphere in Blender

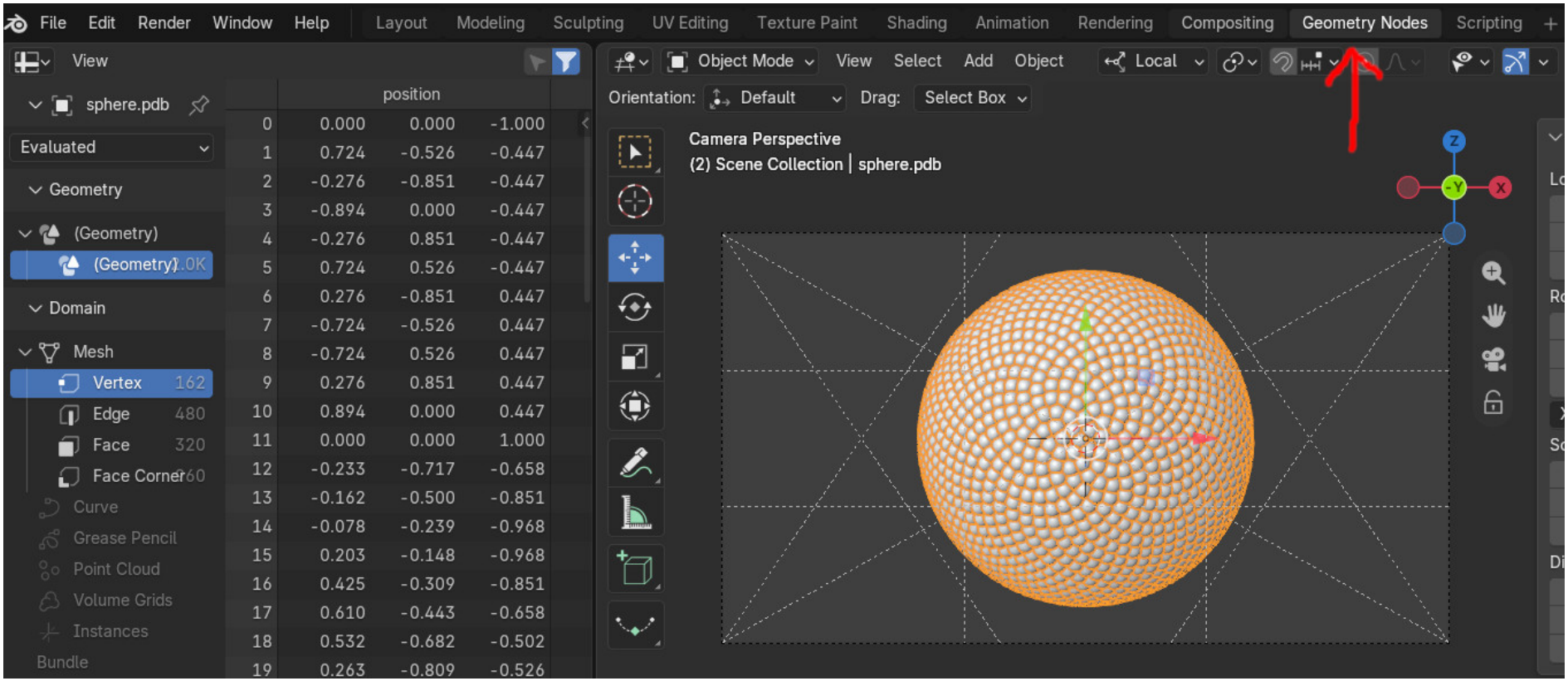
Exercise 1 - Blender

- Open sphere_base.blend
  - Contains the prepared scene:
  - Lighting, Camera, Material, Compositor, Image resolution
- Import sphere.pdb via Molecular Nodes
  - Scene :: Molecular nodes :: Import Local File
- Geometry Nodes tab
  - Click on the sphere object or in menu at sphere.pdb
  - Keep only (Set Color) and (Style Sphere) nodes
  - Style Spheres - Scale 1.0, select instance - quality 3
- Shading tab
  - Select sphere.pdb - it has no material set
  - Set material Default - its already set as simple shader
  - Switch to SSS_Shader_v6
- F12
  - Save

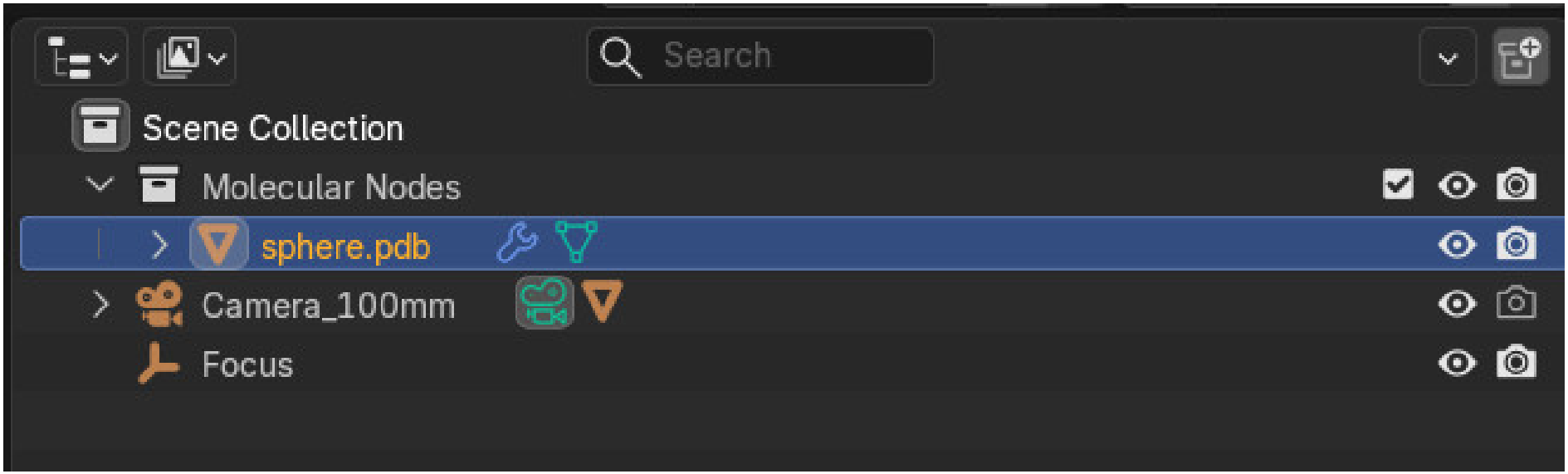
Sphere in Blender - Import pdb



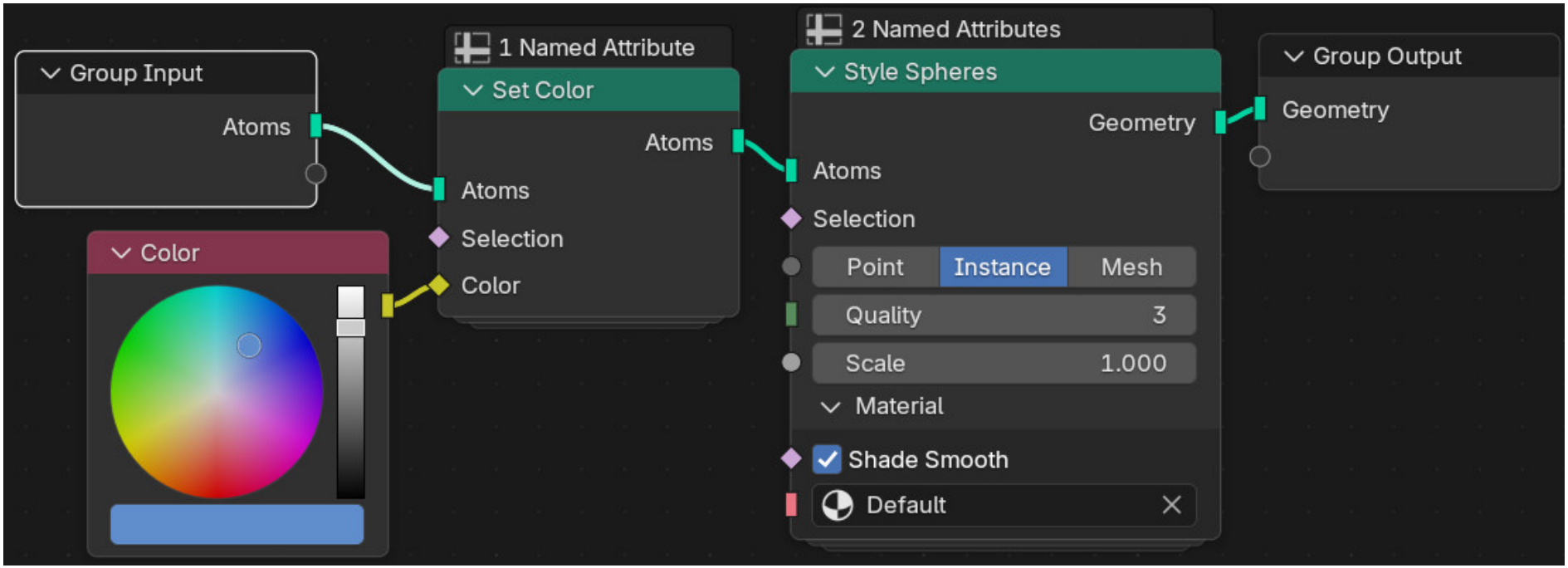
Sphere in Blender - Geometry Nodes Tab



Sphere in Blender - Select sphere.pdb in the menu



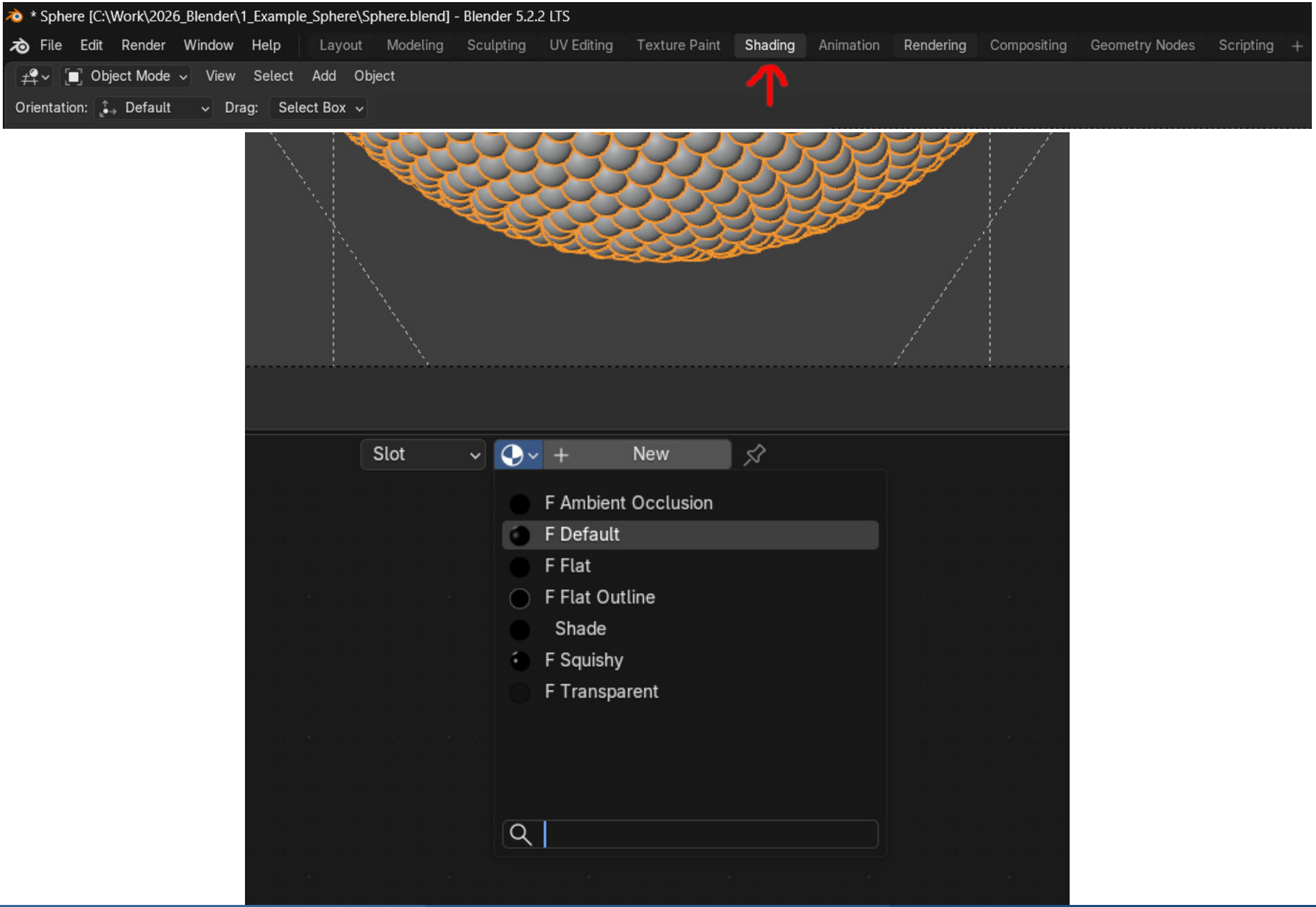
Sphere in Blender - Geometry Nodes Setup



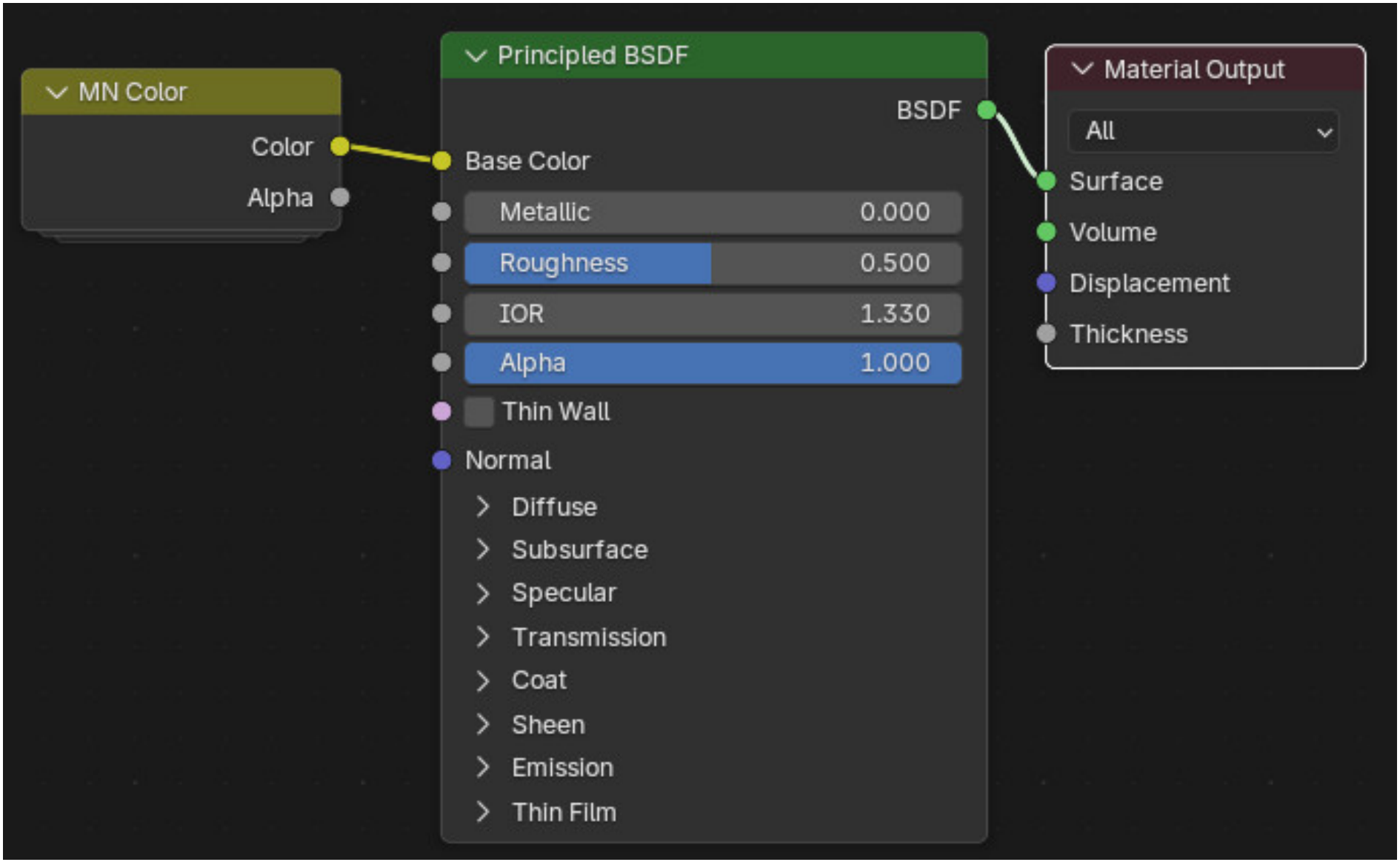
Color - Hue, Saturation, Value

- Simple Shader
  - Color (0.6, 0.53, 0.8)
- SSS shader - subsurface scatering
  - Color (0.6, 0.53, 1.0), setting in material not geometry nodes (intended, but with no reason for it)

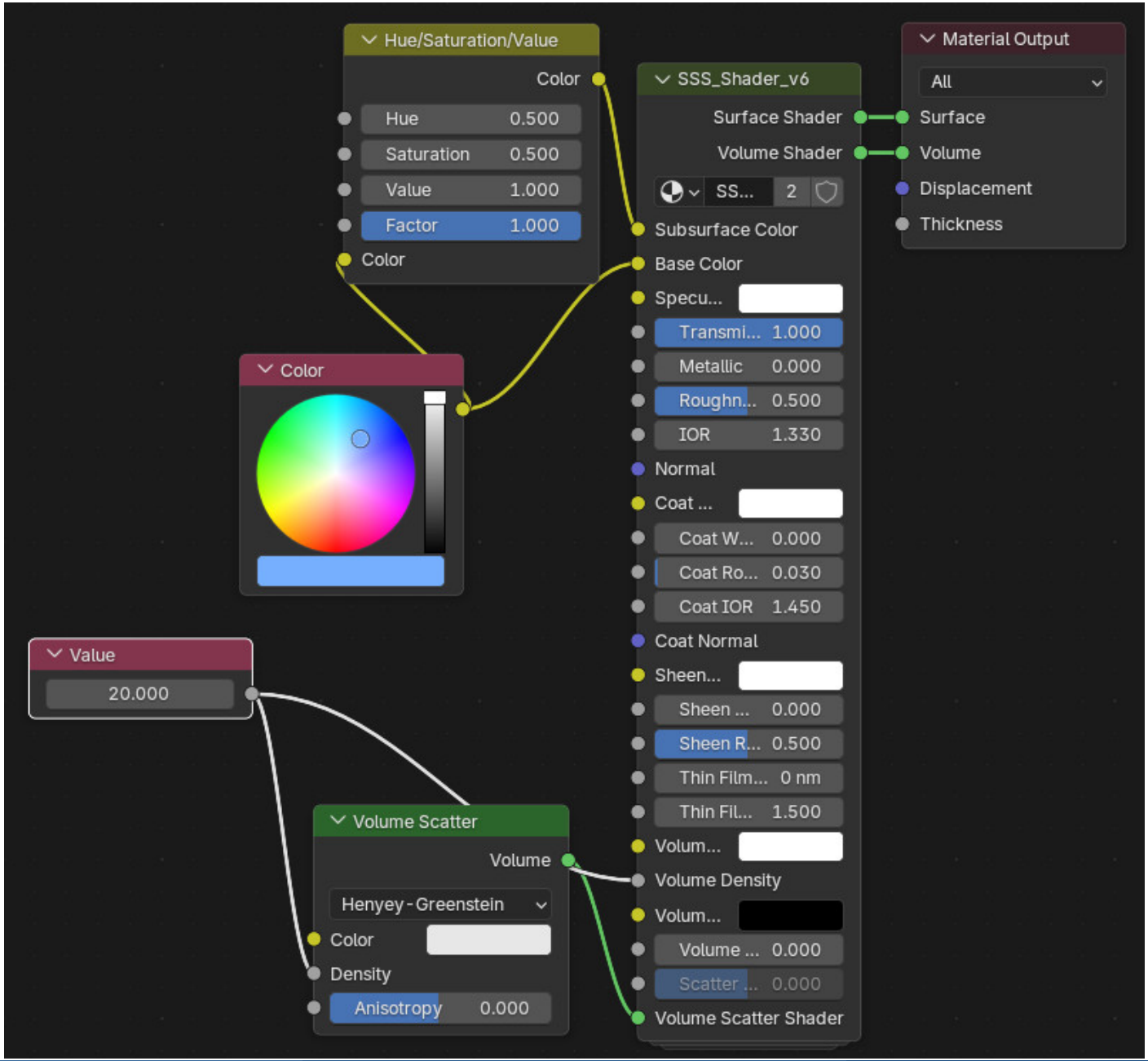
Shading Tab - Select sphere.pdb, set material Default



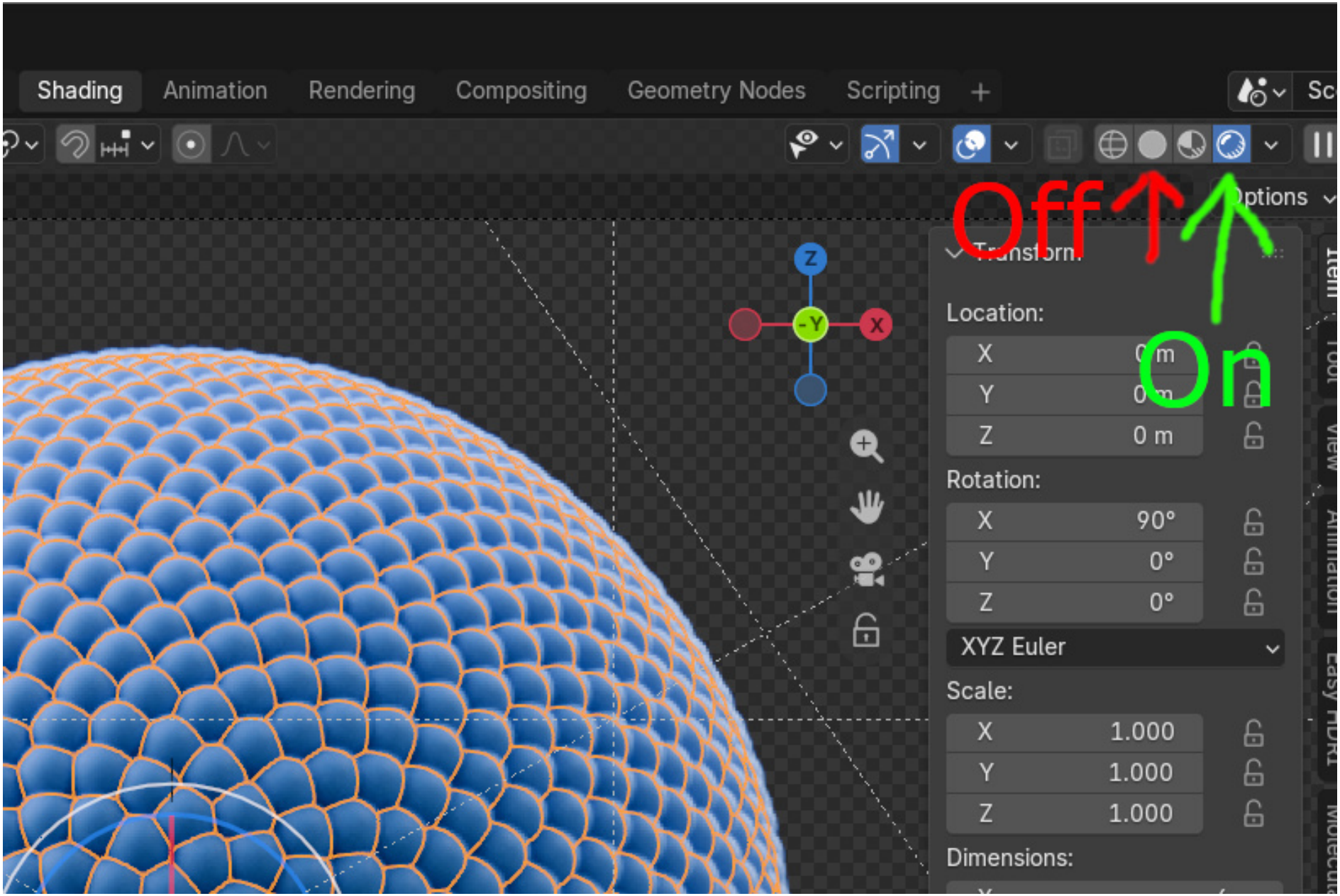
Sphere in Blender - Simple Shader Setup



Sphere in Blender - SSS Shader Setup



Sphere in Blender - Viewport Render, Full Render F12



Recreating Connexin 46/50 - pdb 7JM9

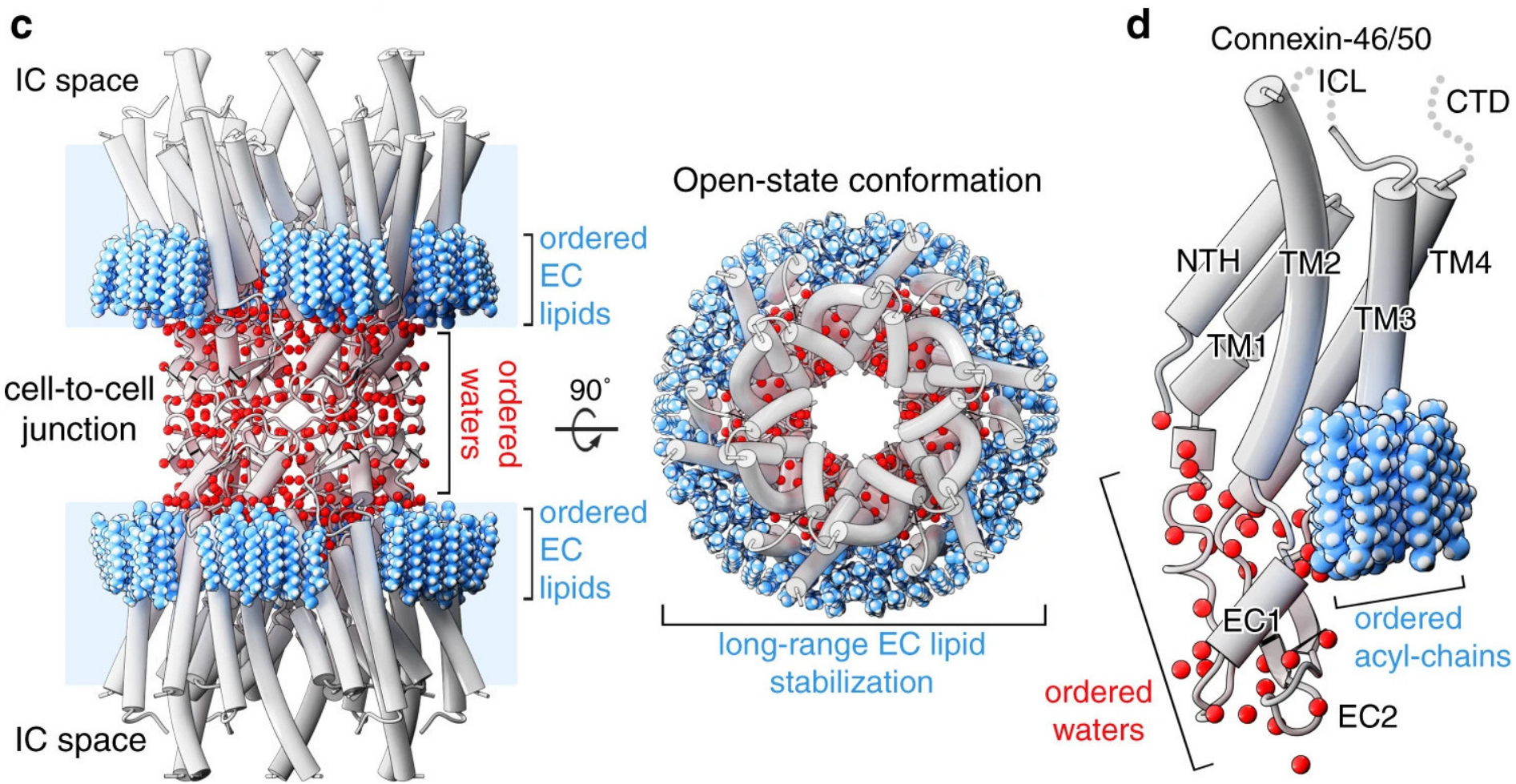
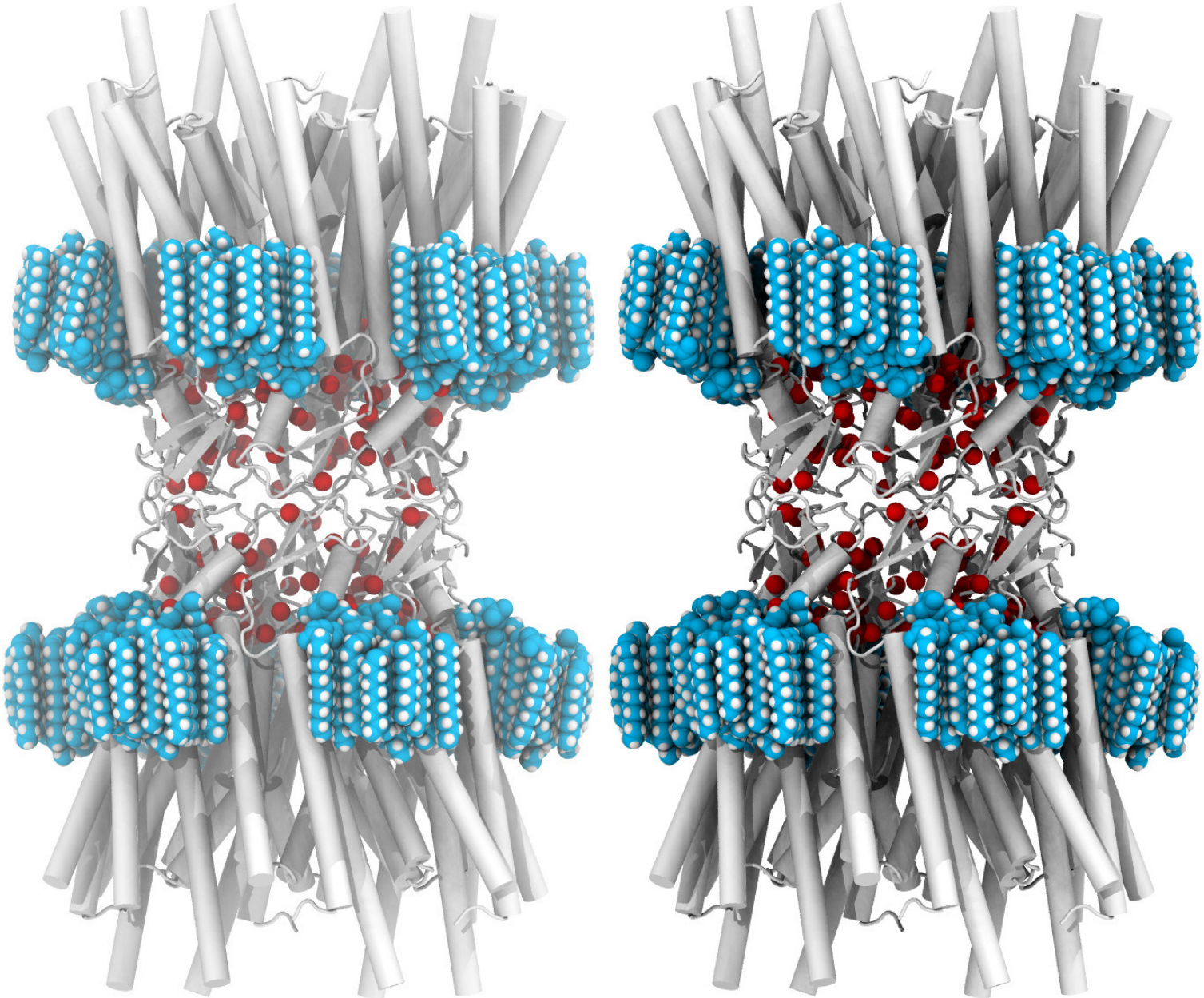
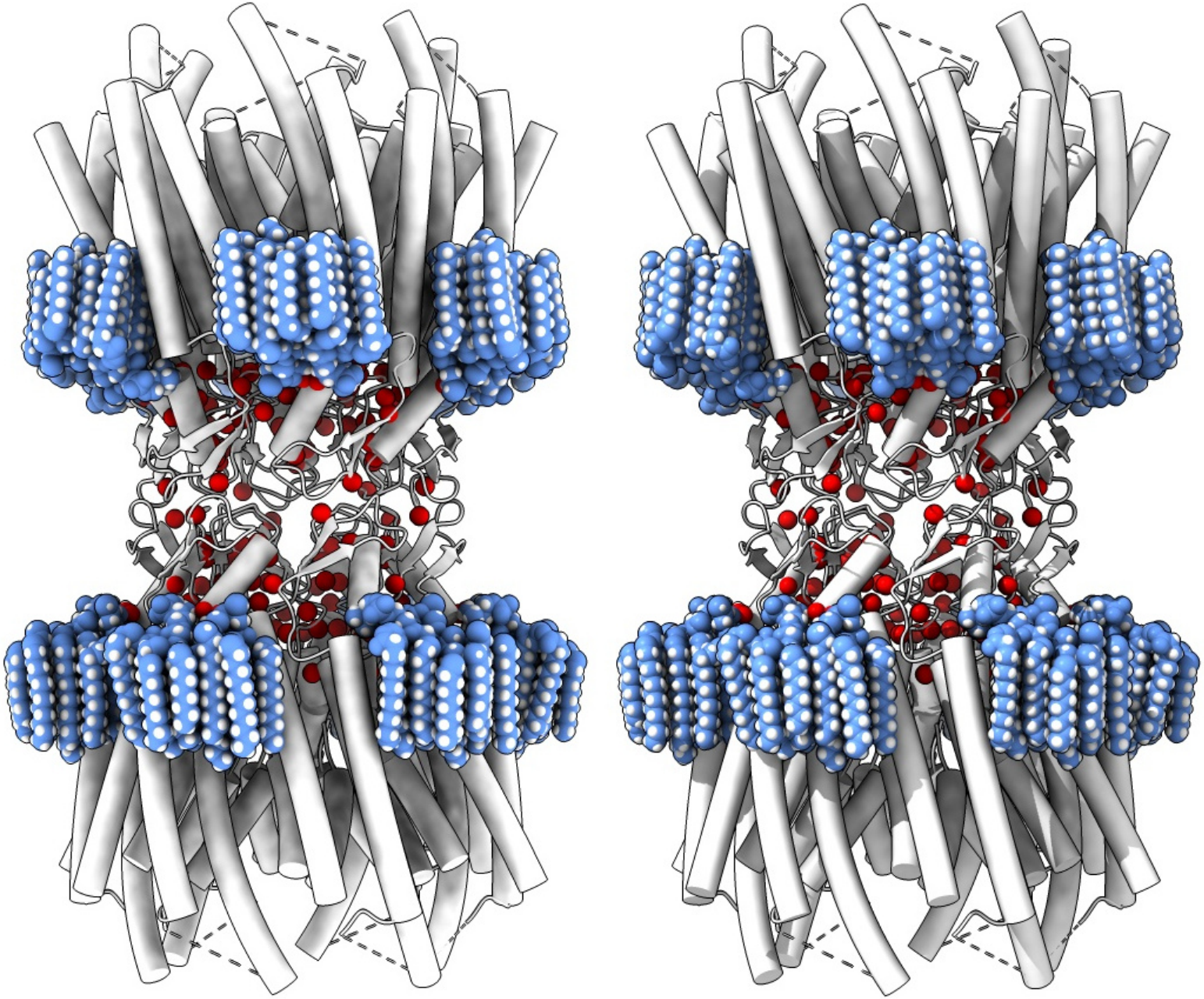


Figure: 10.1038/s41467-020-18120-5

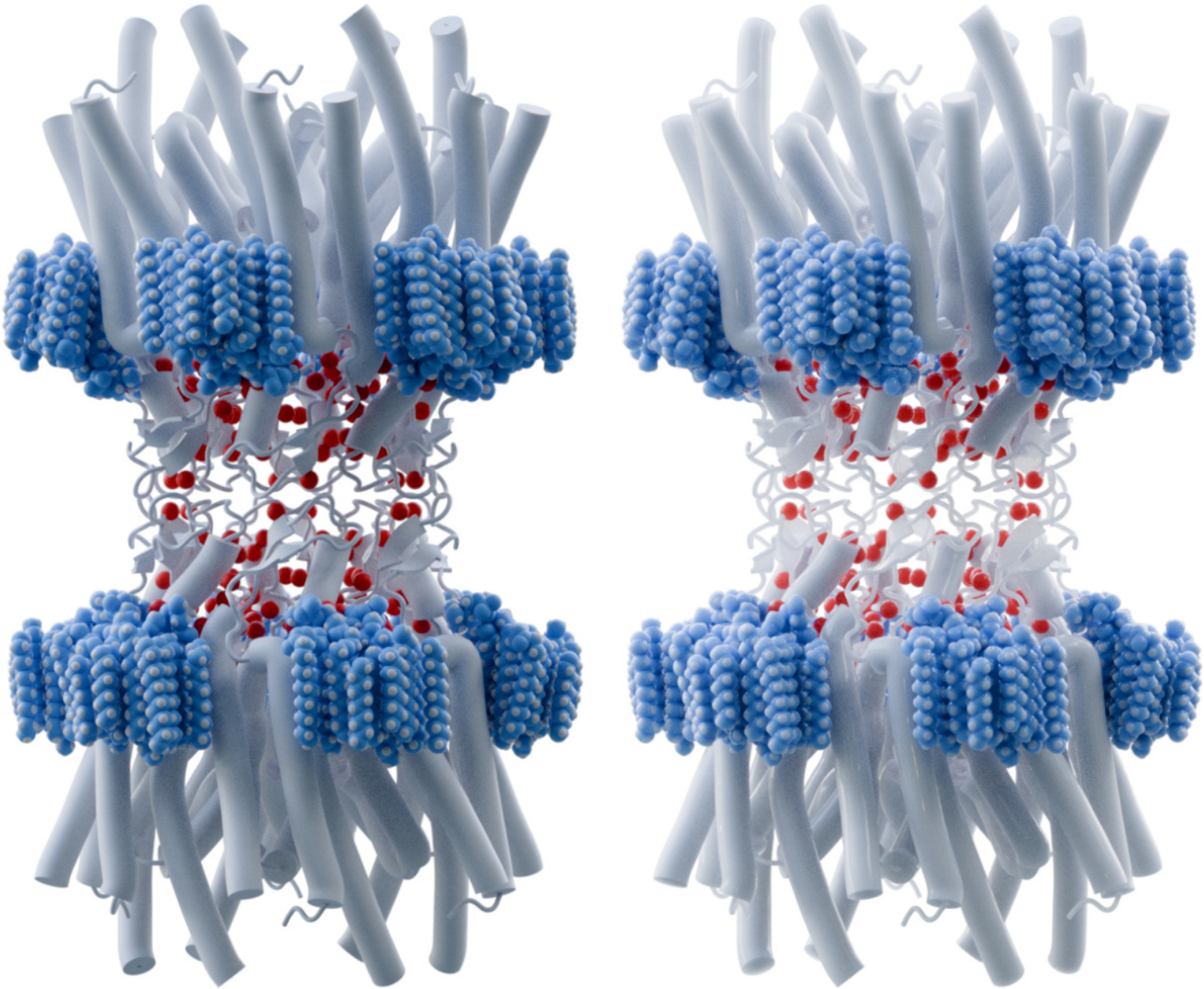
Connexin 46/50 - vmd - Depth Cueing ON vs OFF



Recreating Connexin 46/50 - chimeraX Soft vs Full lighting



Recreating Connexin 46/50 - Blender Simple vs SSS shader



Recreating Connexin 46/50 - pdb 7JM9

Exercise 2 - vmd - part 1

- Rotate the molecule to roughly similar configuration
- Display
  - Orthographic
  - Rendermode - Tachyon RTX RTRT
  - Display Settings... - Shadows ON, Amb. Occl. ON
  - Axes - Off
  - Depth Cueing - ON/OFF (up to you)
- Graphics - Colors - Display - Background - 8 White
- Graphics - Representations
- Create 4 representations with the following "Selected Atoms"
  - water
  - protein
  - not protein && not water && element H
  - not protein && not water && not element H

Recreating Connexin 46/50 - pdb 7JM9

Exercise 2 - vmd - part 2

- except protein, the representation is
  - Coloring Method - ColorID
  - Material - AOChalky
  - Drawing Method - VDW
- protein representation is
  - Drawing Method - Cartoon
- File - Render... , alternatively just printscreen
  - TachyonL-Optix (internal, in-memory, GPU-accelerated)
  - vmdscene.ppm
  - Render Command - empty
- in terminal: convert vmdscene.ppm vmdscene.jpg

Recreating Connexin 46/50 - pdb 7JM9

Exercise 2 - chimeraX - part 1

- Drag and drop —or— File - Open 7JM9.cif
- Molecule Display tab
  - Hide all - Atoms, Cartoons, Surfaces
  - Show Cartoons
  - Actions - Cartoon - Tube Helices
- Rotate the molecule to roughly similar configuration
- Select - All
- Action - Color - White
- Select - Residues - HOH
- Molecule Display tab
  - Show Atoms
  - Styles - Sphere
- Action - Color - Red

Recreating Connexin 46/50 - pdb 7JM9

Exercise 2 - chimeraX - part 2

- Select - Residues - MC3
- Molecule Display tab
  - Show Atoms
  - Styles - Sphere
- Action - Color - Cornflower Blue
- Command line (Command:)
  - select ::name="MC3" & H
  - Its essentially a set intersection of MC3 and all hydrogens
- Action - Color - White
- Graphics tab
  - Lighting & Effects - Silhouettes + Soft or Full
  - Background - White
- Select - Clear
- File - Save (this time preserve aspect)

Recreating Connexin 46/50 in Blender

Exercise 2 - Blender - part 1

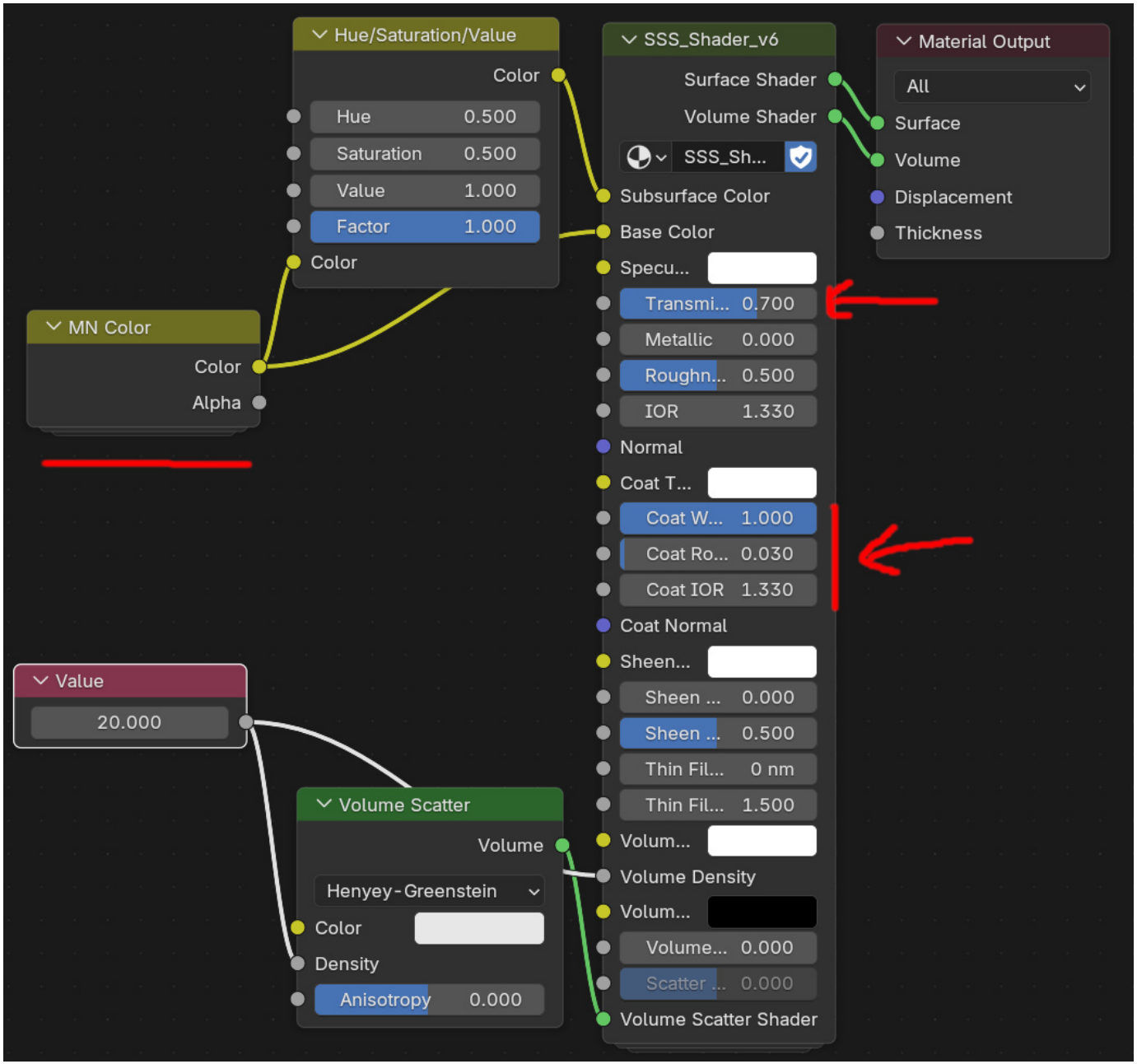
- Open connexin_base.blend
  - Contains the prepared scene:
  - Lighting, Camera, Material, Compositor, Image resolution
- Import 7JM9.cif via Molecular Nodes
  - Scene :: Molecular nodes :: Import Local File
- Shading tab
  - Select 7JM9.cif - it has no material set
  - Set material Default - its already set as simple shader
  - Use and alter the SSS_Shader_v6
  - change transmission to 0.7
  - Coat to 1.0
  - This time, lets set color in geometry nodes
  - So lets connect the geometry nodes output to shader color input

Recreating Connexin 46/50 in Blender

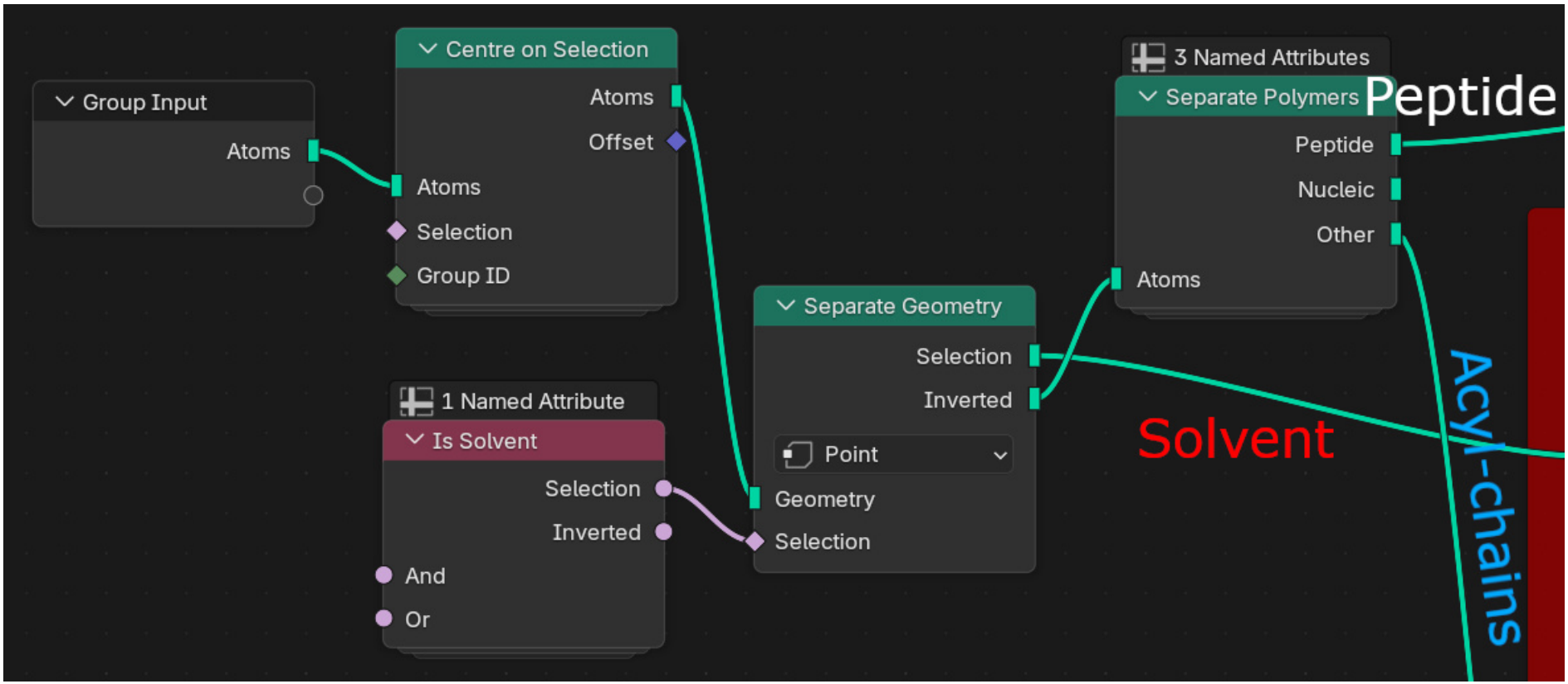
Exercise 2 - Blender - part 2 - Geometry nodes setup

- Geometry Nodes tab - select 7JM9.cif
  - click add and type node name
  - First use (Centre on Selection)
  - Separate water with (Is Solvent) - (Separate Geometry)
  - Separate acyl chains from proteins using (Separate Polymers)
- Then for each setup the representation
  - (Set Color) - (Style Spheres) for solvent
  - (Set Color) - (Style Cartoon) for peptide
- Separate hydrogens in Acyl chains requires, alter their vdw radius
  - (Select Element) to select hydrogens
  - (Named Attribute) to access vdw_radii
  - (Math) to summon satan
  - (Store Named Attribute) to store the multiplied vdw_radii
- Finally use unite geometry (Join Geometry)
- F12
  - Save

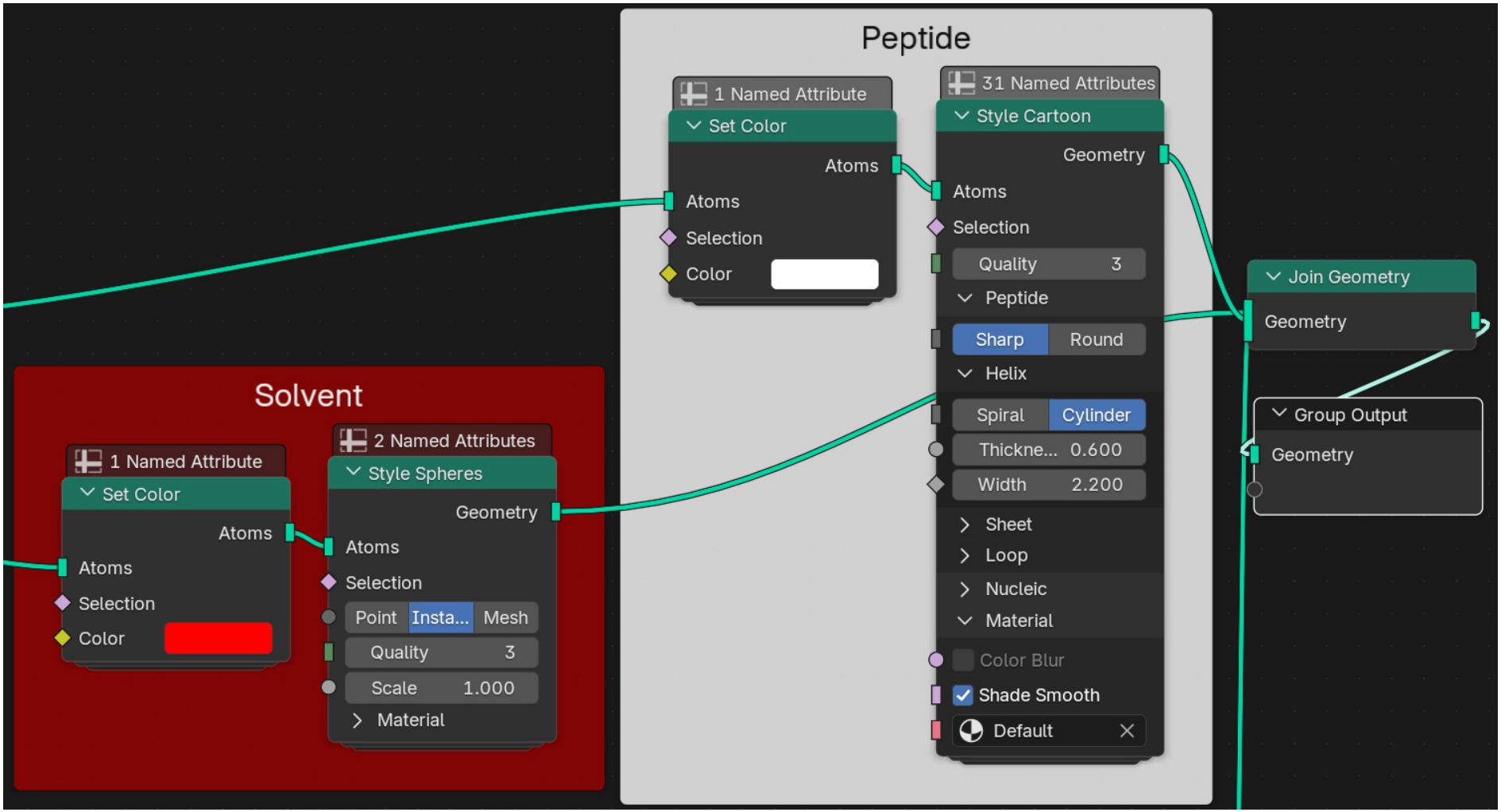
Shading tab, SSS shader setup



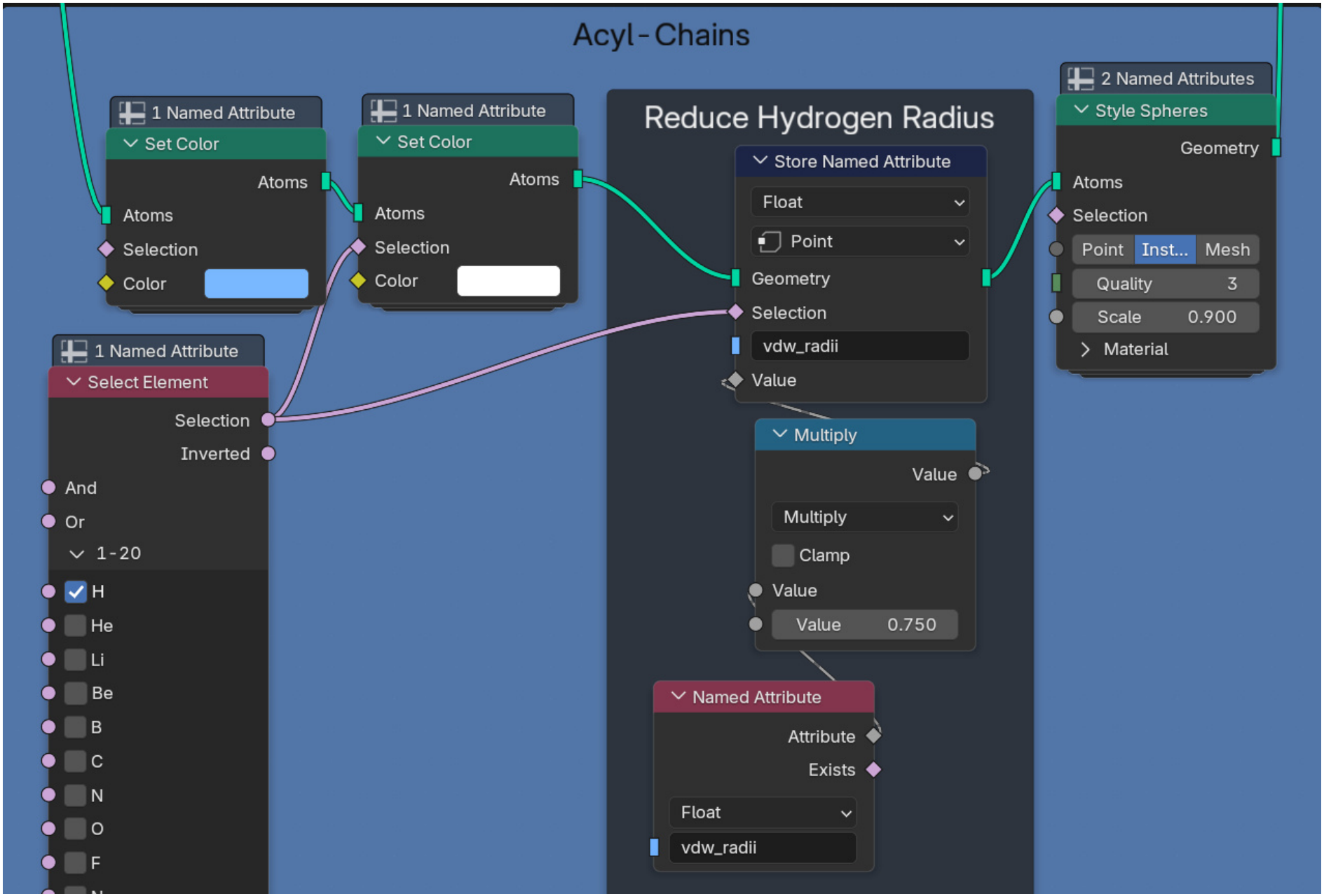
Geometry Nodes tab, Separate Water, Acyl-chains, Peptides



Geometry Nodes tab, Water, Peptides Setup



Geometry Nodes tab, Acyl-Chains Setup



Recreating SNOR - SBDS domain-containing ribosome

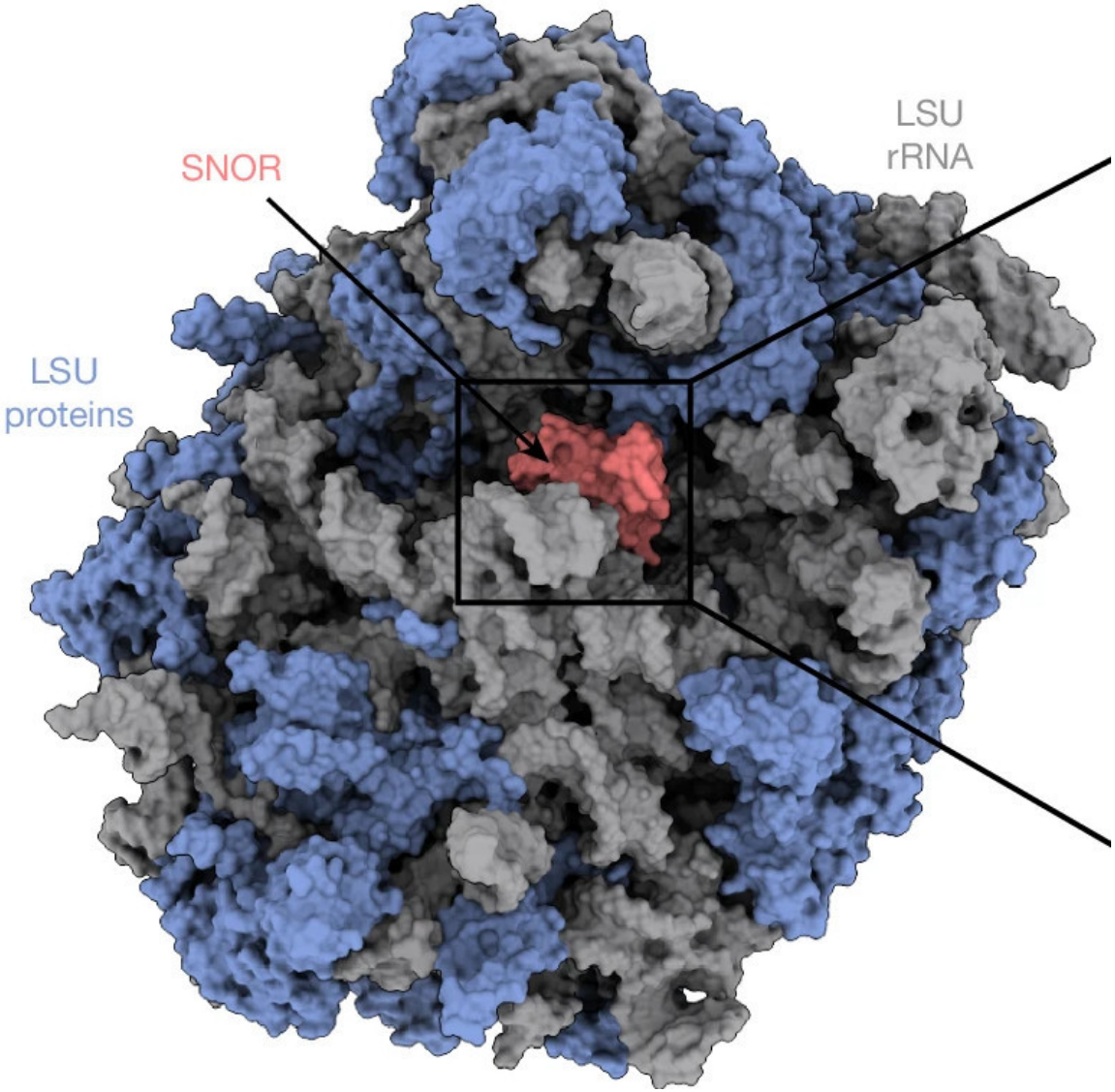
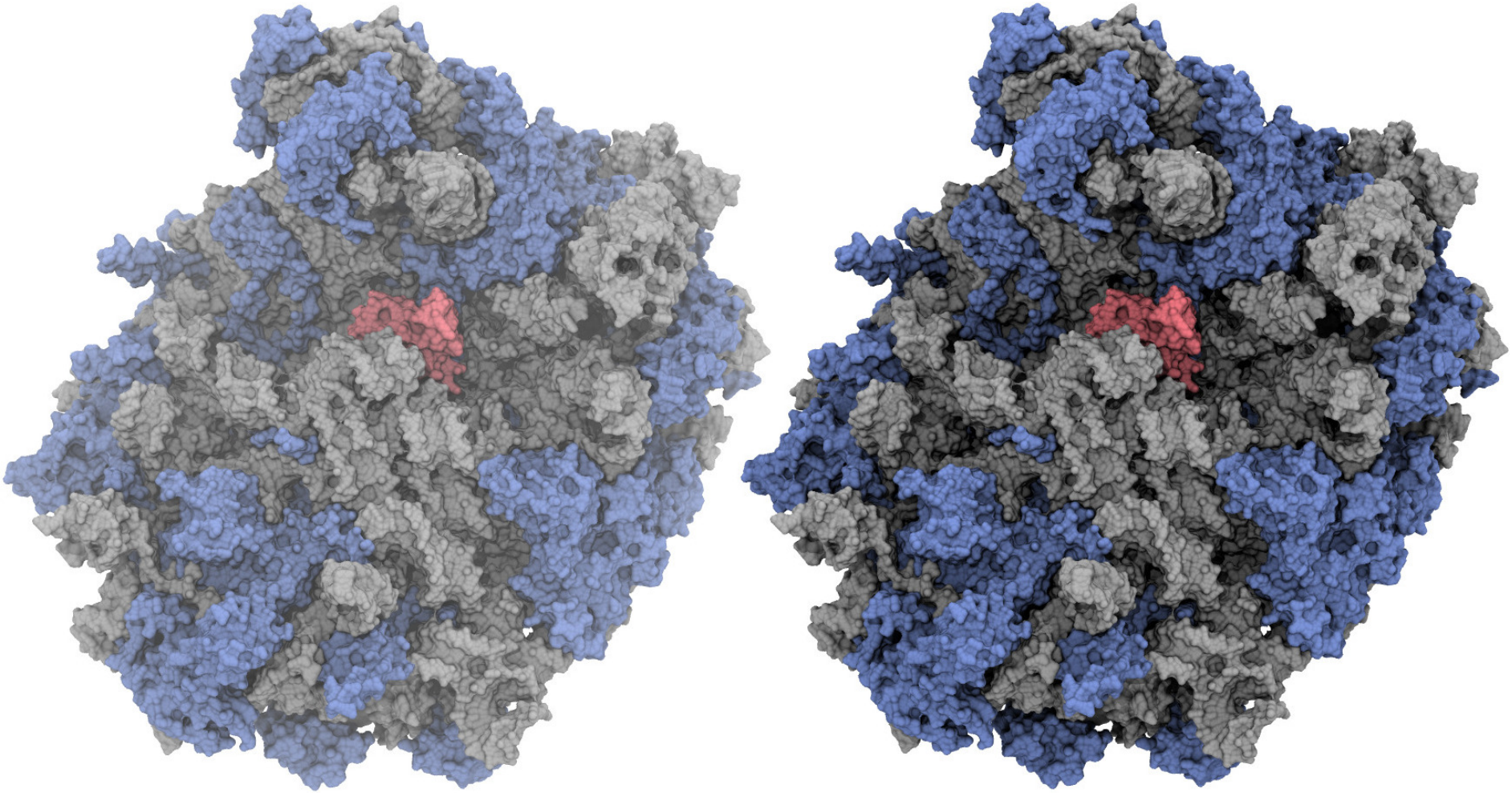
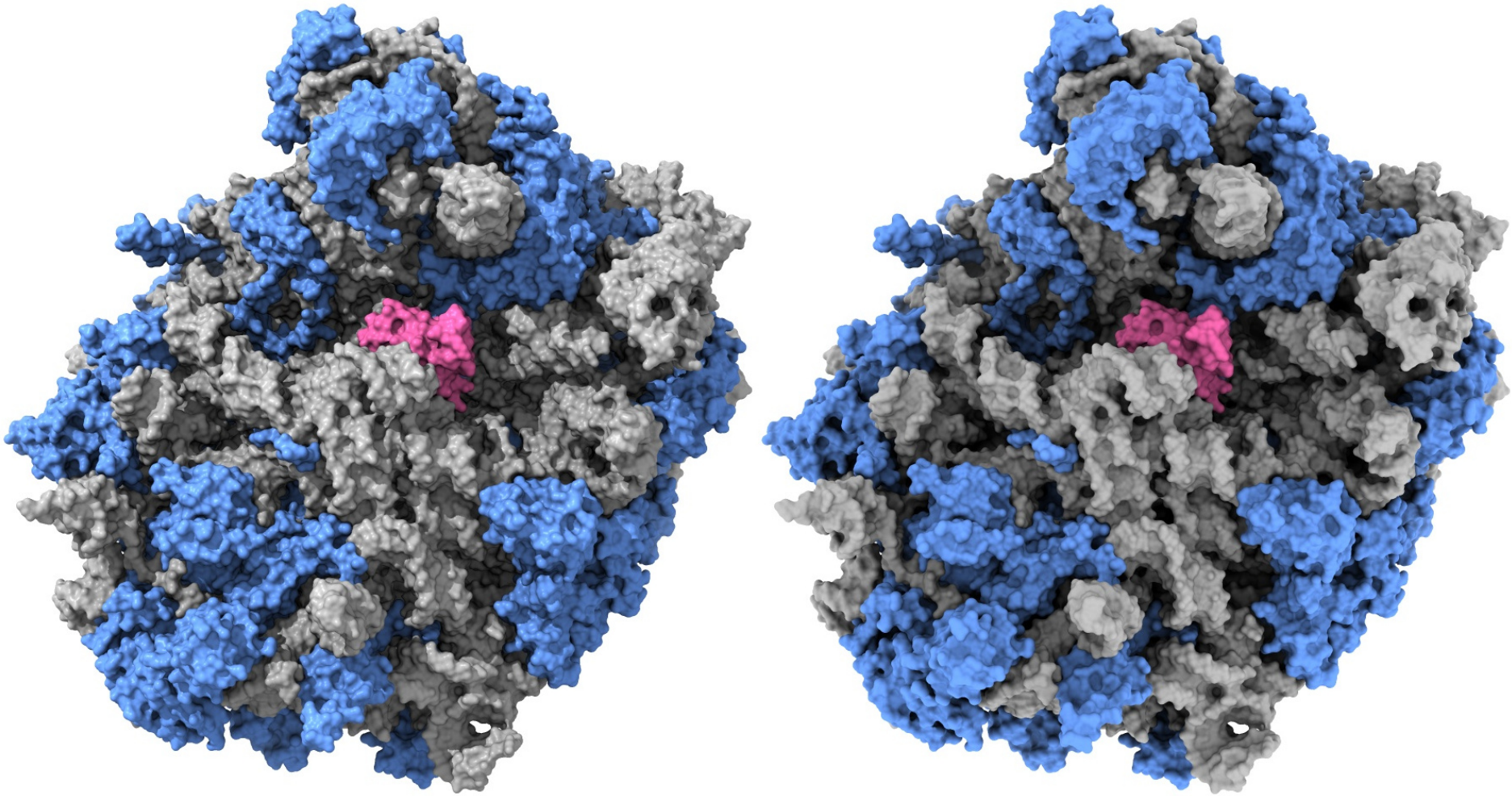


Figure: 10.1038/s41586-026-10530-7

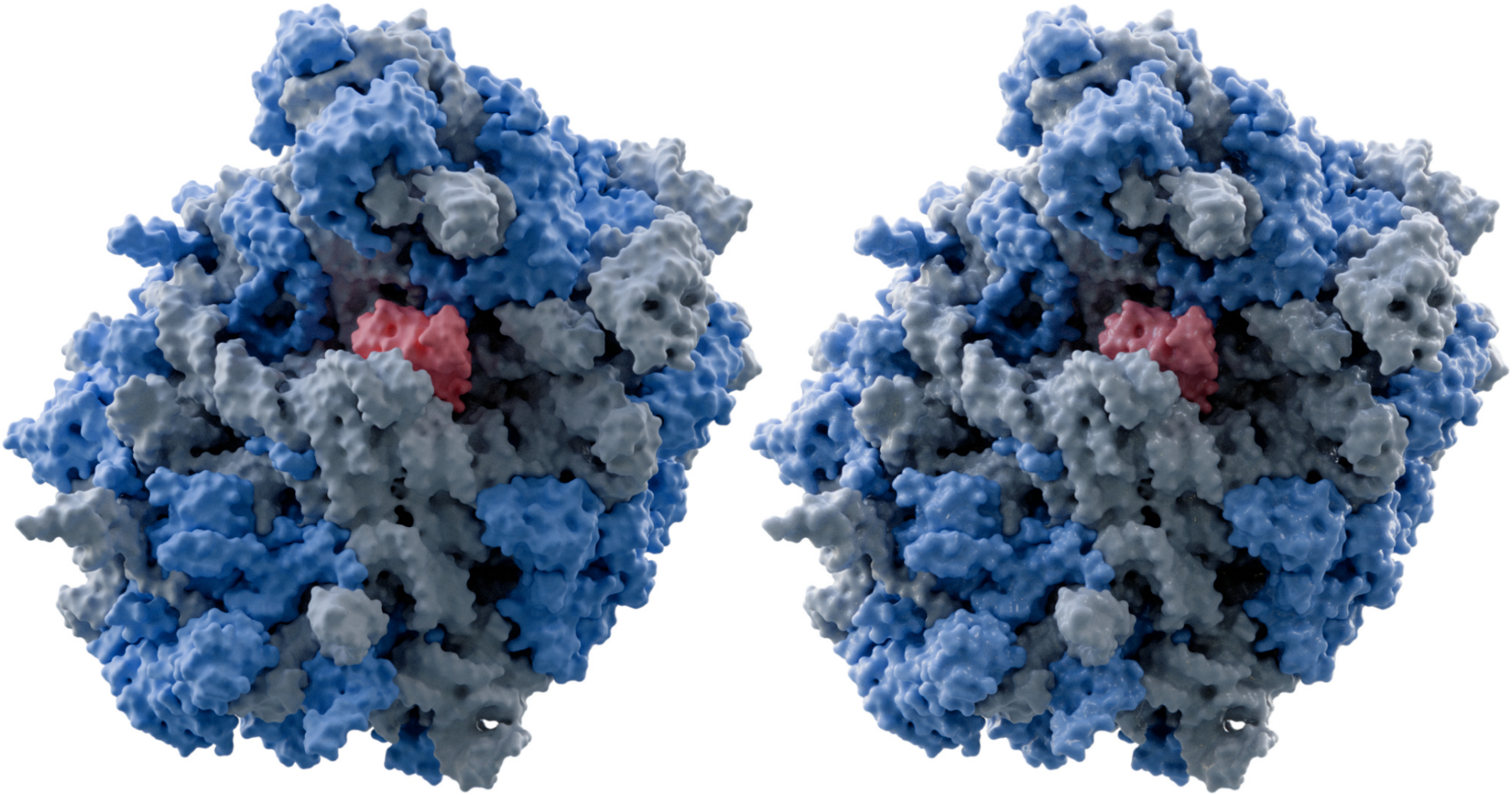
Recreating SNOR - vmd - Depth Cue ON vs OFF



Recreating SNOR - ChimeraX - Full vs Soft

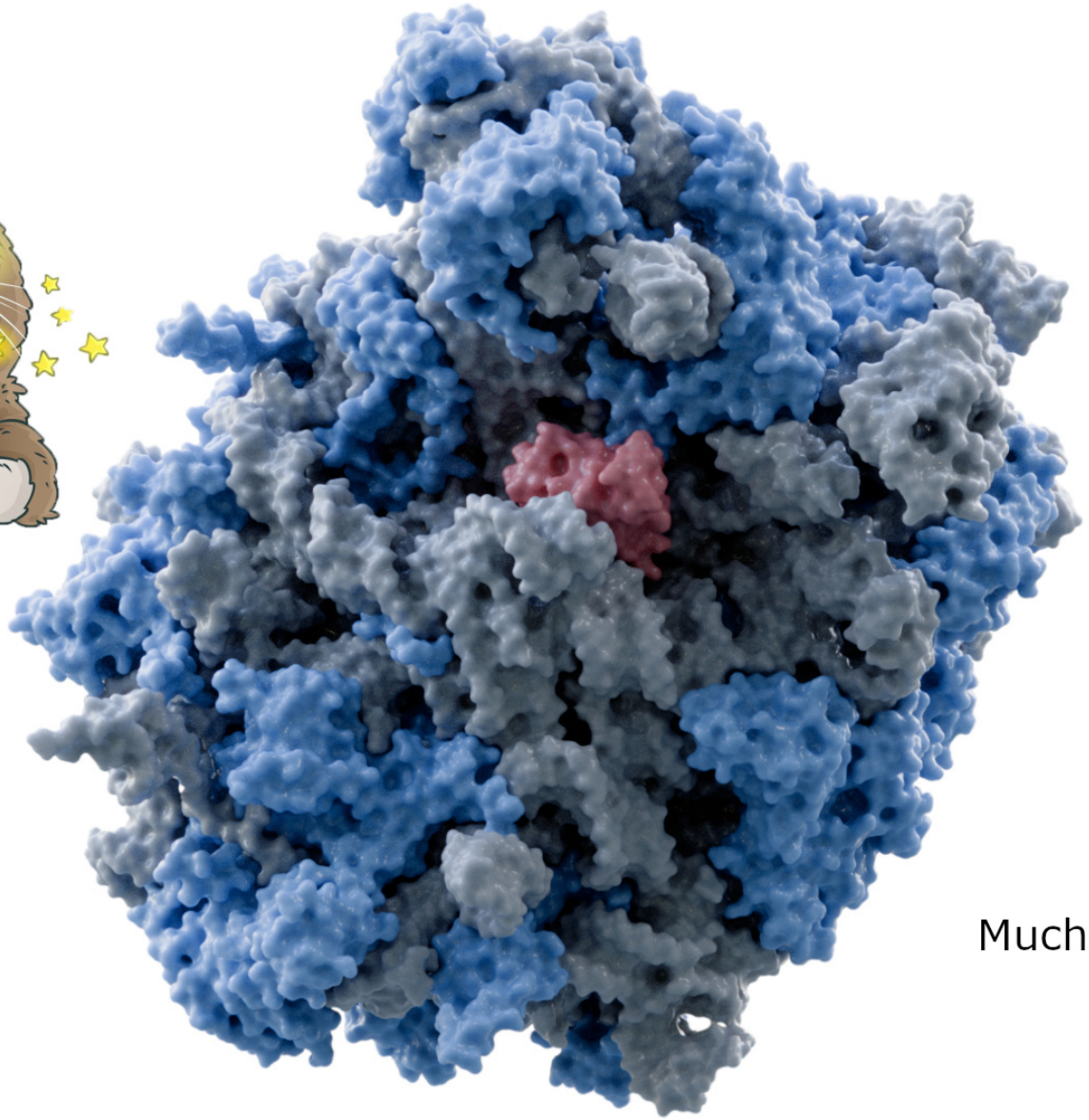


Recreating SNOR - Blender - Aproximate SES



Recreating SNOR - Blender - SES from chimeraX







True SES

Wow

Much Difference

Recreating SNOR - vmd - pdb 9PHC

Exercise 3 - vmd - part 1

- Display
  - Orthographic
  - Rendermode - Tachyon RTX RTRT
  - Display Settings... - Shadows ON, Amb. Occl. ON
  - Axes - Off, Depth Cueing - ON/OFF (up to you)
- Graphics - Colors - Display - Background - 8 White
- Graphics - Representations
- Create 3 representations with the following "Selected Atoms"
  - protein && not chain RA1
  - nucleic
  - all
- Representation protein
  - Material - AOChalky
  - Coloring Method - ColorID
  - Color - 15 Iceblue
  - Drawing Method - MSMS

Recreating SNOR - vmd - pdb 9PHC

Exercise 3 - vmd - part 2

- Representation nucleic
  - Material - AOChalky
  - Coloring Method - ColorID
  - Color - 2 Gray
  - Drawing Method - MSMS
- Rotate the molecule to roughly similar configuration
- Mouse - Label - Atoms
  - MSMS may fail to construct surface of the entire protein
  - Pick beads that lack the SES surface
  - In terminal, vmd will write out their chain ID
  - Chain V, OA1, Y - setup additional representations
  - Chain RA1 is SNOR
- Delete Representation all
- Graphics - Labels - Delete Labels

Recreating SNOR - vmd - pdb 9PHC

Exercise 3 - vmd - part 3

- Representation: chain RA1
  - Material - AOChalky
  - Coloring Method - ColorID
  - Color - 9 Pink
  - Drawing Method - MSMS
- Bonus: Alter Colors in (Graphics - Colors)
  - 2 gray - hsv - 0.5, 0.0, 0.7
  - 9 pink - hsv - 5.907, 0.503, 1.0
  - 15 iceblue - hsv - 3.746, 0.431, 0.85
  - The widget is absolutely terrible i know
- File - Render... , alternatively just printscreen
  - TachyonL-Optix (internal, in-memory, GPU-accelerated)
  - vmdscene.ppm
  - Render Command - empty
- in terminal: convert vmdscene.ppm vmdscene.jpg

Recreating SNOR - chimeraX - pdb 9PHC

Exercise 3 - chimeraX - part 1

- Drag and drop —or— File - Open 7JM9.cif
- Molecule Display tab
  - Hide all - Atoms, Cartoons, Surfaces
  - Show Surfaces
- Rotate the molecule to roughly similar configuration
- Select - Residues - Standart Nucleic Acids
- Action - Color - Dark Gray
- Select - Residues - Standart Amino Acids
- Action - Color - Cornflower Blue
- Select - Chains - SDO1-like protein (the last one in the list)
- Action - Color - Hot Pink

Recreating SNOR - chimeraX - pdb 9PHC

Exercise 3 - chimeraX - part 2

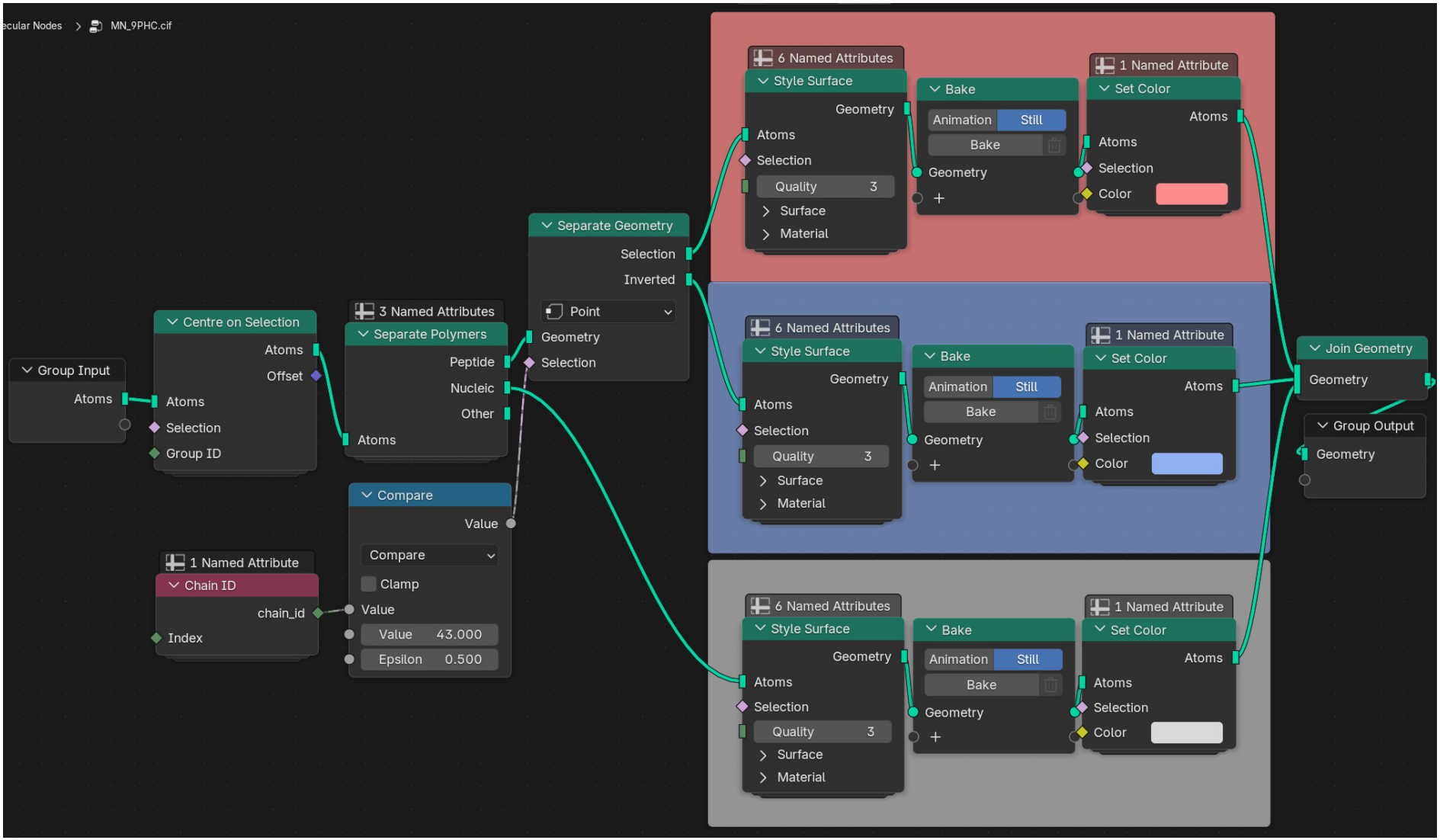
- Graphics tab
  - Lighting & Effects - Soft or Full
  - Background - White
- Select - Clear
- File - Save (this time preserve aspect)

Recreating SNOR - Blender - pdb 9PHC

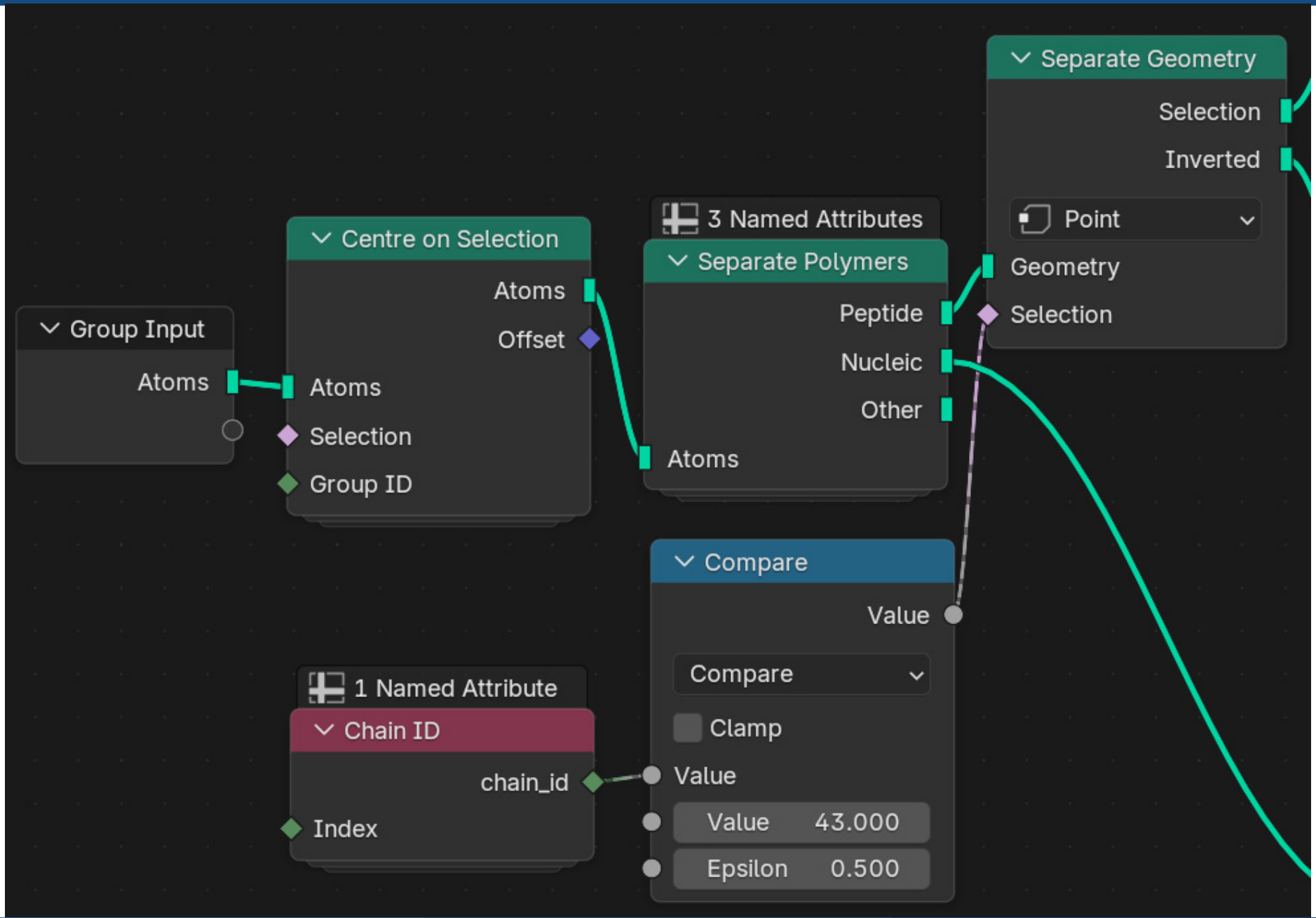
Exercise 3 - Blender

- Open SNOR_base.blend
  - Contains the prepared scene:
  - Lighting, Camera, Material, Compositor, Image resolution
- Import 9PHC.cif via Molecular Nodes
  - Scene :: Molecular nodes :: Import Local File
- Shading tab
  - Select 9PHC.cif - it has no material set
  - Set material Default - its already set as simple shader
- Molecular Nodes tab
  - Setup 3 selections - protein, nucleic acis, SNOR protein
  - Setup Surface Style, Color
  - Bake node to improve performance
- Rotate 9PHC.cif

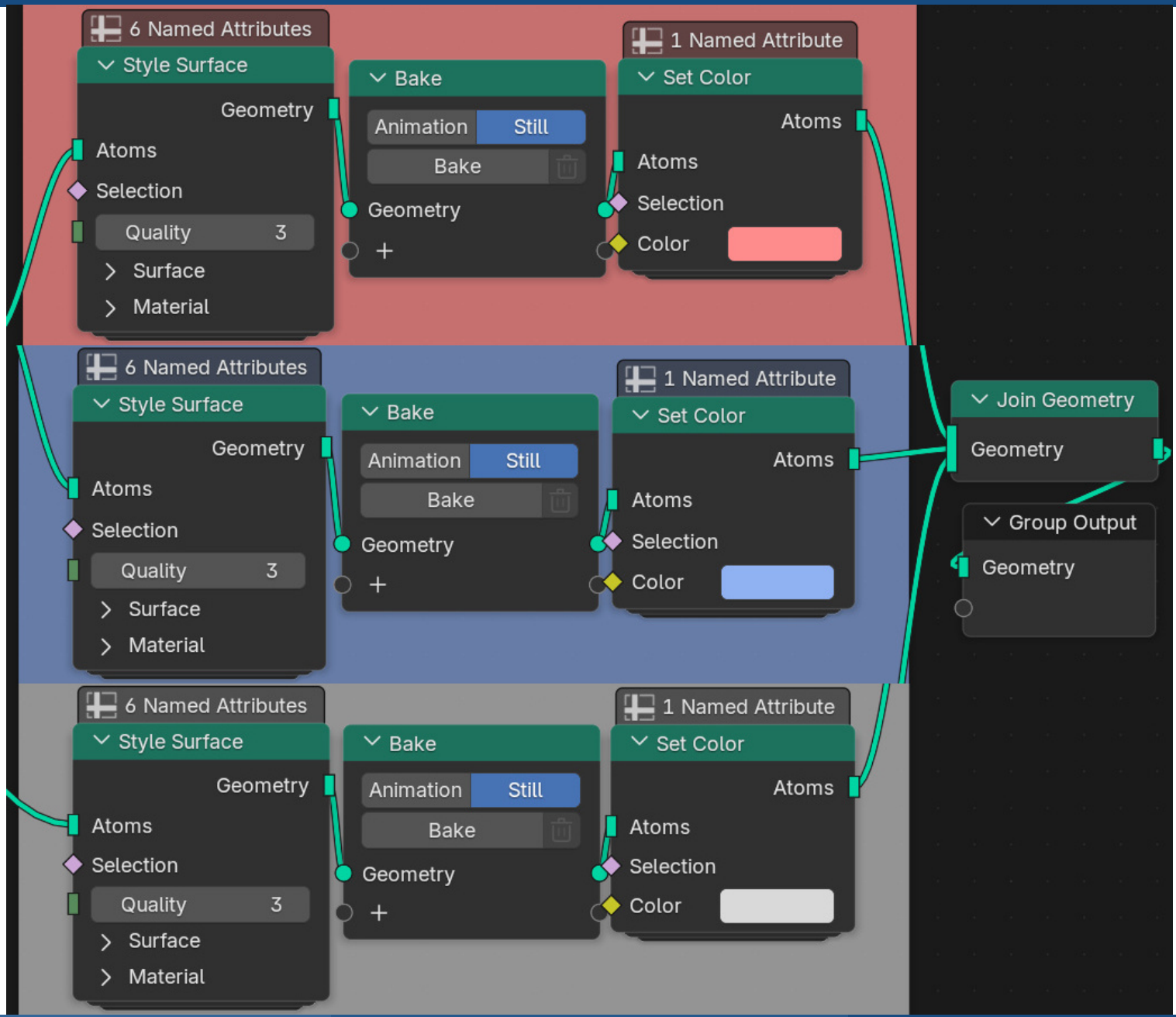
Recreating SNOR - Blender - Full Geo Node Setup



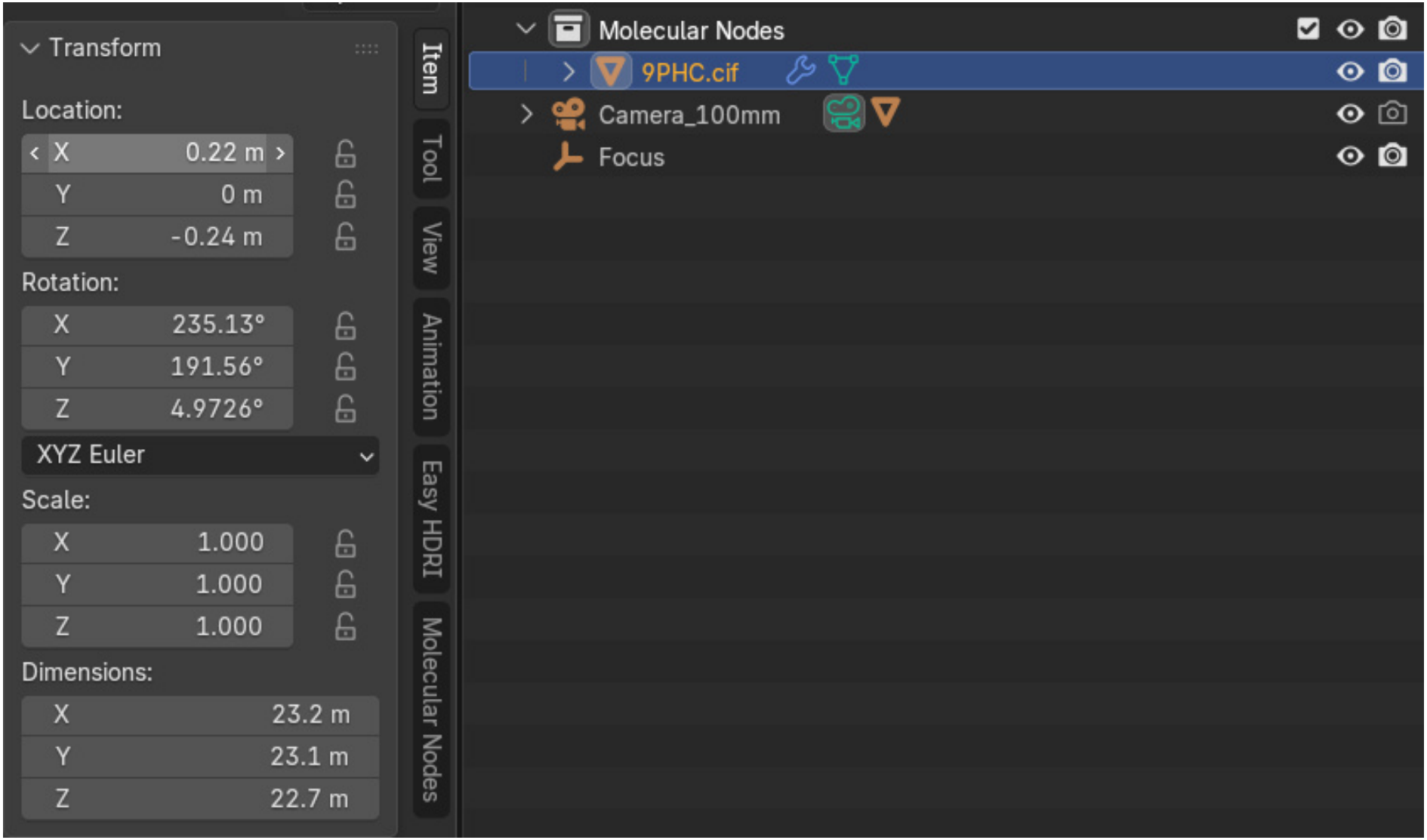
Recreating SNOR - Blender - Full Geo Node Setup



Recreating SNOR - Blender - Full Geo Node Setup



Recreating SNOR - Blender - Rotation



Recreating SNOR - chimeraX + Blender - pdb 9PHC

Exercise 3 - chimeraX for SES

- command: select protein & /HS
- command: surface sel enclose sel
- File - Save - Wavefront OBJ
- Hide Surface
- Select - Chains - SDO1-like protein (the last one in the list)
- Molecule Display - Show Surfaces
- File - Save - Wavefront OBJ
- Hide Surface
- Select - Chemistry - Nuileic Acid
- command: surface sel enclose sel
- File - Save - Wavefront OBJ

Recreating SNOR - chimeraX + Blender - pdb 9PHC

Exercise 3 - blender - part 1 - import SES, transform, remesh

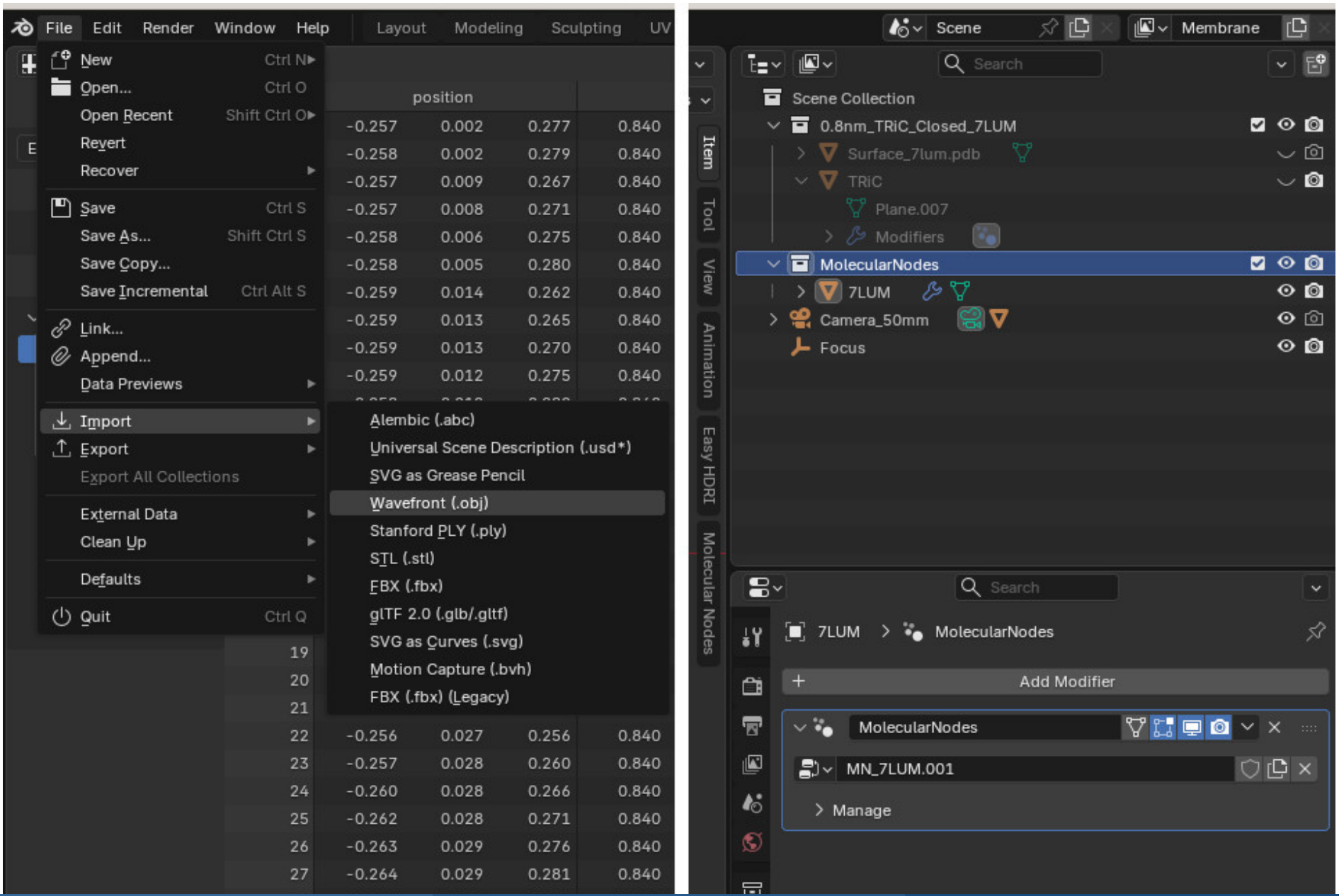
- Open SNOR_base.blend
- Import 9PHC.cif via Molecular Nodes
- Import OBJs - Na, protein, SNOR
- create collection SES, move OBJs into it
- OBJs are 10x larger, rotated by 90deg by default, apply transformation
- Voxel Remesh

Recreating SNOR - chimeraX + Blender - pdb 9PHC

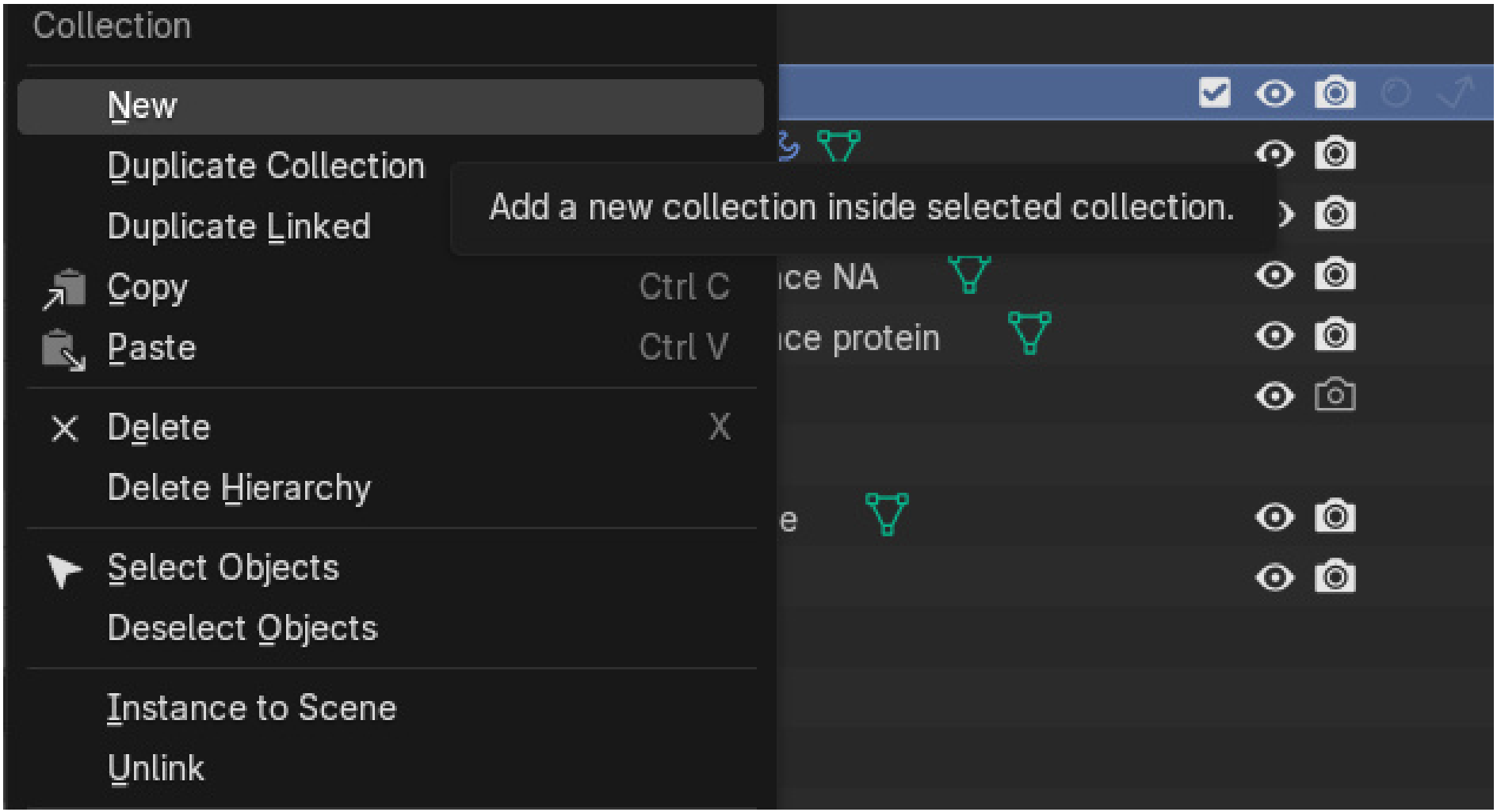
Exercise 3 - blender - part 2

- Shading tab
  - For each OBJ setup a material - unique copy of material Default
  - Setup colors for each OBJ
- Geometry Nodes tab
  - Get position vector from 9PHC.cif (centre selection)
  - (Atribute statistics) - mean
  - (point) - (instance on points)
  - Drag and drop the SES collection into geometry node area
  - Connect its geometry output to instance input of (instance on points)
  - Rotate using the original 9PHC.cif transformation with (transform geometry)

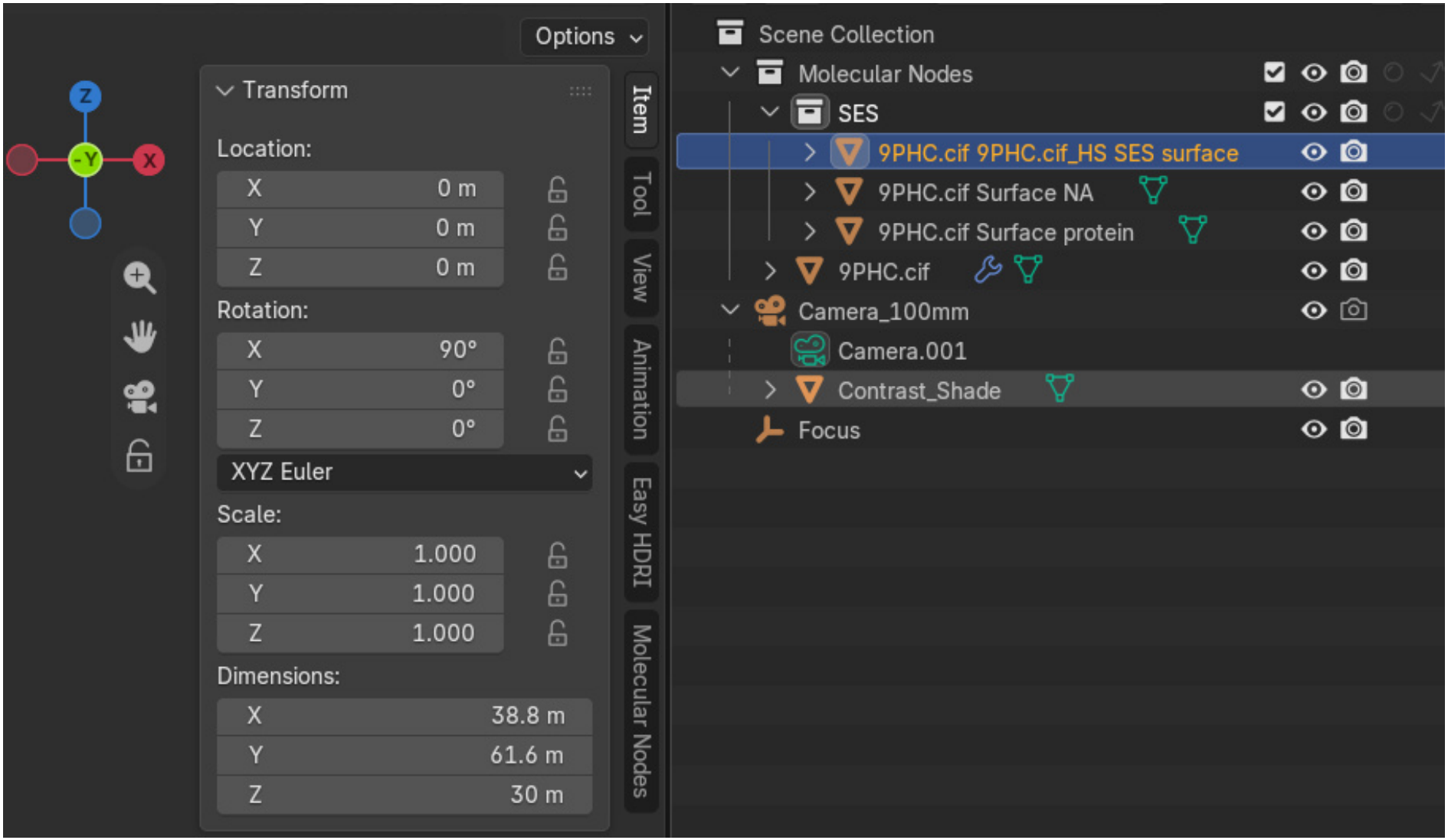
Recreating SNOR - Blender - Import OBJ



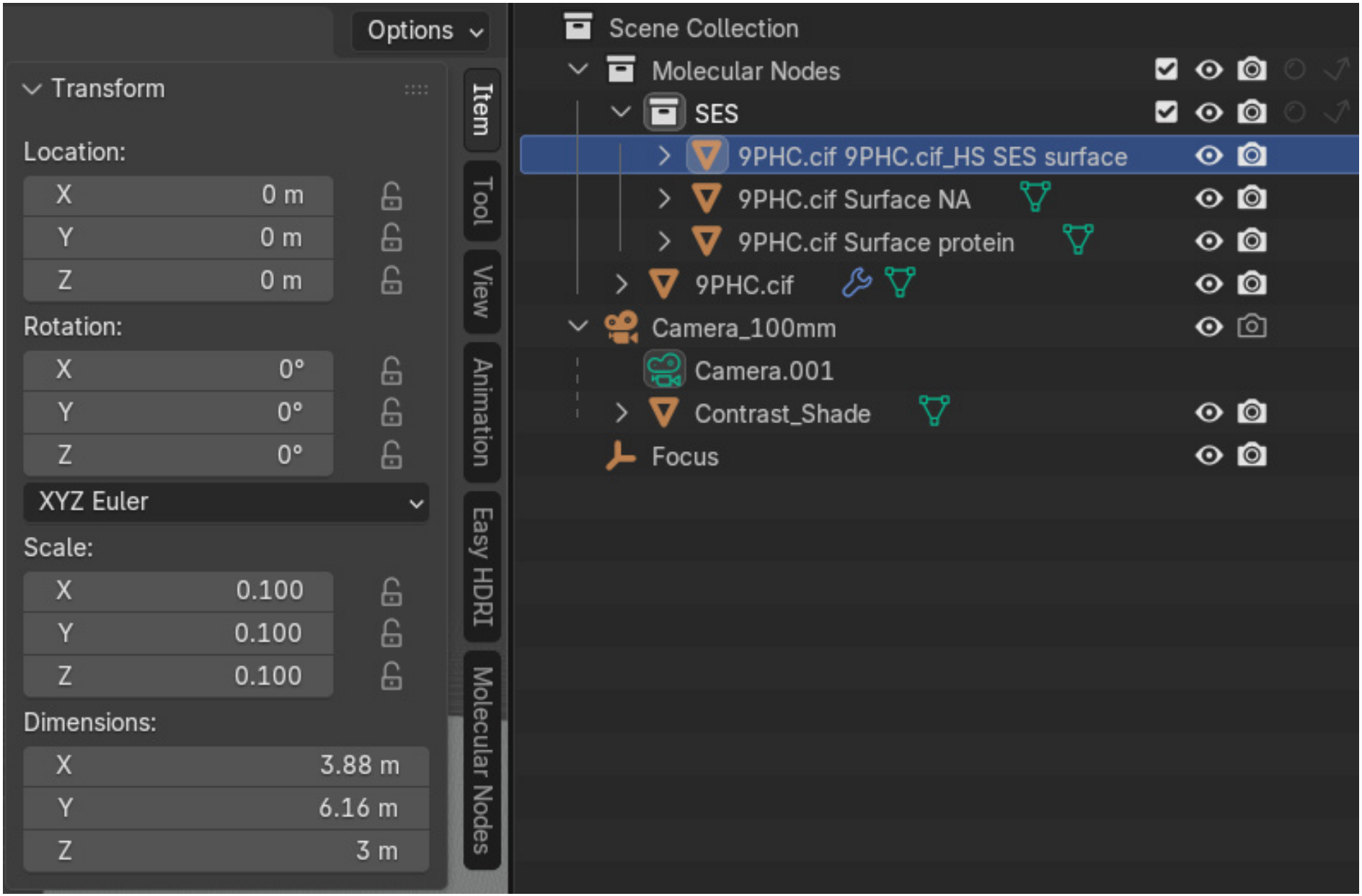
Recreating SNOR - Blender - New Collection



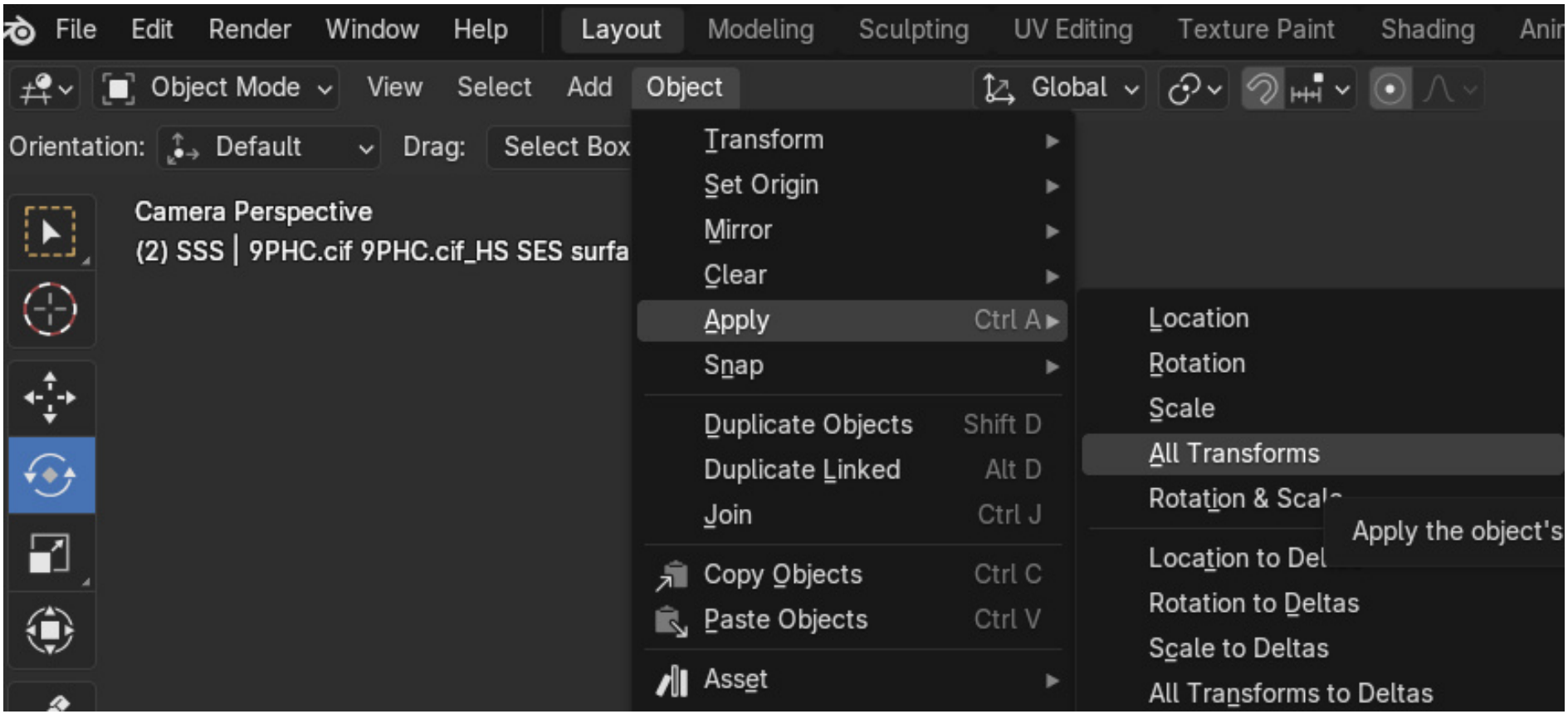
Recreating SNOR - Blender - OBJ initial state



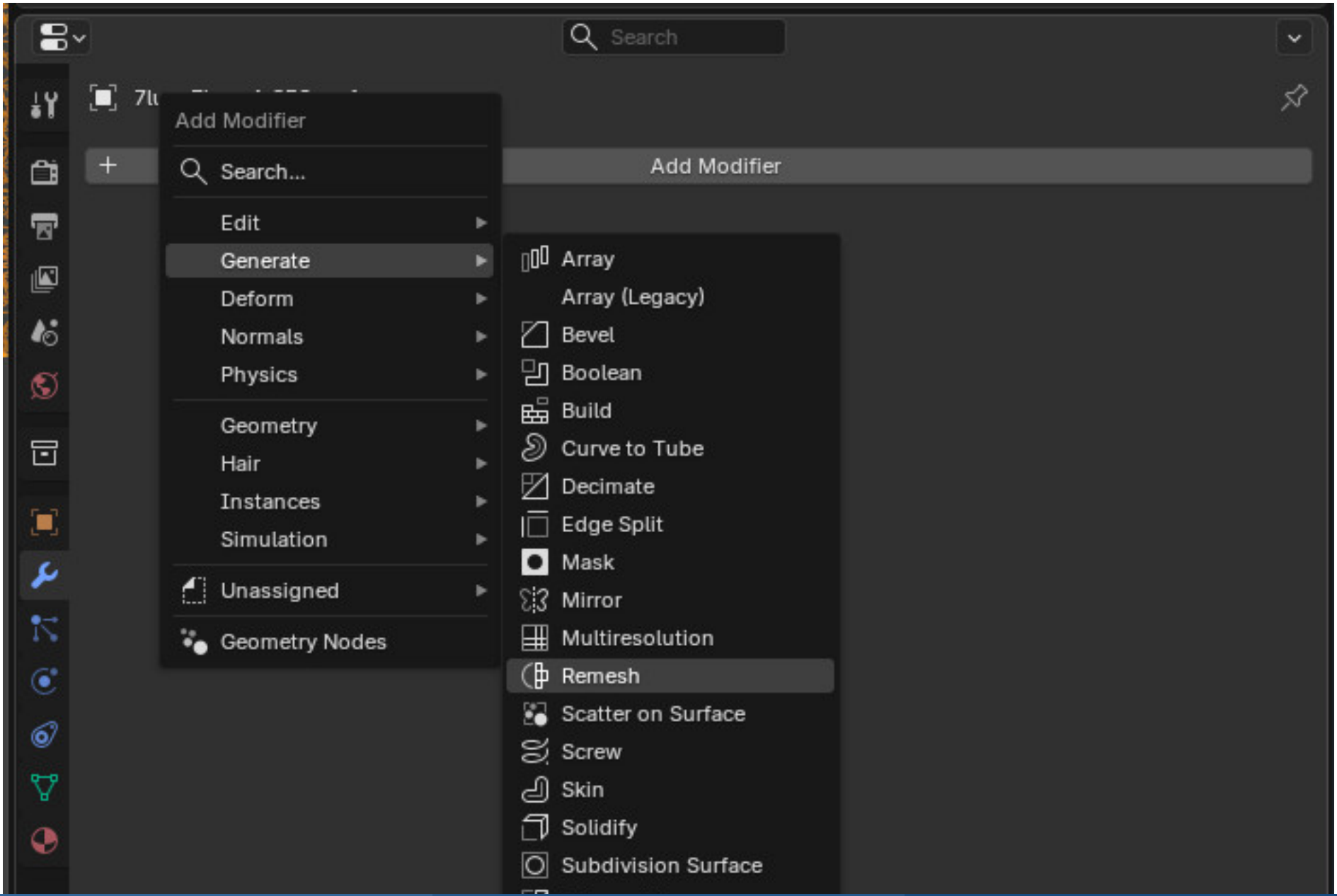
Recreating SNOR - Blender - OBJ new state



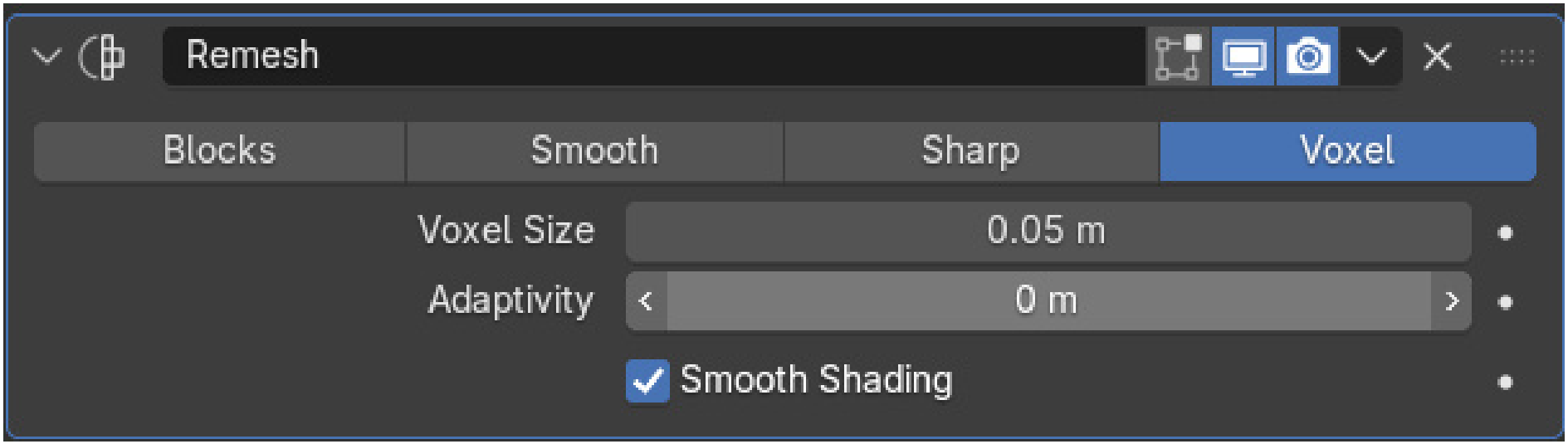
Recreating SNOR - Blender - OBJ apply new state



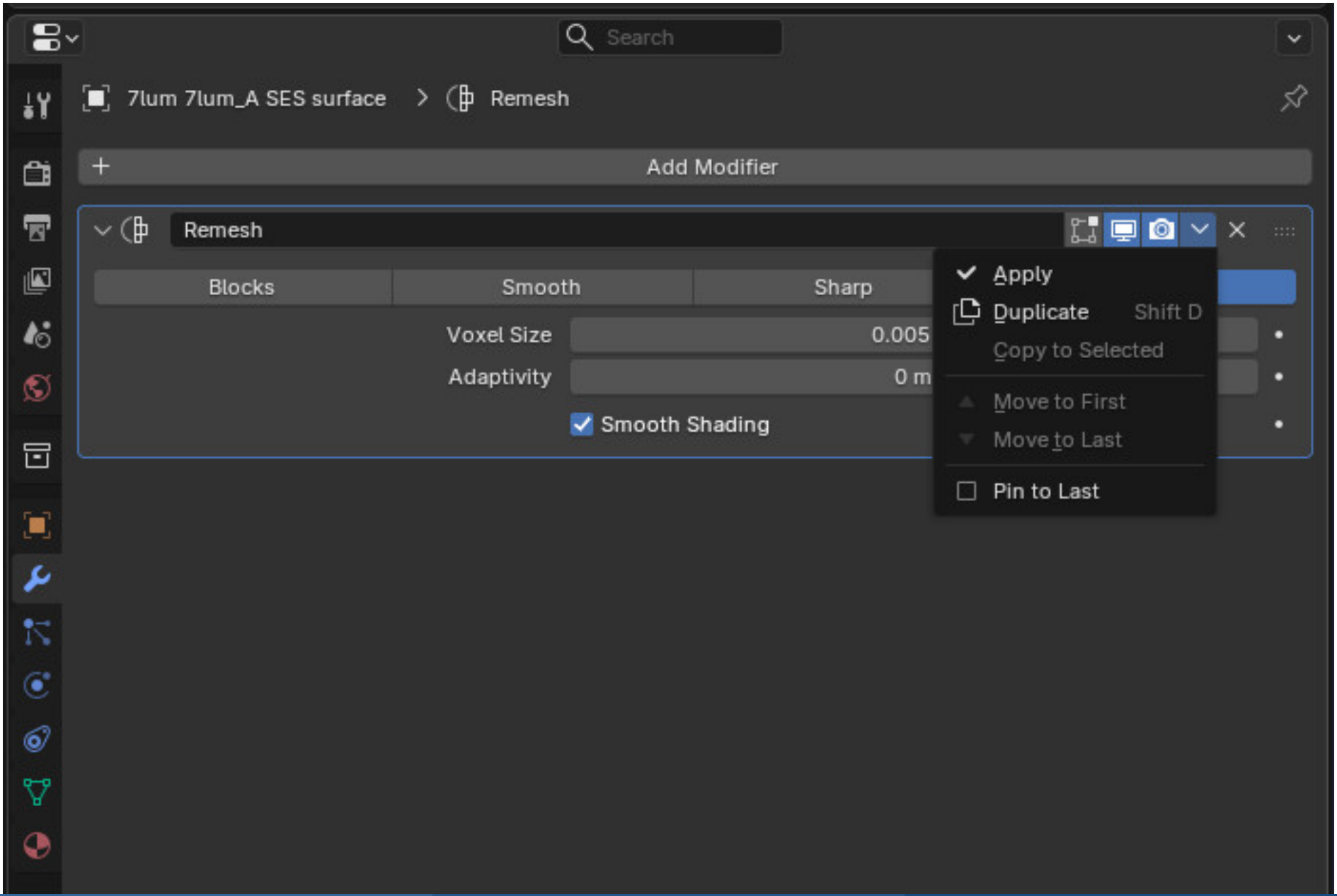
Recreating SNOR - Blender - Voxel Remesh



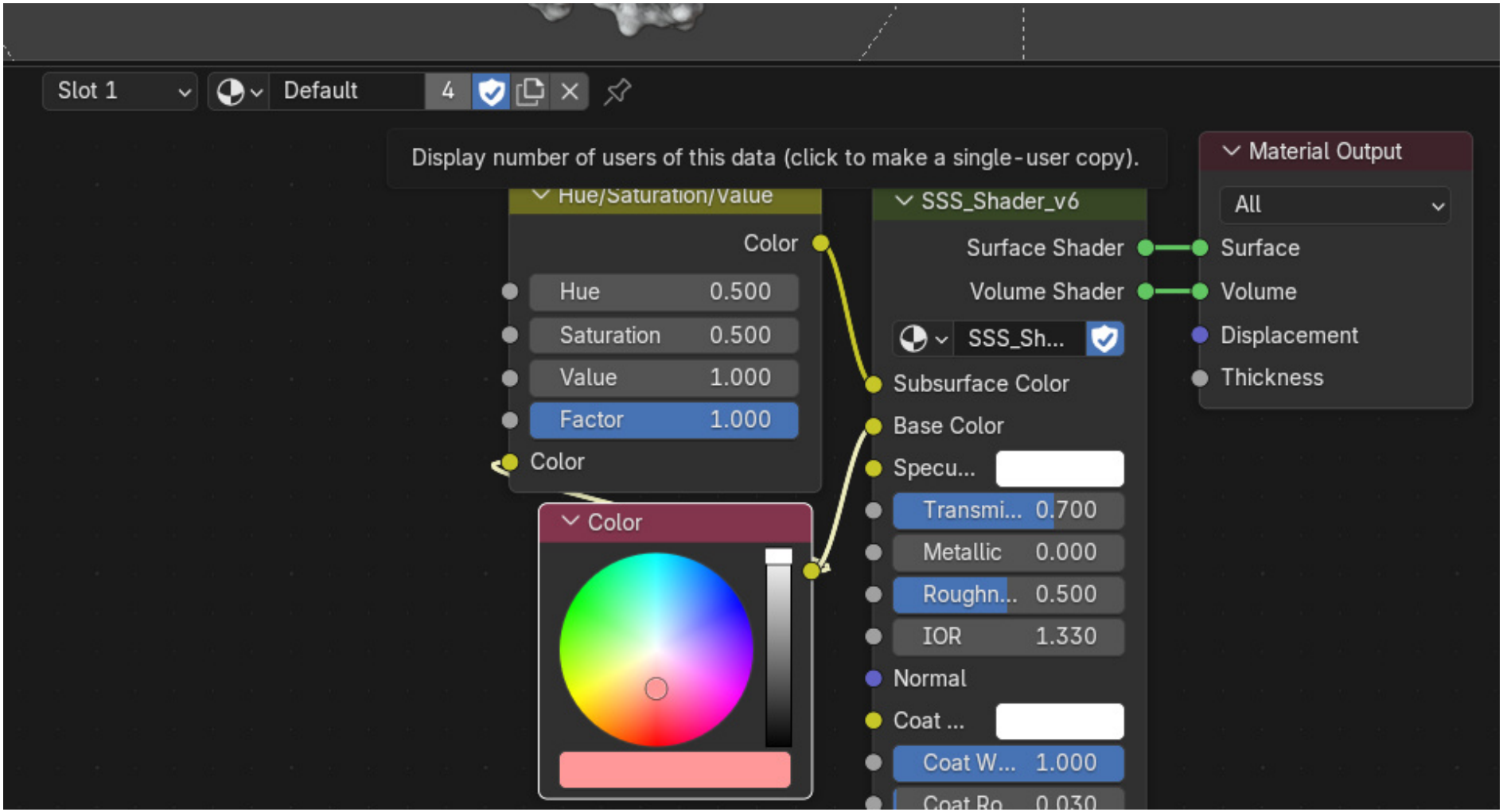
Recreating SNOR - Blender - Voxel Remesh Setting



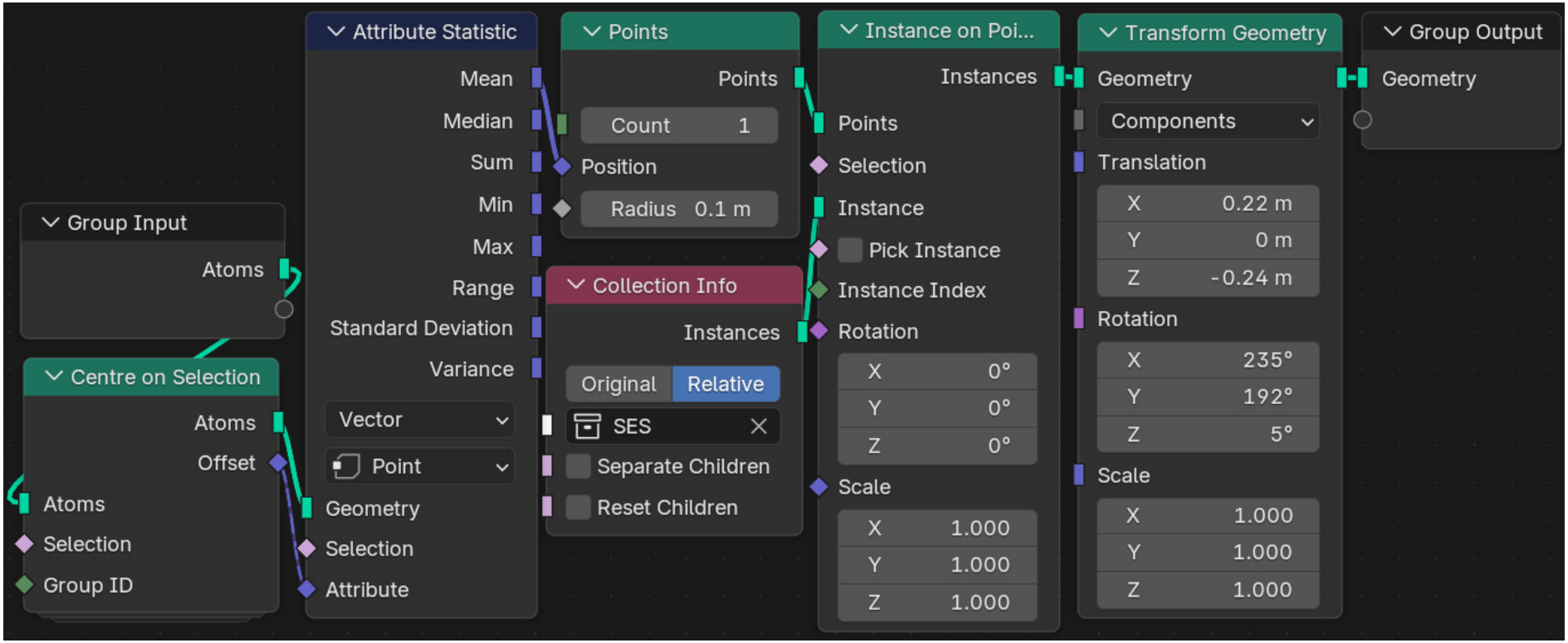
Recreating SNOR - Blender - Voxel Remesh Apply



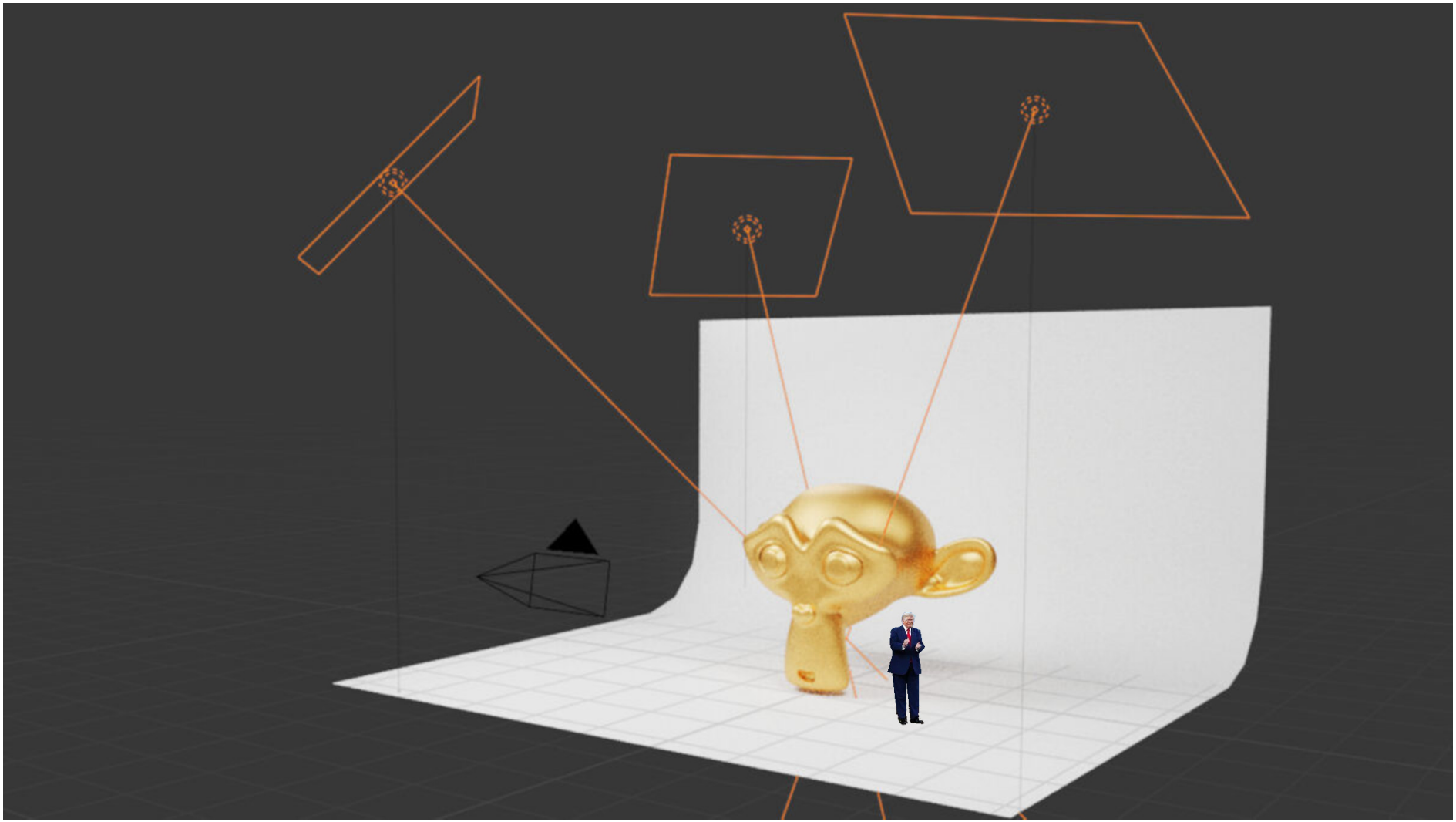
Recreating SNOR - Blender - Unique Material Copy



Recreating SNOR - Blender - Geometry Nodes Setup

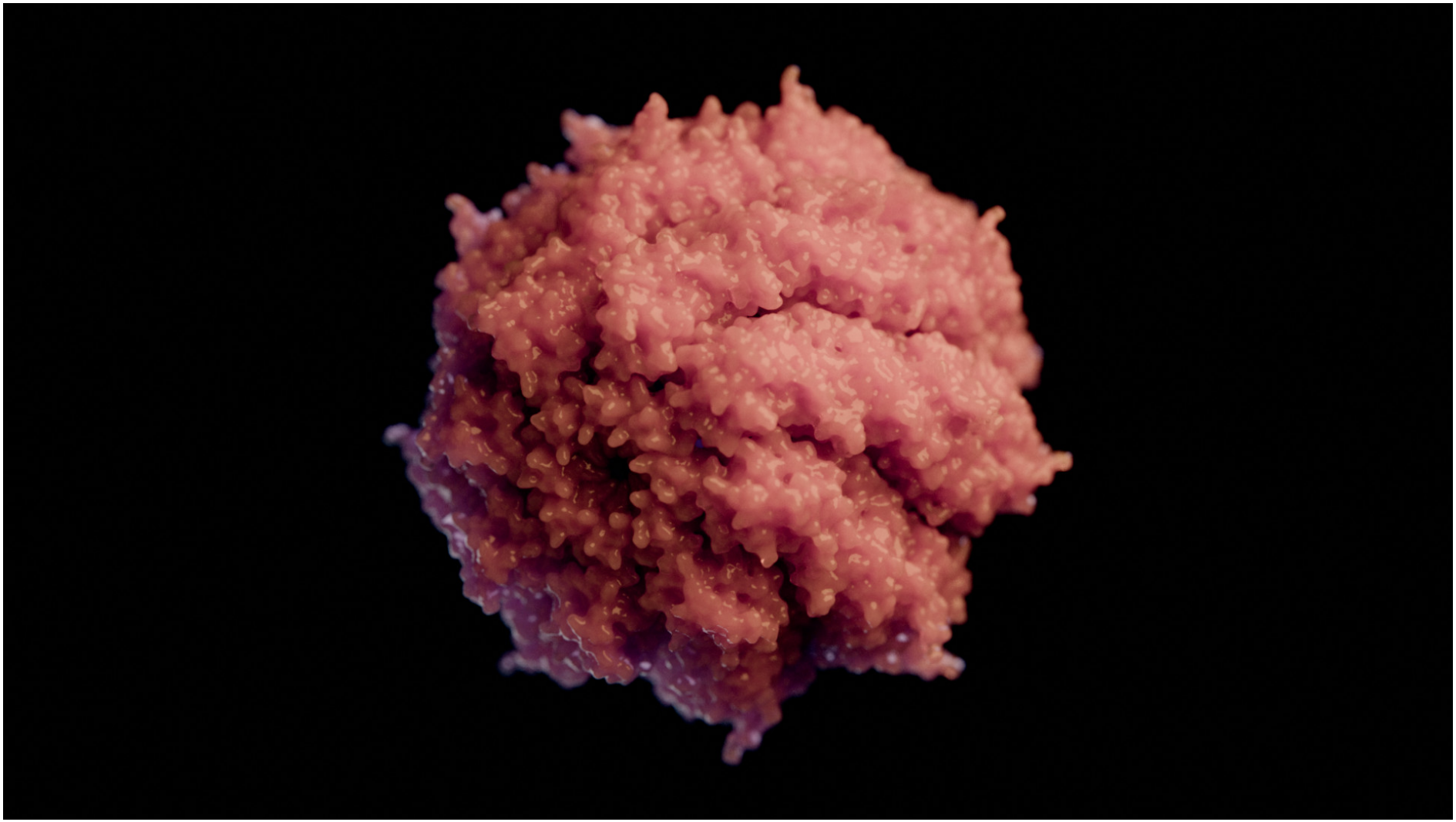


Lighting 1: 3 Point Lighting is tricky



Lighting 2: 3 Point Lighting is Tricky

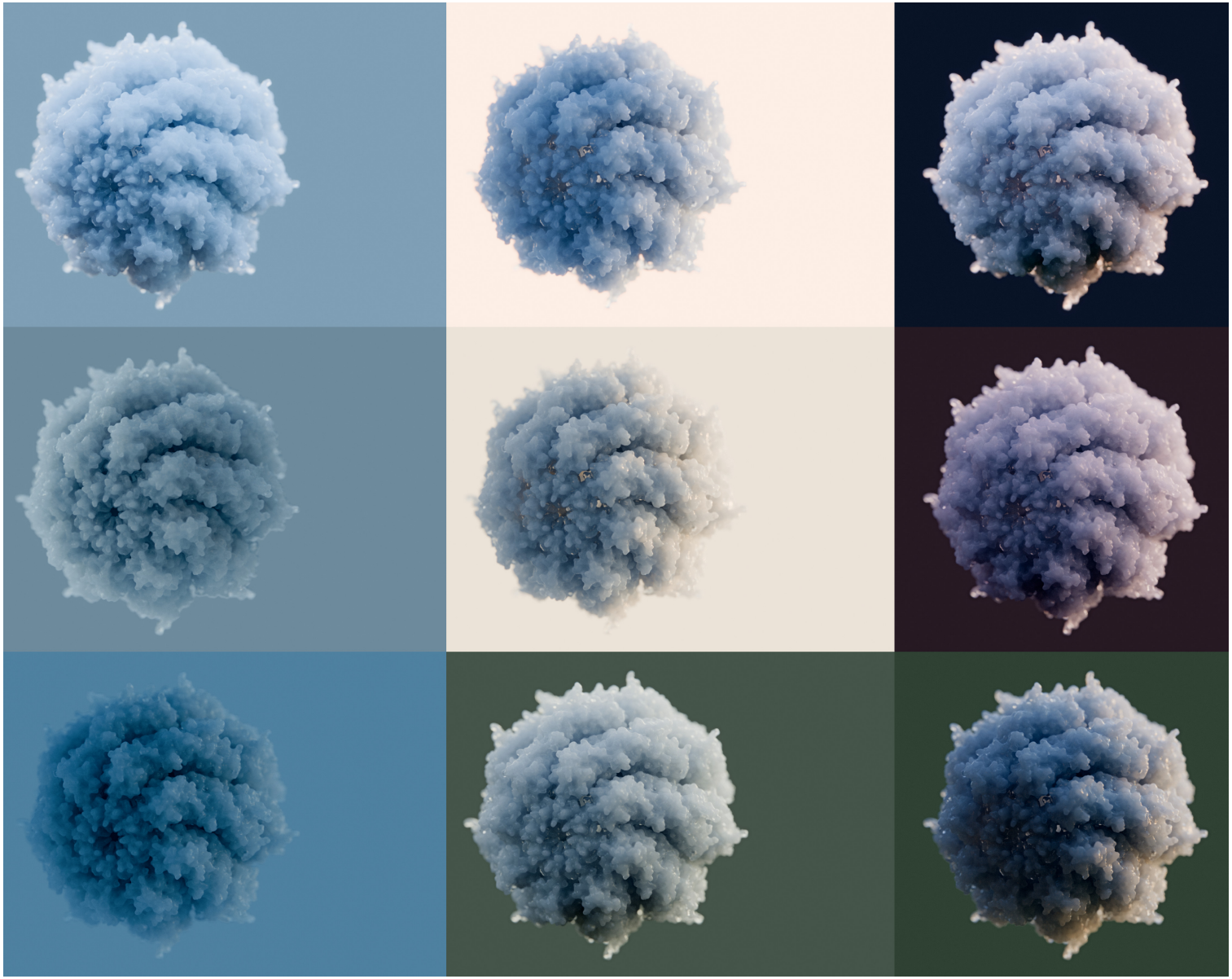
- It takes a lot of experimentation with the light to get a so so results
- 6_Example_7LUM/3_Point.blend



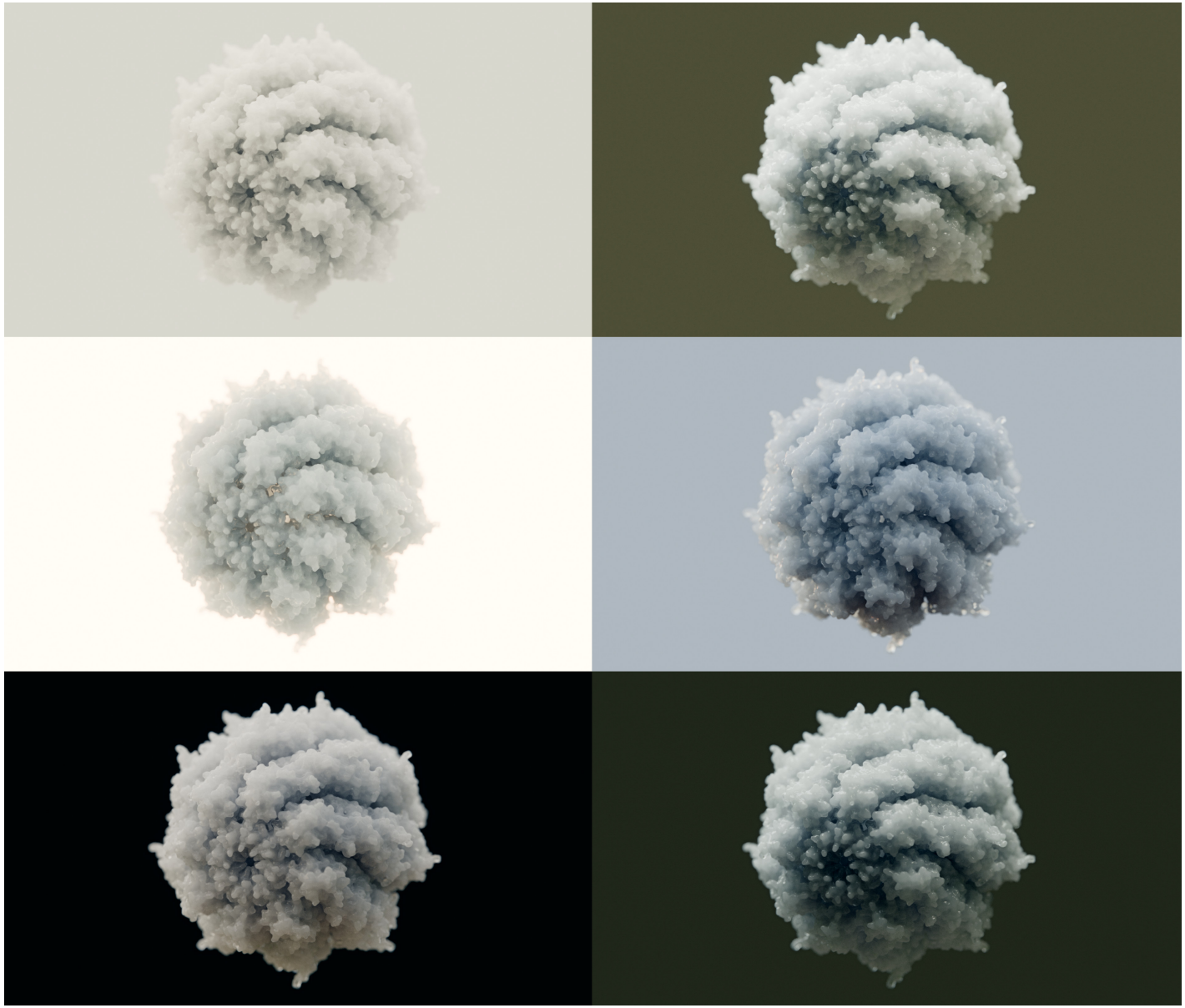
Lighting 3: HDRi are Easy to Use



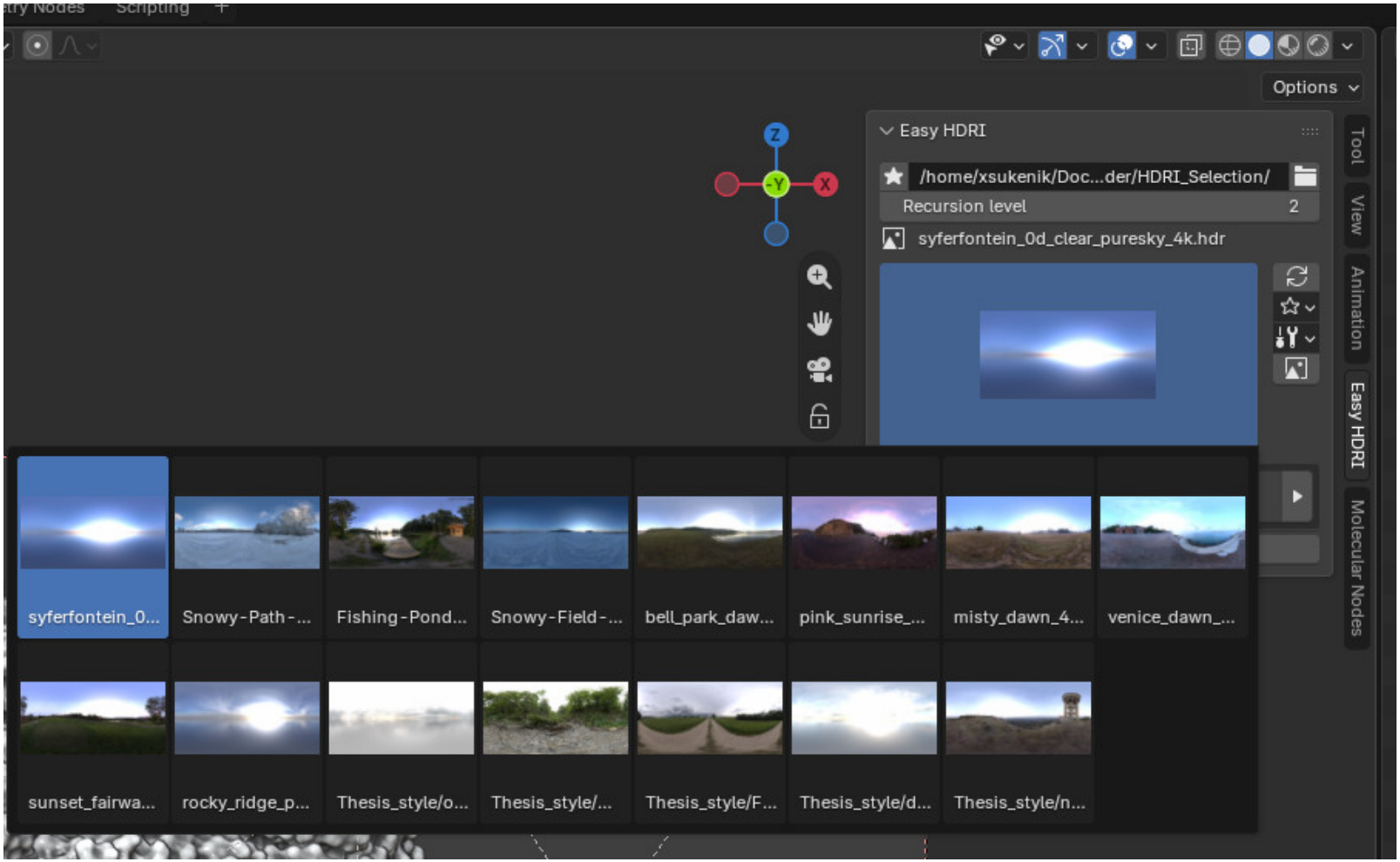
Lighting 4: Cover Style HDRi - Colorful



Lighting 5: Thesis Style HDRi - neutral colors

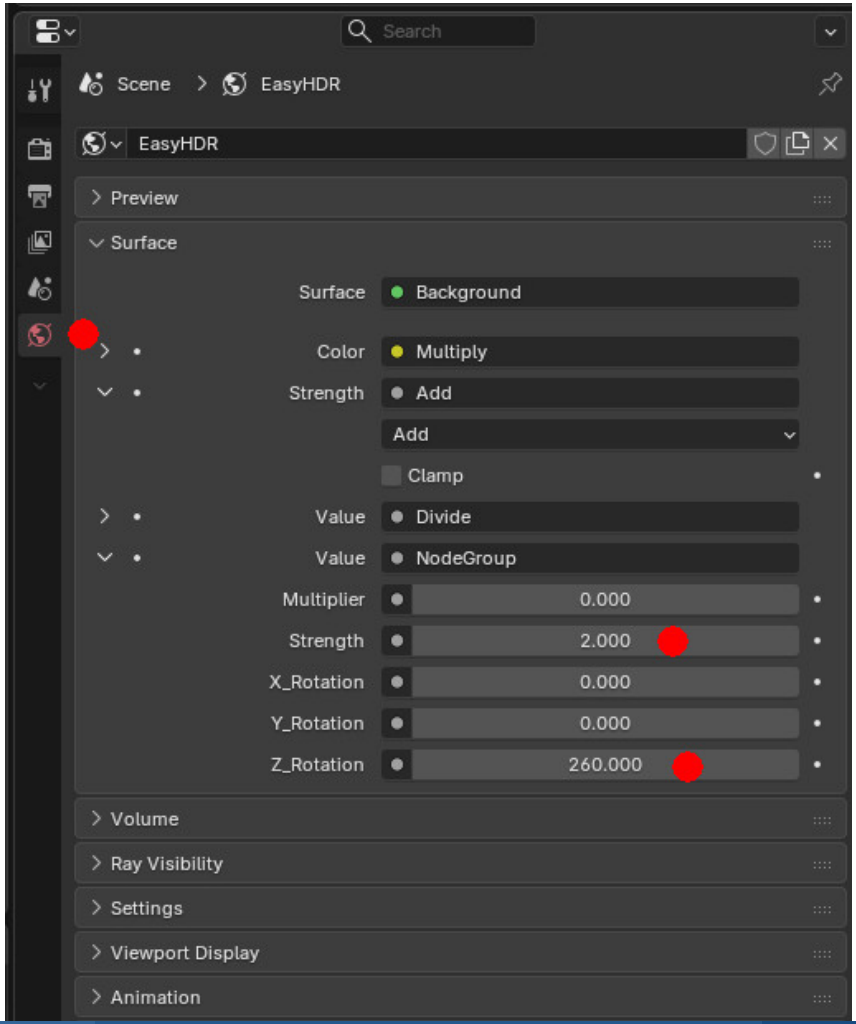


Lighting 6: Selecting HDRi via EasyHDRI

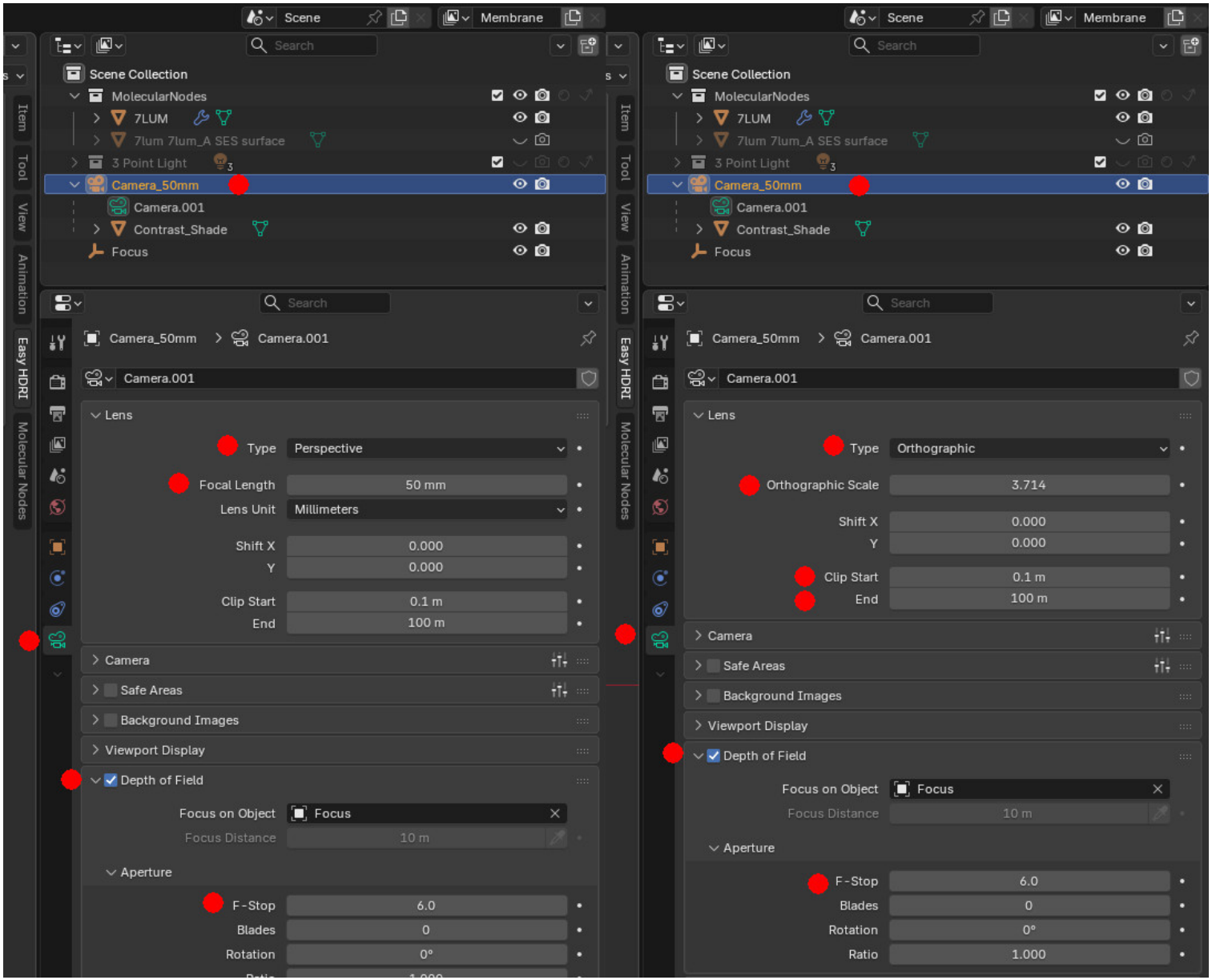


Lighting 7: HDRi Setting

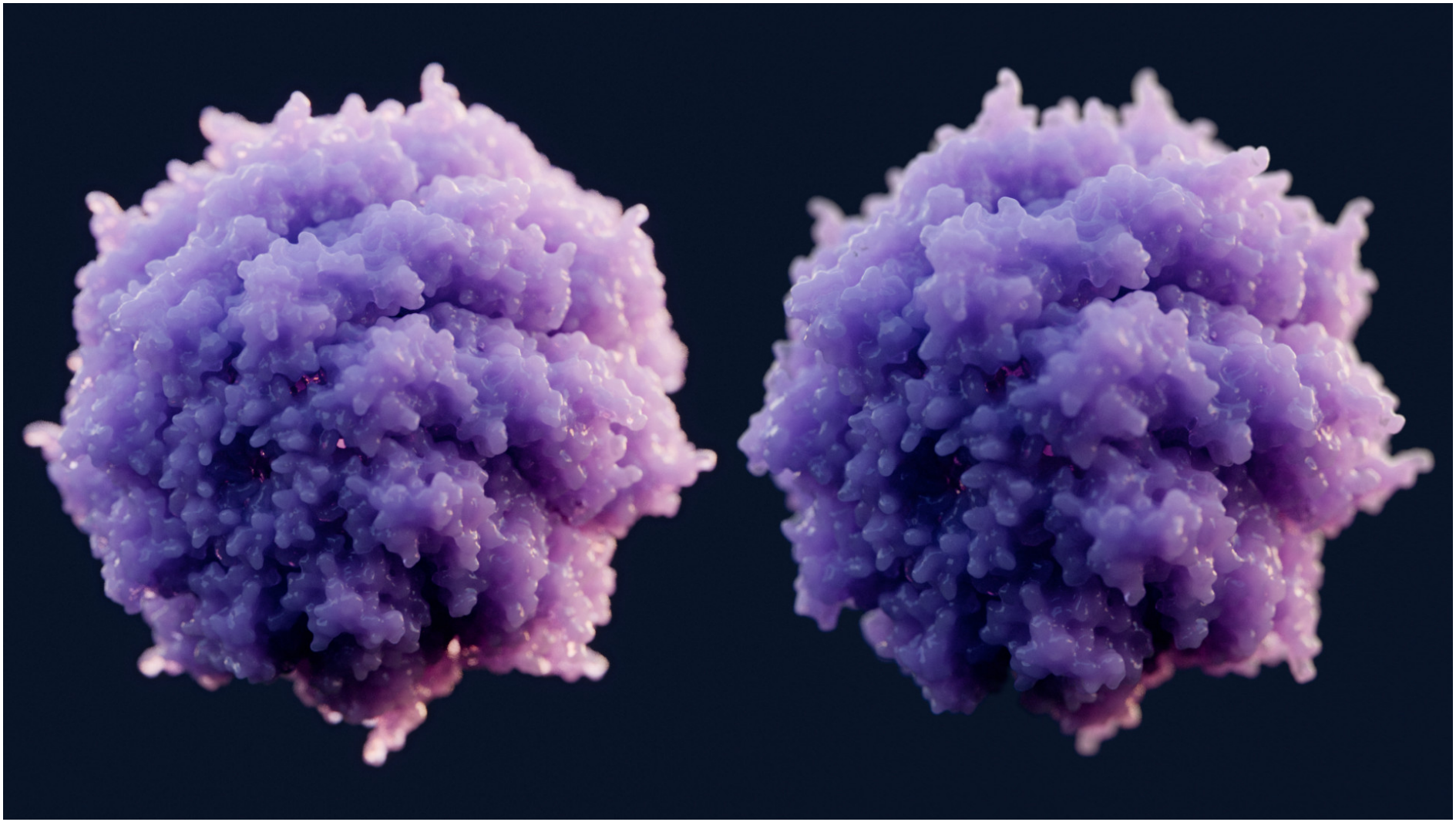
- Try out which HDRi and Setting you like
- Examples are in 7_HDRI_Renders, the setting is part of name
- S for strength, R for z rotation



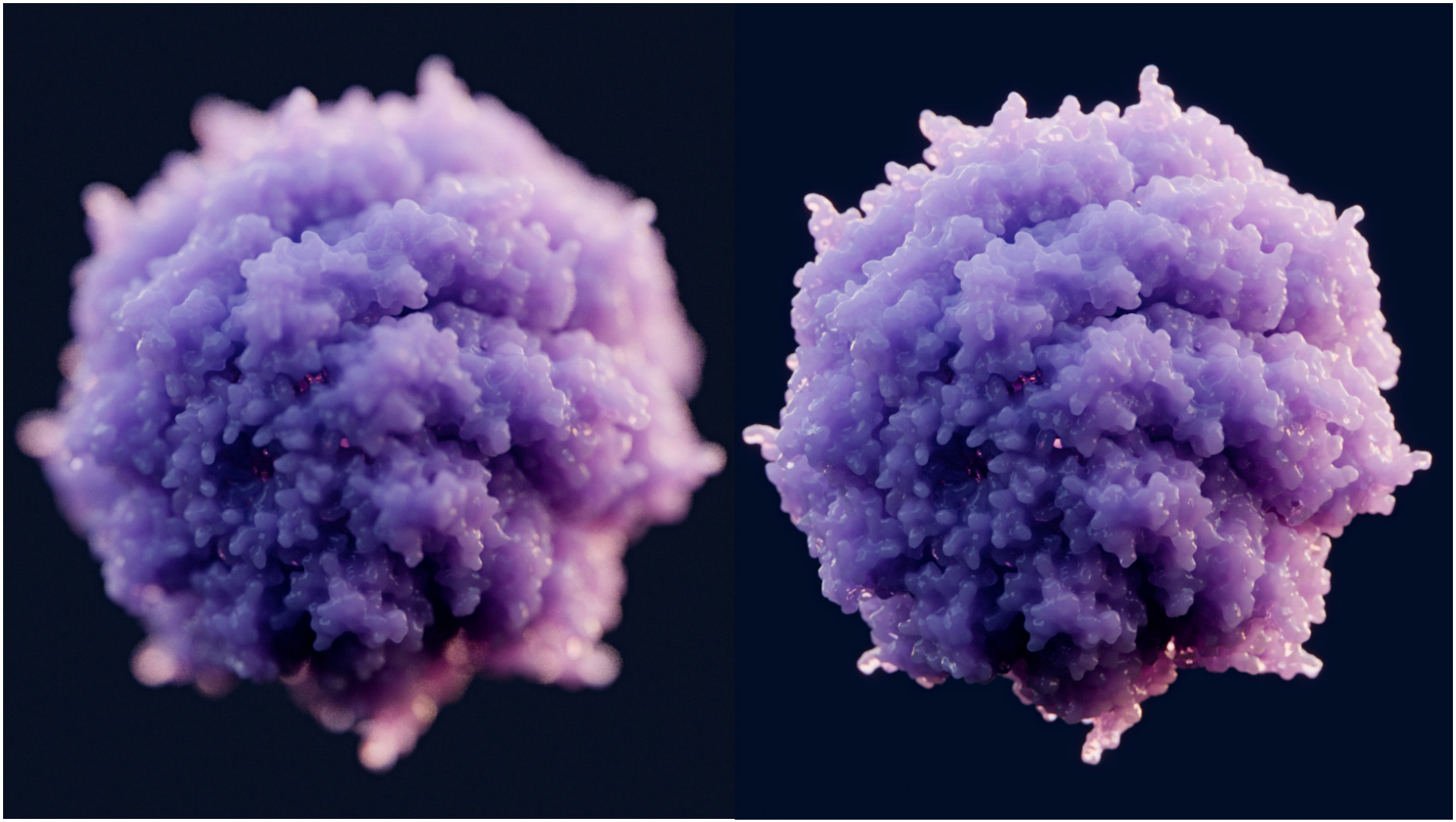
Camera Setting 1



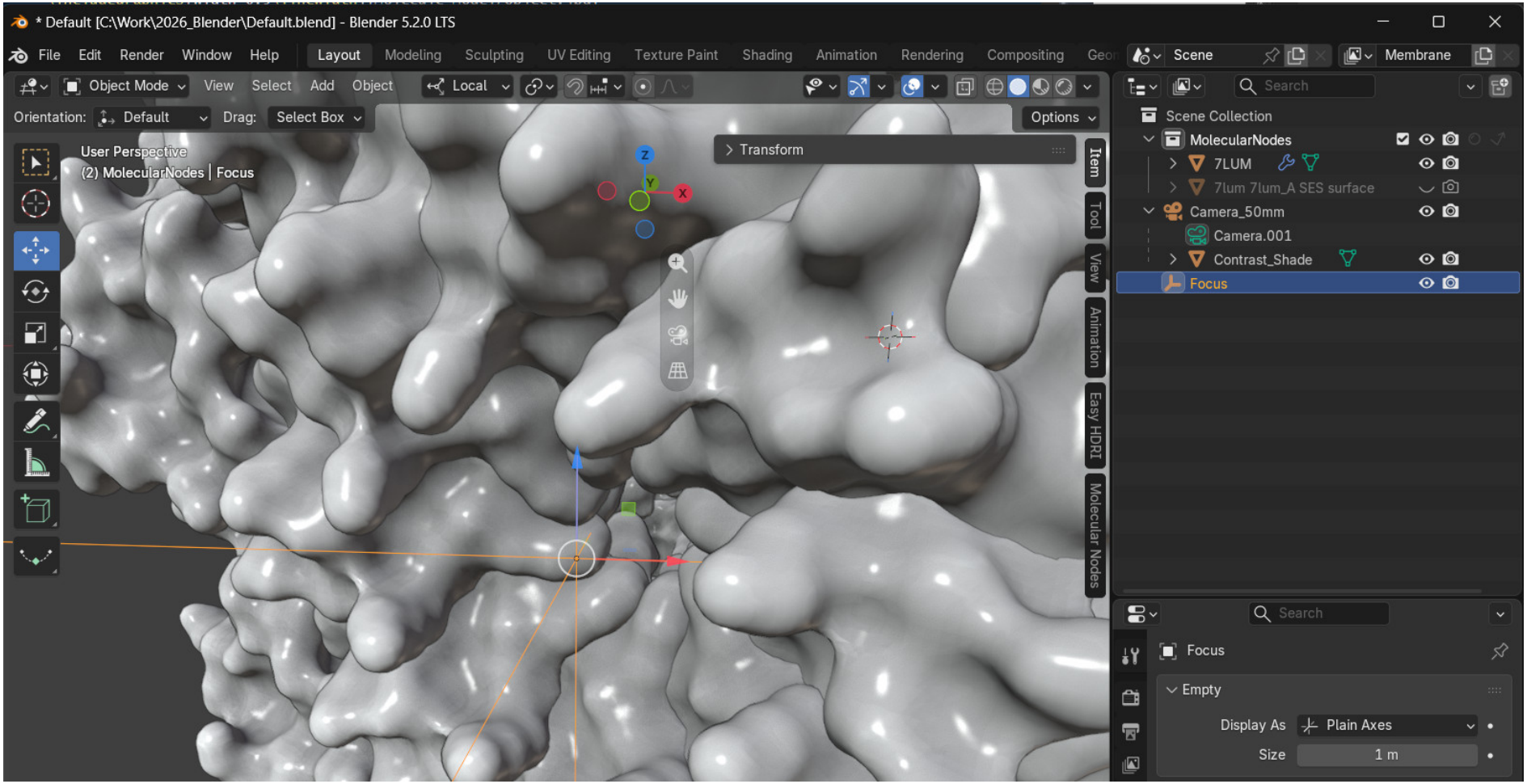
Camera Setting 2: Orthographic vs Perspective 30mm



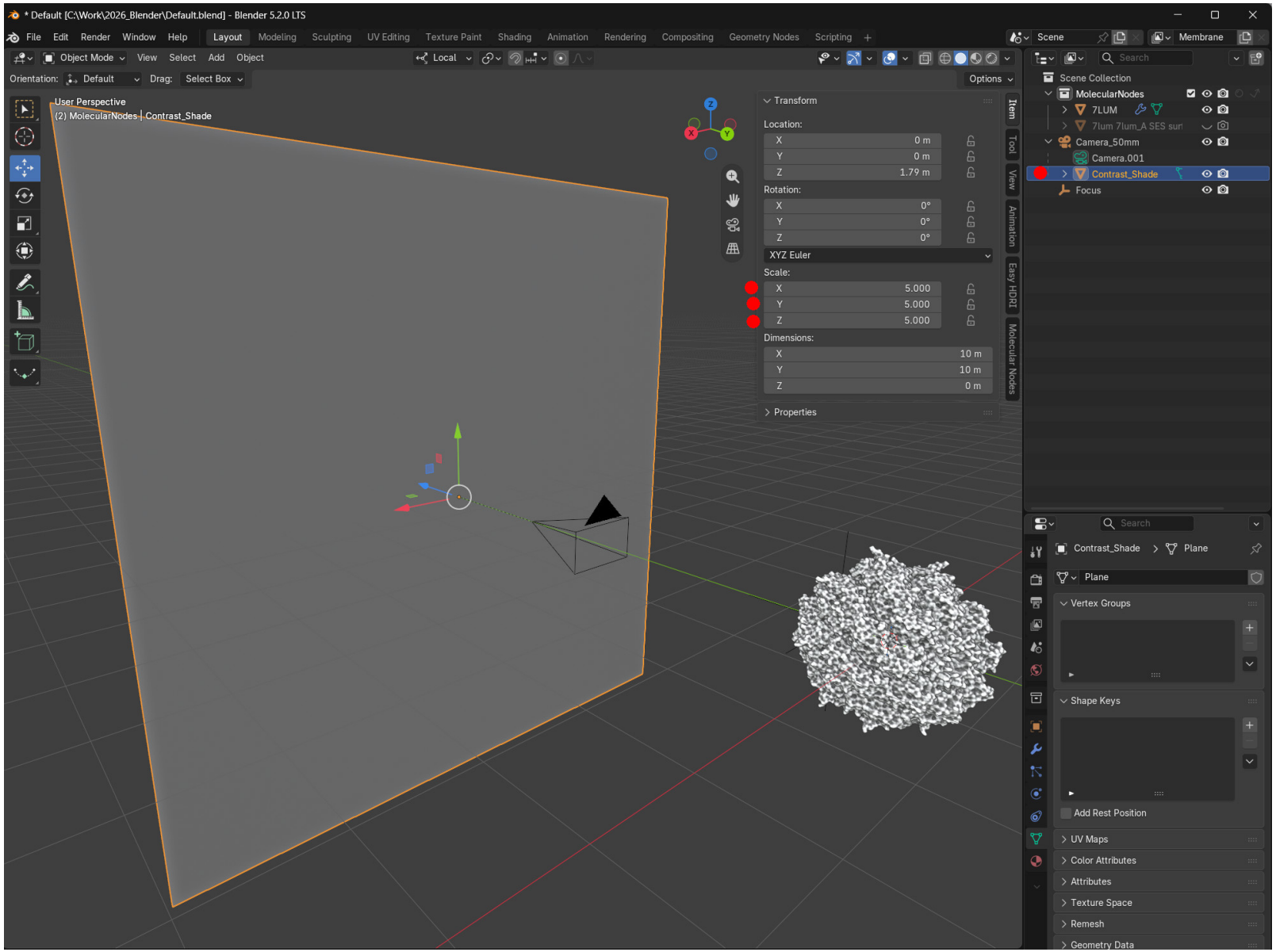
Camera Setting 3: Depth of Field, F-stop 2.0 vs 12.0



Camera Setting 4: Focus Object for Convenience

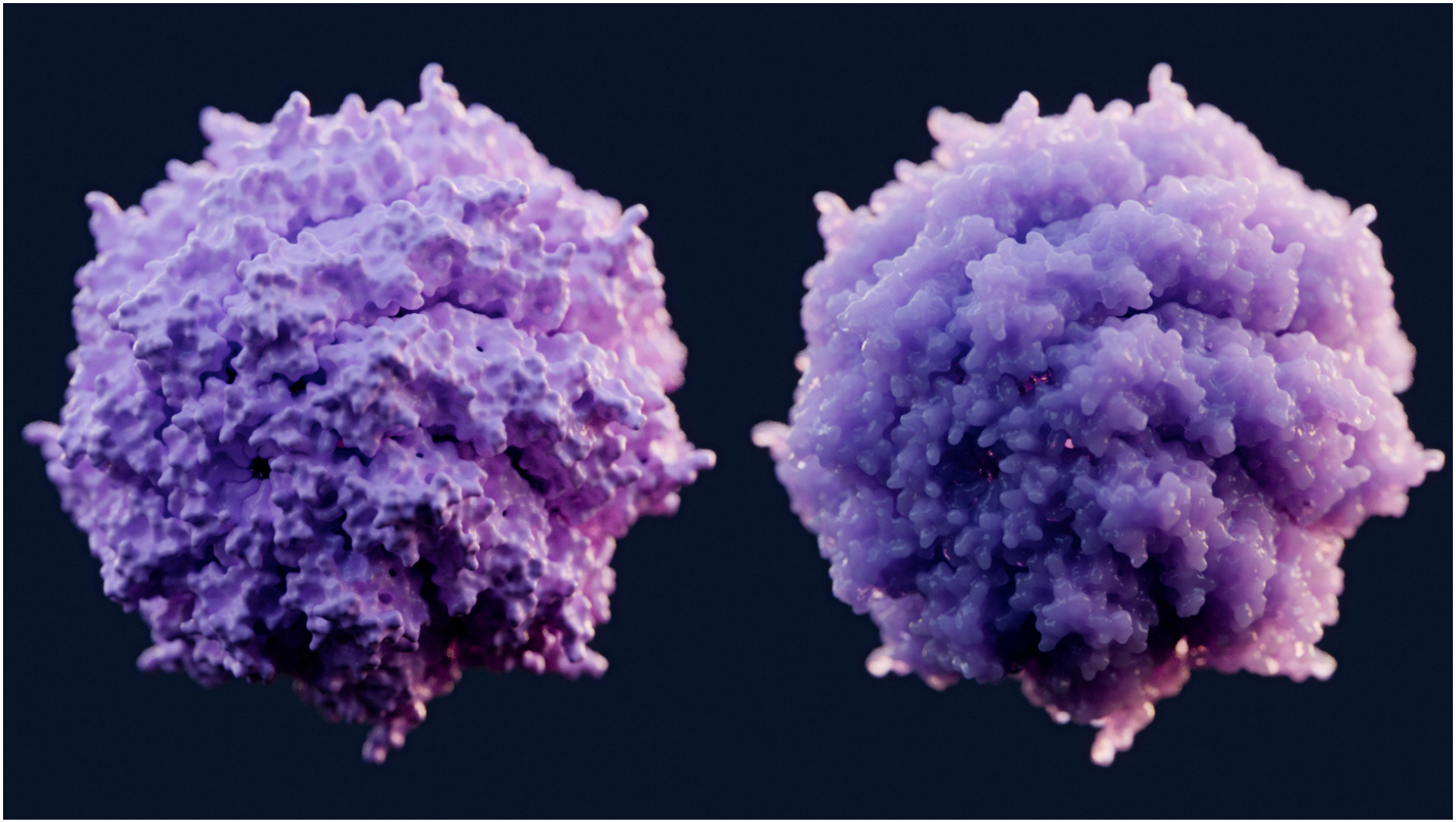


Camera Setting 5: Trick - Contrast Shade behind Camera

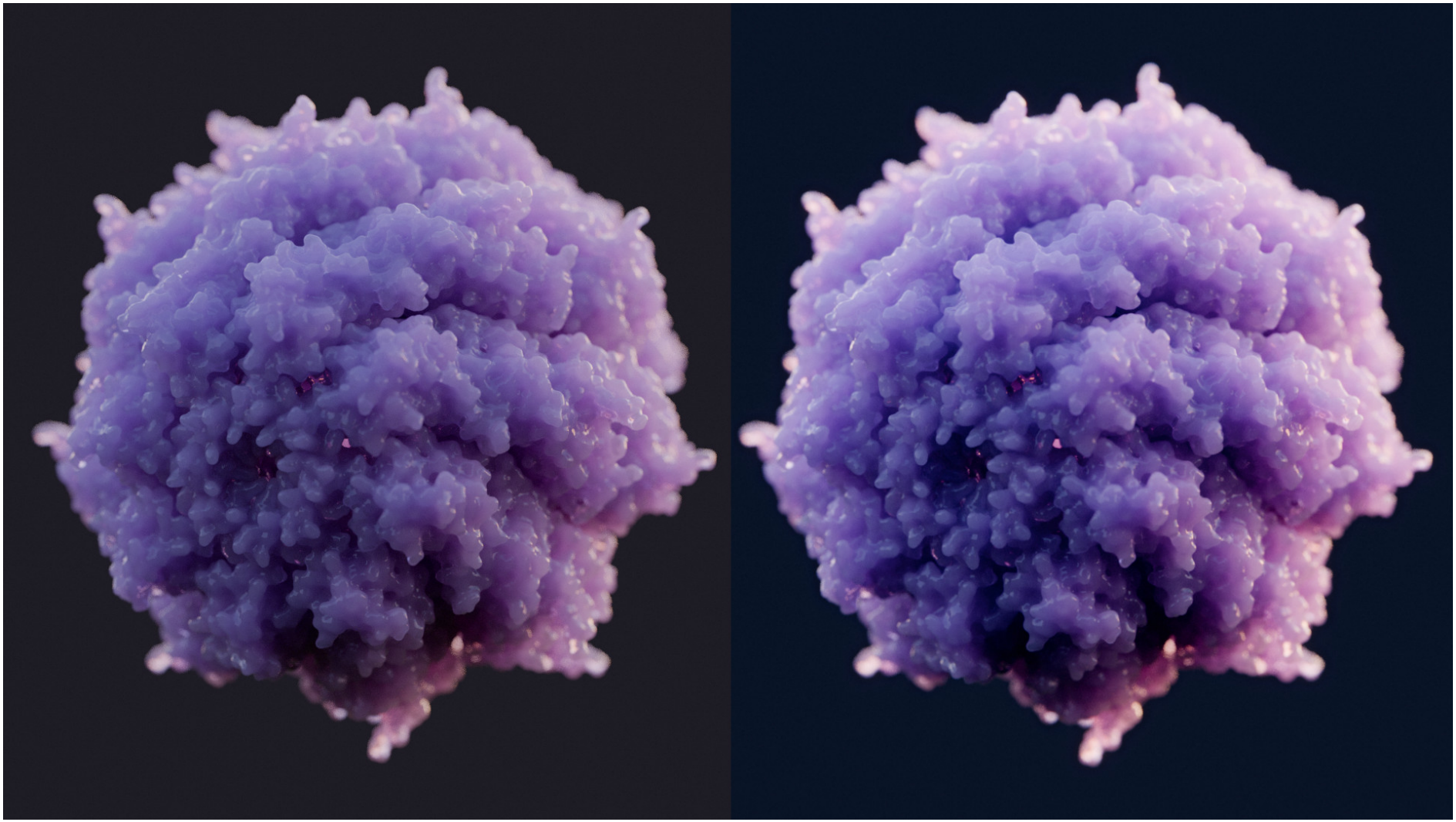


Shaders 1 - Sub-surface Scattering

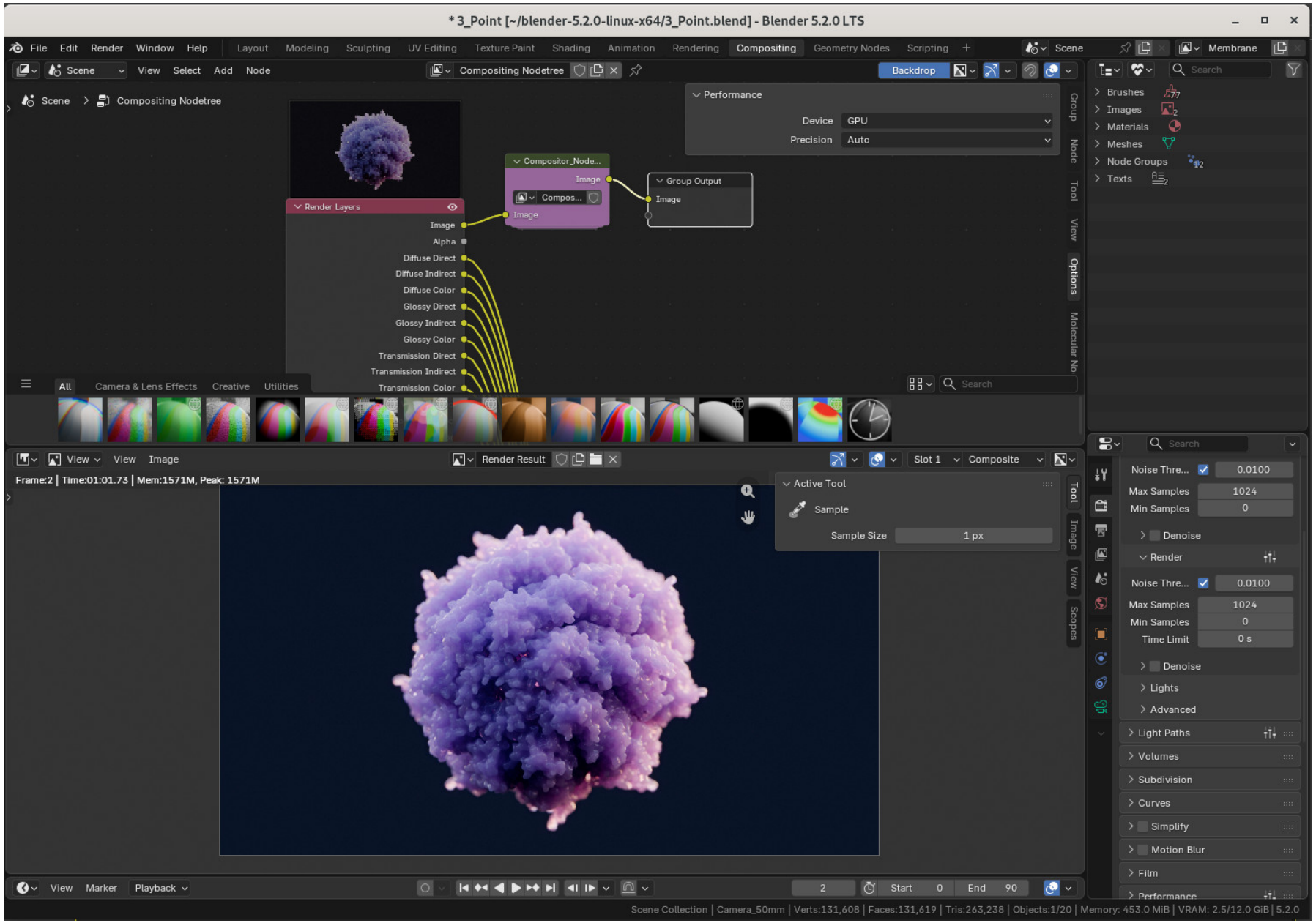
- A trick to good renders of molecules is to use sub-surface scattering



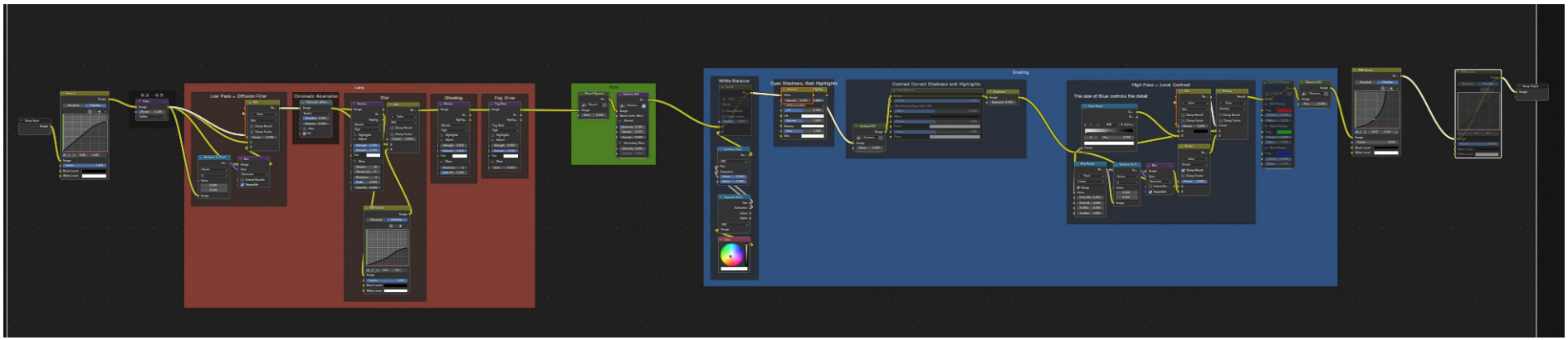
Post-processing 1 - Set and Forget



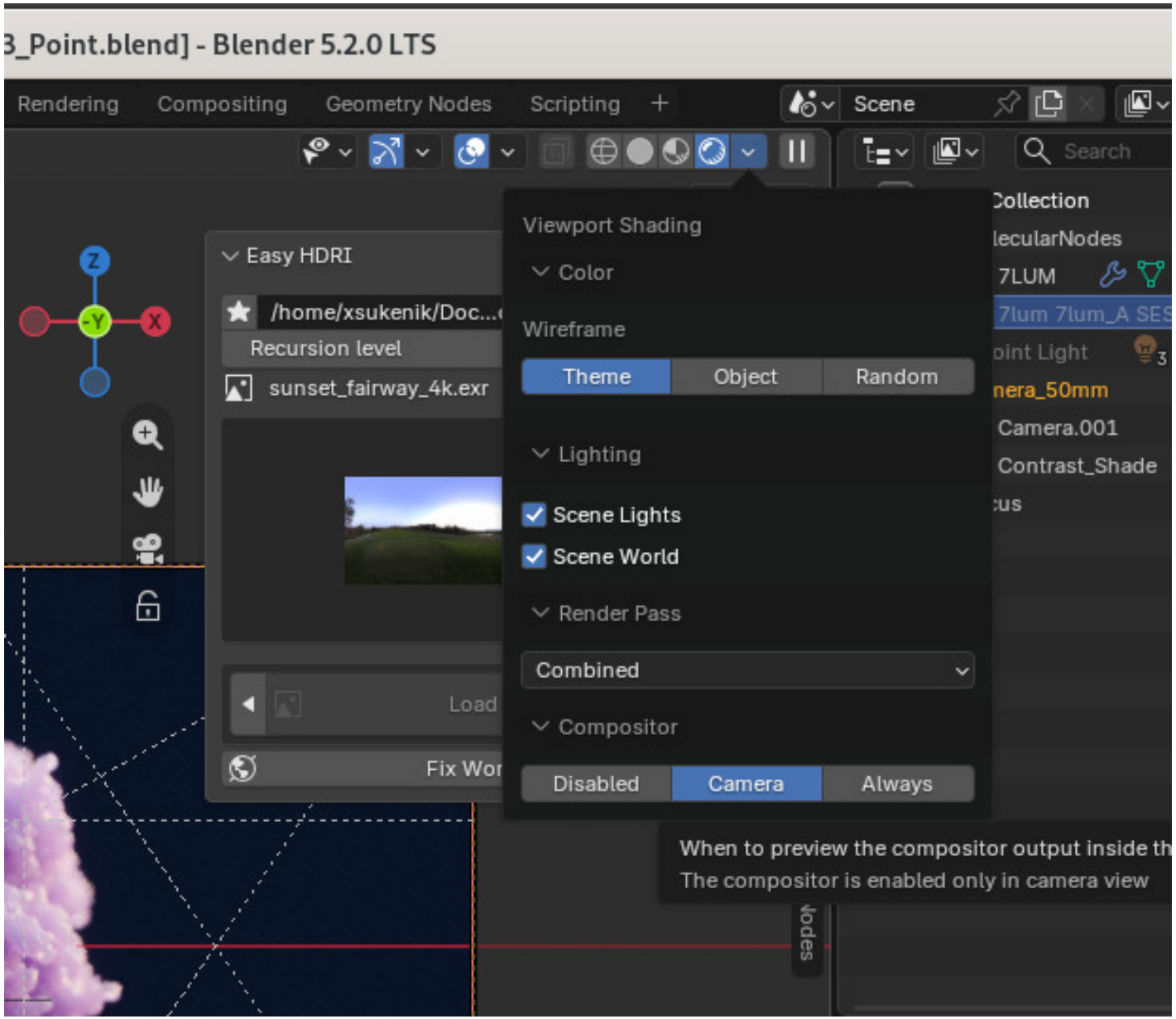
Post-processing 2 - Where it is



Post-processing 3 - Its complicated



Post-processing 4 - Turn it On During Preview



Try Creating your Own Molecule

Voluntary Exercise

- Download a molecule from pdb database
  - Try using some pdb molecule of the month
  - https://pdb101.rcsb.org/motm/motm-by-date
  - Full bacteriophages are large, don't try SES

